

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_7049e1ed-b93\agent-a82ce35ba77fa5f15.jsonl`  
- SHA-256: `090f6e52115e004ca165fa308418dafde553dcd9d2d3dff8e4f32794467ba167`  
- Source modified: `2026-07-02T04:58:36+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_7049e1ed-b93`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T04:45:39.424Z)

You are a senior engineer producing ONE isolated PR. Today is 2026-07-02.

REPO: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer (a shared checkout ? follow the safety rules exactly).

AUDIT FINDINGS YOU ARE FIXING: R1-13, R1-14, R1-01(doc half), critic DATA_IN_GCS + dropped-pool items ? Grep C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md for these IDs and read the full evidence + verifier notes before coding.

TASK BRIEF:

In C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer, ONE docs-only reconcile PR: (1) docs/DOC_STATUS.md:47 cleanup-audit row ? current reality (Phase 2 nearly done; remaining P10 + P16+; add a DECISION_SNAPSHOT_REGRESSION row); (2) docs/README.md:41-42 stale blurbs; (3) docs/NEXT_STEPS.md 'Where we are' rewrite to post-06-21 state ? PRESERVE the owner NO-GO block (line ~8, #404) VERBATIM; (4) docs/BACKLOG.md:39 ByteTrack row ? done (#386) and DELETE its '<0.30.0 cap' sentence (cap already lifted, requirements.txt:18); flip only what the verifier confirmed (lines 22/24 rows are PARTIAL ? annotate rather than flip, see report R1-13 verify notes); (5) docs/CODE_CLEANUP_AUDIT_2026-06-24.md: mark D13 ? (#386), refresh the P10/D1 row status (PR #390 stale/conflicting ? to be re-cut); (6) the stale ByteTrack docstring in backend/pipeline/detectors/person_detector.py (doc-comment only ? no code change); (7) docs/DATA_IN_GCS.md \$5 gap-table rows contradicting \$7 (trajectories/labels migrated 2026-06-22) ? reconcile. VERIFY each edit against code/git before writing it. No behavior changes.

BRANCHING SAFETY (mandatory ? shared checkout, sibling sessions exist):

1. In the MAIN repo checkout run: git fetch origin
2. Create an ISOLATED WORKTREE: git worktree add "C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-<shortname>" -b <branch-name> origin/master (branch explicitly FROM origin/master ? never from the current local branch; use origin/main for repos whose default is main)
3. Do ALL work inside the worktree. Before committing: git branch --show-current + git log --oneline -1 must show YOUR branch on the origin-master base.
4. Stage EXPLICIT paths only (git add <file> <file>) ? NEVER git add -A / . / -u
5. Before opening the PR: git log --oneline origin/master..HEAD must list ONLY your commits.

COMMIT/PR CONVENTIONS:

- Commit message: conventional style matching the repo's history; end the message with a blank line then: Co-Authored-By: Claude Fable 5 <noreply@anthropic.com>
- PR body: what/why, a "## Verification" section stating exactly what you ran locally and what CI covers, and end with: ? Generated with [Claude Code](https://claude.com/claude-code)
- Do NOT merge the PR. Do NOT remove the worktree (a fixer may need it). Return the PR number, branch, worktree path.

QUALITY BAR:

- Read the repo's CLAUDE.md first; for the engine also docs/CODE_MAP.md \$0 (file placement) before adding any new file.
- Add/extend tests for every behavior change. Run the targeted tests you touched (python -m pytest tests/<relevant> -q) + ruff check on changed files; if mypy config exists, run mypy on

touched modules. Full suite is CI's job ? do not run it.

- Default-safe: new knobs default OFF / behavior-preserving unless the task explicitly says otherwise.

- The full audit evidence is in C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md ? Grep it for your finding IDs and READ the verifier notes; they contain corrections (exact file:line, scoping traps) you MUST honor.

If the brief's premise turns out false against current code (something already fixed, file moved), adapt honestly and say so in notes ? never force a pointless diff. If you genuinely cannot complete (env broken, auth missing), open NO PR and explain in notes with pr_number=0.

2. user (2026-07-02T04:45:47.546Z)

master...origin/master

3. user (2026-07-02T04:45:47.640Z)

CLAUDE.md

4. user (2026-07-02T04:45:52.401Z)

```
1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid Gemini as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - docs/README.md ? doc index. docs/CODE_MAP.md ? code navigation + data
9        contracts (the cold-start map; read before touching code).
10     - Authoritative owned-model roadmap: docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md (+
11       the
12       live docs/NEXT_STEPS.md). These SUPERSEDE the 2026-06-07 strategy docs below where
13       they
14       disagree (licensing, Gemma-4 stance, base model, conversational-QA).
15     - Strategy (post-scaffolding): docs/COMMERCIALIZATION.md,
16       docs/PMF_MARKET_RESEARCH.md,
17       docs/PLATFORM_ARCHITECTURE.md, docs/DEFERRED_HARDENING_PLAN.md. Also
18       docs/ROADMAP.md
19       (offline-segmenter narrative) and docs/BACKLOG.md.
20     - Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation
21       in progress):
22       docs/HOW_RALLY_DETECTION_WORKS.md (2-layer mental model) ?
23       docs/MULTI_SIGNAL_FUSION_PLAN.md
24       (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built
25       rally_gate.py
26       "Gate A" not wired into production ? cheapest win) ? docs/EXPERIMENT_HARNESS.md
27       (A/B + shadow
28       eval so cues land flag-gated/default-OFF with zero regression) ?
29       docs/ROORA_LABELING_GUIDELINES.md
30       (v8 vendor labeling spec). docs/NEXT_STEPS.md has the prioritized pick-up for this
31       track.
32     - History: docs/archives/ ? ADRs (decisions/DECISIONS.md), research, and
33       checkpoints/ (latest: 2026-06-strategy-foundation/). Archive policy: only
34       move
35       terminal-state (DONE/REJECTED) work; live docs stay at docs/ top level ? and
36       before
37       archiving, HONESTLY update the doc first (verify claims vs code ? flip status + a
38       completion
39       note ? update inbound refs ? keep the lineage ? then git mv), per the checklist +
40       living
41       lifecycle register in docs/DOC_STATUS.md.
```

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28
29     ## Current state (June 2026)
30     - **Scaffolding complete**; strategy foundation + owned-model roadmap merged. **No
31       platform/owned-model implementation has started.** A 2026-06-10 audit-driven hardening
32       sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate,
API
33       validation, reliability, CI + test coverage) ? see git history.
34     - **Two tracks, both owner-gated:** the **committed next PLATFORM slice = async job
model**
35       (`DEFERRED_HARDENING_PLAN.md` #16, single-tenant); the **owned-model track** starts at
a
36       greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start
37       implementation without explicit owner approval.**
38
39     ## Architecture in one breath
40     - **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers /
validators
41       / sports / annotations ? see `backend/pipeline/segmenters/base.py`. **Future:**
42       `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
43     - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
44     - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
45       user_version` migration ladder in `backend/infrastructure/database.py`.
46
47     ## Committed guardrails (non-negotiable)
48     - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
49       training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
50       code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-
agreed).
51     - **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
**Gemma-4
52       is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
53       needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license).
**Exclude
54       minors** from the training pool; per-user training data needs a separate explicit opt-
in.
55     - **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly
future**
56       (GDPR Art. 9).
57     - **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.
58     - **Compute = GCP ONLY (ADR-013, 2026-06-20):** GPU training + serving run on **GCP**
(`deploy/gcp/`;
59       Cloud Run+GPU for serving per ADR-012). **DigitalOcean is DROPPED for compute** (DO
GPU not enabled
60       on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't
re-evaluate
61       DO/Paperspace for compute without a new ADR; the `$200` DO credits go to **non-compute
only**. The
62       `deploy/digitalocean/` scripts are kept solely as provider-agnostic Linux-GPU setup
reused by GCP.
63
64     ## Workflow conventions
65     - **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
66     - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers`
package requires ≥3.10); `ffmpeg` required.
67     - Never push to the default branch directly; descriptive commit messages.
68     - **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
69     - **Launch / "bit-flip" PRs** (a default-flip that drives a tracked project to terminal
state) MUST, in the SAME PR, run the `docs/DOC_STATUS.md` §2 archival due-diligence on that

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project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR lineage ? `git mv` to `docs/archives/` ? record in DOC_STATUS §3), and fix any internal-link breakage the move causes. Enforced by `.github/PULL\_REQUEST\_TEMPLATE.md` + the `link-check` CI job (`python -m tools.check\_md\_links`).

70 - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to the owner,

71 who has the golden corpus + labelled videos; verify pure-logic paths locally.

72

5. user (2026-07-02T04:45:54.820Z)

38:[Omitted long matching line]
52:### R1-01 [P1/M] tech-debt ? badminton-highlight-indexer
160:### R1-13 [P2/S] doc-rot ? badminton-highlight-indexer
169:### R1-14 [P2/S] doc-rot ? badminton-highlight-indexer
489:[Omitted long matching line]
490:[Omitted long matching line]
495:[Omitted long matching line]
496:[Omitted long matching line]
552:- [P2] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root
C:/Users/avidu/Projects/khelsutra-guru): DATA_IN_GCS.md §5 gap table contradicts its own §7:
trajectories/labels marked 'not in bucket' though migrated 2026-06-22
598:- non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root
C:/Users/avidu/Projects/khelsutra-guru): 0.5 GB duplicate video object in the corpus bucket
awaiting the prescribed verify-dupe-then-delete (hygiene, `badminton-highlight-
indexer/docs/DATA_IN_GCS.md`)

6. user (2026-07-02T04:46:01.169Z)

File content (29176 tokens) exceeds maximum allowed tokens (25000). Use offset and limit parameters to read specific portions of the file, or search for specific content instead of reading the whole file.

7. user (2026-07-02T04:46:09.081Z)

30

31 ### annotator:audit

32 Small, well-kept two-frontend labeling tool: a 1497-line VLC Lua extension (vlc/rally_annotator.lua, v1.7.1) and a Chrome MV3 extension (web/, v0.1.0) sharing a byte-compatible CSV format. Both CI workflows (tests, web-extension CI) are green on HEAD (last 15 runs all success; HEAD = PR #25, 2026-06-22); no open PRs; 2 open issues (#26 CSV provenance, #22 native i18n review) ? both verified still unaddressed. Web unit coverage is enforced (95/82 floors) and there is a real-Chromium per-locale E2E. README install/run instructions are accurate (VLC copy-file install; web npm install + build path is exactly what green CI runs). Debt is concentrated in three places: the content script's always-on full-DOM polling on every page, the locked LD-i8 locale-persistence decision being documented-but-unimplemented on the web side, and machine-draft translation catalogs pending native review. The sibling non-git 'Annotation Setup' dir is NOT a duplicate/predecessor of this repo ? it is the local annotation-data workspace (golden .rallies.csv labels since migrated to GCS, ~18MB trajectory CSVs, and Roora pilot-review artifacts that appear to exist nowhere else).

33

34 ### obsreport:audit

35 Tiny, high-quality private Python lib (9 src modules, 8 commits, all authored 2026-06-07; frozen since). Release hygiene is excellent: tag v0.1.3 == local HEAD == origin/main (1f72c1c), pyproject.toml version == version.py == 0.1.3, and the committed JSON Schema is byte-identical to a fresh ReportEnvelope.model_json_schema() dump (verified read-only). 74 tests pass locally in 0.43s and the code is clean (no eval, closed Literal vocabularies, never-raise IO), but the repo has ZERO CI (gh api: 0 workflows), no issues/PRs, and 3 ruff errors sit unchecked. The engine pins it correctly to tag v0.1.3 (not a floating branch) but via the stale pre-transfer org URL (avidullu, repo now lives in Khelsutra ? works only via GitHub redirect). The worst finding is consumer-side: sports-data-collector declares `obsreport>=0.1.3` as a plain PyPI requirement while the name `obsreport` is unclaimed on PyPI (404) ? a dependency-confusion window. The engine?obsreport integration tests never run in any CI (engine CI deliberately omits the private dep; reporting tests importorskip).

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36
37     ### nongit:dirs
38     Two intentional but unversioned dirs. "Annotation Setup" (4.9 GB) is the golden-corpus
staging area: Golden Labelled/ (6 .rallies.csv) and Trajectories/ (20 WASB CSVs, 18 MB) are
VERIFIED safely migrated to gs://khelsutra-rally-corpus (15 labels + 15 trajectories uploaded
2026-06-22, listed live), so the dominant loss risk is NOT the eval corpus ? it is the
Archive/Roorra pilot record (3 raw vendor videos 4.86 GB + CVAT deliverables + feedback +
corrected golden labels), which exists nowhere else (0 roorra objects in the bucket; the engine
repo archives only the guidelines). "khelsutra-screencast" (107 MB) is the documented artifacts
sink of khelsutra's committed e2e screencast harness (playwright.config.mjs writes
../../khelsutra-screencast/_raw), but it also holds 8 ad-hoc Playwright QA scripts and doc-cited
DoD evidence that are unversioned; only 2 of its videos are backed up to the promo bucket, and
the 5 numbered checkpoint screencasts cited in STUDIO_MEDIA_LIFECYCLE.md are absent from this
machine entirely. Engine docs reference the Annotation Setup dir by three different, all-
nonexistent paths, and DATA_IN_GCS.md's $5 gap table contradicts its own $7 done-markers.
Overall: data-loss exposure is concentrated in one 4.9 GB vendor-pilot folder plus moderate doc-
rot; no secrets found in either dir.
39
40     ### cross:repo
41     Workspace of 5 git repos (all just pulled to origin master/main, zero extra worktrees)
plus 2 non-git data dirs ("Annotation Setup" 4.9G raw video/labels ? the n=6 golden labels ARE
versioned inside sports-data-collector's committed data/sports_metadata.db per commit 9e6f204,
so the dir is raw-media overflow; khelsutra-screencast 107M e2e artifacts). The engine<->SPA
runtime API contract is coherent in source: every SPA-called endpoint exists in backend/main.py
+ backend/api/uploads.py, and the new fps/resolution fields landed on both sides (engine
main.py:516-517 via #413; SPA types.ts:44-45 + lib/video.ts:35-36 via #140). The real cross-repo
drift is the COMMITTED vendored SPA bundle (backend/ui @ khelsutra ed33a515, 2026-06-22) now 10
web/ commits behind, with a skew guard that by design only catches engine-side API_VERSION
bumps. CI maturity is very uneven (engine: 4 guard lanes; khelsutra: excellent incl. coverage
floor + layout/a11y; sdc: pytest-only; rally-annotator: ok; obsreport: none), and docs in both
engine and khelsutra still describe the DO droplet deploy as LIVE a week after its 2026-06-25
retirement. Trees are clean except sports-data-collector (uncommitted .gitignore WIP + a stale
full nested self-clone), and 20 merged branches linger across remotes/locals.
42
43     ### engine:issues-triage
44     Engine repo is active and healthy on master (last merge #413 on /api/videos; CI trusted
green; ~1500 tests, coverage floor 70 enforced in both CI lanes). 13 issues open (not 14) plus
one PR. Triage found one live P0 cost bug (#340: /api/process re-runs the paid Gemini segmenter
for an already-processed video_id ? verified no dedup short-circuit exists in current code), the
D1 god-module-split PR #390 now CONFLICTING and 6 days stale while main.py grew to 2198 lines,
and several issues with partially-stale premises (#388 half-done since PR #234, #331/#326/#328
still citing the retired DO droplet, #175 referencing a design doc that does not exist in the
workspace). No fully fixed-but-open issues; the debt level is moderate and mostly well-tracked,
but the issue text drift means several tickets need scope-correcting comments before anyone acts
on them.
45
46     ### engine:perf-resource
47     Engine repo at origin/master e410136 (clean tree). Performance/resource-lifecycle
posture is mixed: the deliberately-built safeguards are genuinely wired and healthy ? SQLite
goes through a single choke point with WAL + busy_timeout=30000 + connect timeout=30s
(backend/infrastructure/database.py:48-57; the only sqlite3.connect in backend/, so the engine
does NOT have sports-data-collector's contention gap), keep_frames defaults False with post-
success frame/stage deletion in all three detector runners (config/models.py:61,98;
wasb_runner.py:210; tracknet_runner.py:225; native_wasb_runner.py:315), Gemini chunk files are
removed after analysis (segmenters/gemini.py:289,375-377), the stitcher clips live in a
TemporaryDirectory (compiler.py:419), and no unwaited Popen exists (no zombie risk). The debt is
concentrated where blocking work meets the max_workers=1 job executor: every serving-path ffmpeg
call runs without a timeout (extraction, audio, trim, per-clip encode, concat) so one hung
ffmpeg wedges all job processing; ExecutionPolicy/run_bounded protects only the Gemini generate
call. All four known perf/lifecycle issues (#349 single-threaded streaming decode, #331 CPU-
bound phases, #330 MD5 reuse, #301 scheduled retention sweep) are verified still open and
unimplemented in code, and one untracked leak was found: remote-input scratch copies (rally-
input-*/rally-worker-* mkdtemp dirs) are never deleted and are invisible to
tools/cleanup_caches.py.
48

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49
50     ## Confirmed findings ? round 1 (43)
51
52     ### R1-01 [P1/M] tech-debt ? badminton-highlight-indexer
53     **PR #390 (D1/P10 main.py split) is stale and CONFLICTING while main.py grew 52% since
it was cut**
54     - anchor: `backend/main.py:1` · tracked in: PR #390;
docs/CODE_CLEANUP_AUDIT_2026-06-24.md $6 row P10 · finder: engine:audit-doc-reconcile
55     - evidence: gh pr view 390: state=OPEN, mergeable=CONFLICTING, mergeStateStatus=DIRTY,
updatedAt=2026-06-26T14:37Z, 3 files (backend/main.py, backend/orchestration.py, mypy.ini).
Since then >=8 PRs modified backend/main.py (#400, #405-#409, #412, #413 per git log
--since=2026-06-25 -- backend/main.py). wc -l backend/main.py = 2198 vs the audit doc's '1448
lines' claim (docs/CODE_CLEANUP_AUDIT_2026-06-24.md:279); 18 @app. routes and 45 print() calls
now live in it; backend.main still mypy-quarantined (mypy.ini:15). No backend/orchestration.py
exists on master.
56     - why it matters: This is the last HIGH-severity item of the cleanup audit (P10, the
only non-Phase-3 ? row in its $6 tracker) and it is compounding: every new endpoint PR lands in
the god-module, widening the conflict surface and making the split more expensive each week. The
3-file extraction in #390 is now unmergeable.
57     - proposed fix: Close #390 and re-cut the orchestration.py extraction fresh from
origin/master (small, ~1 session), or rebase it now. Then continue the split (CLI handlers ->
backend/cli/, new /api/source|api/assets|api/capabilities routes -> backend/api/) before more
endpoints accrete. Update the audit doc's stale '~1450 lines' sizing while at it.
58     - verifier notes: Minor sharpenings only: (a) git log --since=2026-06-25 --
backend/main.py actually shows 10 commits (adds #387 repo-wide ruff format ? itself a likely
source of #390's conflicts ? and #380), so ">=8 PRs" is understated; (b) the mypy quarantine
block is mypy.ini:15-16 ([mypy-backend.main] header at 15, ignore_errors at 16); (c) the audit
doc's "1448 lines" statement is in the SChangelog 2026-06-25 "Accuracy fixes" bullet (~line
278), and the stale "~1450" sizing also appears at lines 69, 198, and 247 ? all four spots need
the size refresh the fix proposes.
59     - verifier reasoning: Every load-bearing claim reproduced. (1) `gh pr view 390` returns
state=OPEN, mergeable=CONFLICTING, mergeStateStatus=DIRTY, updatedAt=2026-06-26T14:37:18Z,
exactly 3 files: backend/main.py (+19/-595), backend/orchestration.py (ADDED, +635), mypy.ini
(+6). (2) Current working tree (branch master, HEAD e410136 = PR #413): `wc -l backend/main.py`
= 2198; 1448?2198 is +51.8% ? the claimed 52%; the 1448 baseline is documented in
docs/CODE_CLEANUP_AUDIT_2026-06-24.md changelog ("main.py size '~1300' ? '~1450' (now 1448
lines)", 2026-06-25 entry, ~line 278-279). (3) Not fixed: backend/orchestration.py does not
exist in the tree, and git log --since=2026-06-25 -- backend/main.py shows the file only grew ?
commits for PRs #413, #412, #409, #408, #407, #406, #405, #400 (the claimed 8; the log actually
lists 10 incl. #387 ruff-format and #380). (4) grep counts on backend/main.py: 18 lines matching
`^@app\.` and 45 `print(` occurrences ? exactly as claimed. (5) mypy.ini:15-16 = `[mypy-
backend.main]` / `ignore_errors = True` (quarantine intact). (6) Audit doc $6 tracker line 198:
P10 row "D1: Split main.py god-module (~1450 lines ? smaller modules)" is ?; the only other ?
row is line 205 (P16+ Phase 3 bucket), so P10 is indeed the sole non-Phase-3 open item; D1 is
rated HIGH at doc line 69. Severity P1 is honest for a tech-debt audit: the concrete failure is
already materialized (the 635-line extraction in #390 is unmergeable today), and it compounds ?
8 endpoint-adding PRs landed in the god-module in the week since the PR was cut, all inside a
mypy ignore_errors module, so each week of delay increases rework cost and unchecked-type
surface. Not owner-gated: internal refactor, no serving go-live / corpus-promotion / spend /
licensing implications ("P10" here is the audit tracker row, not Studio P10). Proposed fix
(close-and-recut or rebase from origin/master, then continue the split per the doc's own plan at
line 247, and fix the stale ~1450 sizing) is sane and correctly scoped; effort M is right given
the fix includes continuing the split beyond the re-cut.
60
61     ### R1-02 [P1/M] critical-fix ? badminton-highlight-indexer
62     **/api/process re-runs the paid Gemini segmenter for an already-processed video (re-bill
on refresh/'Try again') ? still unfixed at HEAD**
63     - anchor: `backend/main.py:1172` · tracked in: Khelsutra/badminton-highlight-indexer#340
(also #330) · finder: engine:correctness
64     - evidence: backend/main.py:1171-1172 computes video_id and `was_reingest =
db.get_video(video_id) is not None`, but was_reingest is only used for the report/destroy-after-
confirm; there is NO reuse/skip branch ? the segmenter always runs (main.py:1251-1255). Verified
against gh: issue #340 OPEN ('re-runs the full (paid) segmenter anyway'), plus #330 (MD5
segmentation reuse) OPEN.
65     - why it matters: A page refresh or SPA 'Try again' on an already-indexed video (same

```

MD5) re-bills Gemini (~\$0.10/8-min match) and burns a multi-minute compute run for zero new information. Live-triggered once in the alpha (2026-06-22, per #340).

66 - proposed fix: In `_execute_processing`, when `db.get_video(video_id)` exists with `status='processed'` and existing `play_segments` (and the request doesn't explicitly force reprocess), return the stored segments as the job result instead of dispatching the segmenter; add a ``force`` flag to `ProcessRequest` for deliberate re-ingest.

67 - verifier notes: Severity P1 is honest: concrete failure scenario is a user-reachable one-click ("Try again" after the SPA's 10-min poll ceiling, live-triggered 2026-06-22 per #340) that silently re-bills a metered Gemini key ? and per #340 the M8 cost ledger does not meter Gemini yet, so the re-bill is invisible. Proposed fix is sane and correctly scoped: the skip condition must require `status=="processed"` AND existing `play_segments` (a prior "failed" record ? `main.py:1196-1201` ? must still re-run), and the added ``force`` flag preserves the existing `destroy-after-confirm` re-ingest path (`main.py:1233-1235, 1278`). Not owner-gated: the fix reduces real cloud spend rather than incurring it, and the owner already specified the desired behavior + acceptance criteria in OPEN issue #340. Effort M is right (endpoint logic + `ProcessRequest` field + a no-Gemini-call regression test). One duplicate-tracking note: #330 covers the same reuse mechanism as an optimization; a fix should close both.

68 - verifier reasoning: Verified against origin/master at HEAD e410136 (pulled, "Already up to date"). (1) Code behaves as claimed: `backend/main.py:1171-1172` computes ``video_id` = compute_video_id(local_video)` then `was_reingest` = db.get_video(video_id) is not None`; the ONLY other use of `was_reingest` in `_execute_processing` is main.py:1327, where it is passed to `emit_indexer_report` as a report field. There is no reuse/skip/short-circuit branch: after validation passes (main.py:1210-1214 upserts status "processed"), the flow unconditionally dispatches segmentation ? cloud via `_maybe_dispatch_remote` (main.py:1241-1248) or in-process `segmenter.process_video` (main.py:1251-1255). ProcessRequest (main.py:313-324) has no `force`/`reprocess` field (only video_path, player_pool, max_frames, segmenter, locale, compute). (2) Not fixed: `git log --oneline -15 -- backend/main.py` shows only unrelated commits (#413, #412, #409, #408, #407, #406, #405, #400, ...); no idempotency/reuse change. (3) Cost claim is real in served config: config.json:26 sets "default_segmenter": "gemini" (the paid path is the production default; the "motion" fallback at main.py:1181-1183 is only the code-level default). (4) Tracking verified via gh: issue #340 OPEN (title and body match the finding verbatim, including the 2026-06-22 live trigger and ~$0.10/8-min cost) and #330 OPEN (MD5 segmentation reuse).`

69

70 `### R1-03 [P1/M] critical-fix ? badminton-highlight-indexer`

71 `**/api/source` re-downloads the ENTIRE remote video from the bucket on every HTTP request (each browser Range request) into a fresh, never-deleted temp dir**

72 - anchor: ``backend/main.py:646`` · tracked in: ? · finder: engine:correctness

73 - evidence: `backend/main.py:644-651: get_source` calls `storage.acquire_local_input(rec['filepath'], ...)` on EVERY request; `storage/__init__.py:89-90:` for a non-local ref this does ``tempfile.mkdtemp(prefix='rally-input-')`` + full fetch ? a new dir + full download per call, never removed. Proxy warming is explicitly local-only (`main.py:655-657`), so a remote-source video never gets a proxy. `FileResponse` under `fastapi>=0.111` (`pyproject.toml:28`) serves Range ? a browser scrubbing a <video> issues many separate requests to the same URL.

74 - why it matters: Direct cloud-egress cost leak plus a disk-fill denial-of-service of the appliance, reachable through a supported flow (remote-registered assets + P8 annotate-in-UI).

75 - proposed fix: Cache the fetched local copy keyed by `video_id` (e.g. under <output>/proxies or a per-video workspace source/ dir) and reuse it across requests; extend proxy warming to remote sources; delete or reuse the scratch dir instead of `mkdtemp-per-request`.

76 - verifier notes: Path correction: the storage citation is `backend/storage/__init__.py:89-90` (not `storage/__init__.py`). Sharper failure scenario: store a 4 GB match via `POST /api/assets` with an `s3://gs://` medium (P4, #409), open the P8 annotate UI, scrub ~10 times ? ~40 GB bucket egress (~\$4-5 real cost) and ~40 GB of never-deleted `rally-input-*` temp dirs per session; each seek also stalls behind a full multi-GB download, so the feature is effectively unusable for remote sources. Additional color for the fix: the engine free-space guard (`require_free_bytes`, #407) is not applied to this download path, and the cached copy should live under a guarded root; the finding's proposed fix (cache keyed by `video_id`, reuse across requests, extend proxy warming to remote) is sane and correctly scoped.

77 - verifier reasoning: `backend/main.py:646` calls `storage.acquire_local_input(rec["filepath"], ...)` on every `GET /api/source/{video_id}` when no proxy exists (proxy check `main.py:641-642`). `backend/storage/__init__.py:89-90:` non-local refs get a fresh `tempfile.mkdtemp(prefix="rally-input-")` + full fetch per call; docstring (lines 80-82) admits the copy "relies on OS temp reclamation" (nonexistent on Windows %TEMP%), and a

repo-wide grep for "rally-input" shows no cleanup path. Remote filepaths are a supported state: POST /api/assets registers the cloud stored_uri as filepath (main.py:838, 855; comment at main.py:1680 lists gs://, s3://, gdrive://). Proxy warming is explicitly local-only (main.py:655-657, docstring 628-629 "remote-source proxying is a follow-up"), so remote sources never escape the full-fetch path. Installed fastapi 0.136.1 / starlette 1.0.0 FileResponse honors Range, so a scrubbing <video> issues many independent requests, each a full-object download. git log -15 -- backend/main.py (through #413) shows no fix after #406 introduced this. Not owner-gated: the fix is plain engineering that reduces cloud spend.

```
78
79     ### R1-04 [P1/M] tech-debt ? badminton-highlight-indexer
80     **Import cycles: extracted API modules reach back into backend.main globals at call
time; config layer imports pipeline layer**
81     - anchor: `backend/api/job_worker.py:75` · tracked in:
docs/CODE_CLEANUP_AUDIT_2026-06-24.md P5 note + D1/P10 follow-up (line 221) · finder:
engine:structure
82     - evidence: AST scan of backend/ found 5 strongly-connected components. (1) backend.main
<-> backend.api.job_worker <-> backend.api.uploads: job_worker.py:75,98,214,234,251 and
uploads.py:47 do call-time 'import backend.main as _main' to read main.py's module globals
db_instance/global_config (main.py:279-280); main.py:292 re-exports job_worker names back onto
itself. Documented as a transitional shim (CODE_CLEANUP_AUDIT P5 note, line 221) but it makes
main.py load-bearing for every extracted module and keeps backend.main AND
backend.worker.__main__ mypy-quarantined (mypy.ini:15-23). (2) Layering inversion:
backend/config/models.py:249 imports segmenter_registry from backend.pipeline.segmenters.base
inside a validator, while base.py:15 (_normalize_config) imports AppConfig back from
config.models ? config validation now depends on pipeline registration side effects. (3-5) eval-
package cycles (golden_fixtures<->golden_real_fixtures, ablation<->ablation_pdf,
annotations<->gt_loader) are lazy and explicitly documented (golden_fixtures.py:1005-1028) ?
lower priority.
83     - why it matters:
84     - proposed fix: Kill the shim as the concrete step-2 of D1: move
db_instance/global_config onto app.state (create_app already exists per P5) and pass config/db
explicitly to job_worker/uploads, then lift backend.main and backend.worker.__main__ out of mypy
quarantine. For the layering inversion, move the segmenter-name validation out of the pydantic
validator into a startup check (backend/config/startup.py) so config/ no longer imports
pipeline/.
85     - verifier notes: Mechanism correction to the evidence: job_worker's five call-time
imports do NOT read db_instance/global_config ? job_worker already takes config/db as explicit
params (job_worker.py:89,205,230,247; mypy.ini:14 notes it was lifted from quarantine) and
resolves FUNCTION seams (ProcessRequest, _execute_compile, _execute_processing, JobWaiting,
_arm_wake_timer, _get_executor, _MAX_DRAIN_WAKE_SEC) through the backend.main module object per
the documented test-monkeypatch contract (job_worker.py:8-16, main.py:286-291). Only
uploads.py:49 reads a main global (global_config). So the app.state migration for db/config is
mostly about uploads + main.py itself; the harder remaining work is redesigning the monkeypatch
seam contract and its tests ? effort is top-end M. Sharper failure scenario: any alternative
entrypoint (new console_script, uvicorn factory) that builds the app without running main()
leaves backend.main.global_config=None and uploads silently falls back to default upload
dir/size limits (uploads.py:49-52), ignoring operator security config ? same class as the
already-documented python -m backend.main incident (main.py:2188-2195) that hung every queued
job. Layering inversion is slightly worse than claimed: the validator's lazy import
(models.py:249) transitively executes segmenters/__init__.py:1-7, importing ALL concrete
segmenters, so constructing AppConfig anywhere drags the full pipeline in; and
SegmenterRegistry.register (base.py:66-70) re-enters AppConfig construction during registration
with errors swallowed by base.py:19's bare except.
86     - verifier reasoning: Verified in current tree (HEAD e410136, pulled clean). Cycle 1:
backend/main.py:292-302 imports backend.api.job_worker at module load (re-exports);
job_worker.py:75,98,214,234,251 do call-time 'import backend.main as _main' (grep-confirmed all
5); uploads.py:47-49 call-time imports and reads _main.global_config; globals defined
main.py:279-280, set at main.py:101-103 and 1981-1982; mypy.ini:15-16 quarantines backend.main,
:22-23 backend.worker.__main__. Cycle 2: config/models.py:244-256 validator imports
segmenter_registry from pipeline/segmenters/base; base.py:14-18 imports AppConfig back in
_normalize_config; importing base initializes segmenters/__init__.py:1-7 which imports every
concrete segmenter (motion.py:243 etc. register as import side effects), so AppConfig
construction depends on pipeline import health ? and register() (base.py:66-70) re-enters
AppConfig via segmenter_cls(None, {}) with failures swallowed by base.py:19 bare except. Eval
cycles lazy+documented (golden_fixtures.py:1008-1013). Not already fixed: git log for the four
```


files shows only feature commits (#405-#413). Severity honest: the failure class already bit ?
main.py:2188-2195 documents the python -m backend.main dual-module incident
(backend.main.db_instance=None, drain crashed silently, jobs hung 'queued' forever), patched
only for that entry path (main.py:2196-2198); uploads.py:49-52 getattr-defaults mean any
entrypoint that skips main() silently runs uploads with default dir/limits. Tracked_in accurate:
audit doc line 221 is verbatim the P5 shim note. Not owner-gated: only owner question is D1
phase sequencing, none of the listed gates. Fix sane: create_app exists (main.py:2184), handlers
already use app.state (mypy.ini:12-13), backend/config/startup.py exists as the startup-check
home.

87

88 ### R1-05 [P1/M] tech-debt ? badminton-highlight-indexer

89 **Path-jail rejects gs://, s3://, gdrive-api:// before scheme resolution ? bucket ingest
impossible on the exposed alpha box (issue #327 still open)**

90 - anchor: `backend/main.py:1376` · tracked in: <https://github.com/Khelsutra/badminton-highlight-indexer/issues/327> · finder: engine:security-config

91 - evidence: main.py:1376 calls `validate\_input\_path(req.video\_path, allowed\_root=allowed\_root)` BEFORE `storage.resolve\_input\_ref` at main.py:1379;
backend/utils/paths.py:45-55 `resolve\_under\_root` realpaths the raw URI so `gs://bucket/x.mp4`
escapes the jail -> 400. Startup check startup.py:106-119 makes the jail mandatory when exposed
(EXPOSED_NO_VIDEO_ROOT), so an exposed engine can never ingest from a bucket. gh: issue #327
OPEN (verified 2026-07-02), proposing scheme-aware validation.

92 - why it matters: Blocks the alpha CUJ (large-video ingest via gs:// instead of 500 MB
browser uploads) and is a stale-open thread since before 06-26; the fix must be done carefully
to avoid reopening the arbitrary-local-file hole the jail exists for.

93 - proposed fix: Implement #327 option (b): jail only after resolve_input_ref, applying
resolve_under_root exclusively to backend=='local' refs; keep traversal/symlink/absolute
rejection for local paths and add both allow (bucket URI accepted) and reject (local traversal
still 400) tests, updating

tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri.

94 - verifier notes: Minor corrections: (1) startup check path is
backend/config/startup.py:106-119, not startup.py (lines exact). (2) The "500 MB browser
uploads" figure in why_it_matters comes from issue #327's body; the repo's actual browser-upload
cap is security.max_upload_mb default 4096 MB (backend/api/uploads.py:51,
backend/config/models.py:339) ? 500 MB is the Gemini provider per-clip cap
(backend/providers/gemini.py:23). Immaterial to the finding (multi-hour matches exceed 4 GB
anyway; docs/VIDEO_LOCALITY_MODEL.md:26 "The 10 GB problem is real"). (3) Fix scope additions:
there is a second resolve call-site (CLI --video-path, see tests/test_api_endpoints.py:922) ?
verify whether it needs the same scheme-aware treatment; ensure unknown schemes still
ValueError?400 via resolve_input_ref as the sole scheme gate. (4) Caution, not a gate: the
change relaxes a deliberately pinned security contract on the exposed box; #327 records the
owner's explicit want, but the option (a) vs (b) choice and the PR should get owner sign-off
before landing since it alters exposed-serving posture.

95 - verifier reasoning: Personally reproduced from current code. (1) Ordering:
backend/main.py:1376 calls validate_input_path(req.video_path, allowed_root=allowed_root) BEFORE
storage.resolve_input_ref at main.py:1379-1381; ValueError maps to HTTP 400 at
main.py:1382-1383. (2) Mechanism: backend/utils/paths.py:58-68 routes any request with
allowed_root set through resolve_under_root, which os.path.realpath()s the RAW string
(paths.py:47) ? 'gs://bucket/x.mp4' realpaths to a cwd-relative path, commonpath != root
(paths.py:50-54) ? ValueError ? 400. The code itself documents this at main.py:1387-1389: "a
configured allowed_video_root (hosted mode) rejects a non-local URI as out-of-root". (3) Pinned
by test: tests/test_api_endpoints.py:903-912 asserts gs://, s3://, gdrive:// all ? 400 when
allowed_video_root is set ("Pins the security contract"). (4) Exposure mandate:
backend/config/startup.py:106-119 emits LEVEL_ERROR EXPOSED_NO_VIDEO_ROOT when edge_auth_enabled
and allowed_video_root unset ? so an exposed engine must have the jail, making bucket ingest
impossible there. (5) Not fixed: git log -15 -- backend/main.py (head e410136, #413) shows no
commit addressing this; behavior present in current tree. (6) gh issue view 327: state OPEN,
created 2026-06-22, title/body match the finding exactly, including the owner's explicit want
("ingest large videos via gs:// on the hosted box") and options (a)/(b). Concrete failure:
exposed alpha box (edge_auth_enabled=true, allowed_video_root=/jail per boot requirement)
receives POST /api/process {"video_path":"gs://bucket/match.mp4"} ? 400 "path ... escapes the
allowed root"; only ingest path left is the capped browser upload (backend/api/uploads.py:51,
security.max_upload_mb) ? blocks the tracked alpha large-video CUJ. Severity P1 honest (owner-
wanted CUJ blocked, open since 2026-06-22, workaround capped/painful per
docs/VIDEO_LOCALITY_MODEL.md:19-26). Proposed fix (option b: jail only backend=='local' after
resolve_input_ref, keep null-byte check universal and traversal/symlink jail for local, add

allow+reject tests, update the pinned test) matches the issue's own proposal and is correctly scoped; effort M is right.

96

97 ### R1-06 [P1/S] doc-rot ? khelsutra (avidullu/khelsutra)

98 **Docs claim the retired droplet alpha is LIVE (droplet deleted 2026-06-25)**

99 - anchor: `NEXT\_STEPS.md:12` · tracked in: partially in issue #97 (item 2: update DEPLOY_ALPHA.md); the NEXT_STEPS.md droplet claim is untracked · finder: khelsutra:issues-triage

100 - evidence: NEXT_STEPS.md:12-13 ('Where we are (updated 2026-07-01)') still says the site is 'live + browsable on the D0 droplet -> http://168.144.126.176:8787'; deploy/DEPLOY_ALPHA.md:3 says 'Status: LIVE (deploy live-verified on the droplet 2026-06-22)' and :29 'This runbook ships the split topology'; docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 'The alpha stand-up is now LIVE (2026-06-22)'. Project memory (MEMORY.md:63, session-handoff.md:19): droplet claude-do-rally-test/168.144.126.176 RETIRED+deleted 2026-06-25; alpha paused pending Cloud Run. nslookup shows api./app.khelsutra.guru still resolve via Cloudflare to a tunnel with no origin (pending CNAME/Access cleanup).

101 - why it matters:

102 - proposed fix: One doc PR: mark DEPLOY_ALPHA.md status RETIRED/PAUSED with a pointer to the Cloud Run plan, fix NEXT_STEPS.md 'Where we are' (droplet gone, site is CF-Workers-only), and correct LOCAL_APPLIANCE_ARCHITECTURE.md:176. Also note the single-origin repoint that superseded the split topology (issue #97). The 2026-07-01 refresh (#139) touched docs but kept the stale claim, so this will keep misleading ramp-up reads.

103 - verifier notes: Minor anchor correction: the 'Where we are (updated 2026-07-01)' header is NEXT_STEPS.md:5; the droplet-live claim spans lines 12-13 (the finding's line 12 anchor is correct for the URL text). Severity P1 stands: NEXT_STEPS.md's 2026-07-01 stamp is NEWER than the handoff's 2026-06-25 correction, so ramp-up readers reconciling the two will trust the wrong doc; worse, DEPLOY_ALPHA.md \$1 (line 35) directs deploying the engine + secrets onto 168.144.126.176 ? an IP the owner no longer controls and D0 can recycle. Fix is sane, doc-only, and not owner-gated (it records already-made owner decisions: retirement 2026-06-25 + single-origin repoint per issue #97); one caution ? the fix should point to the Cloud Run move as 'paused pending' rather than declare a new serving decision, which would be owner-gated. Did not independently re-verify the finder's nslookup/DNS claim; the finding stands without it.

104 - verifier reasoning: Working tree at HEAD f9eea14: NEXT_STEPS.md:5 'Where we are (updated 2026-07-01)' + :12-13 'live + browsable on the D0 droplet -> http://168.144.126.176:8787 (systemd khelsutra-site)'; deploy/DEPLOY_ALPHA.md:3 'Status: LIVE (deploy live-verified on the droplet 2026-06-22...)' + :29 'This runbook ships the split topology'; docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 'The alpha stand-up is now LIVE (2026-06-22): the engine booted on the droplet (168.144.126.176)'. Ground truth: memory/khelsutra-site-droplet-parity.md:3,10-14 (droplet DELETED 2026-06-25, data archived, doctl delete done, monitor removed via PR #117), session-handoff.md:19 ('the prior LIVE alpha on the droplet is GONE ? alpha paused pending the Cloud-Run move'), MEMORY.md:63. Not fixed: git log shows the last commit touching all three files is 9223ba9 (#139, the 2026-07-01 refresh) and the stale text persists at HEAD. Tracking verified: gh issue 97 is OPEN, item 2 covers only DEPLOY_ALPHA.md (single-origin update); the NEXT_STEPS.md droplet claim is untracked. Issue #97 also confirms split topology was superseded by the 2026-06-22 single-origin repoint, so DEPLOY_ALPHA.md:29 is doubly stale.

105

106 ### R1-07 [P1/M] critical-fix ? khelsutra (avidullu/khelsutra)

107 **Issue #111 still fully valid: refresh loses the in-flight job; progress static for minutes**

108 - anchor: `web/src/hooks/useIngestJob.ts:117` · tracked in: khelsutra#111 · finder: khelsutra:issues-triage

109 - evidence: useIngestJob.ts:117-128 ? job_id lives only in the poll closure (no localStorage/URL persistence); App.tsx:71-73 seeds state only from ?video= (set on done, App.tsx:153). A refresh mid-job lands the user back on the ingest form while the server job keeps running. Progress is the coarse free-text job.progress (useIngestJob.ts:106); no heartbeat/animation beyond a spinner chip (IngestPanel.tsx:337-358).

8. user (2026-07-02T04:46:17.939Z)

158 - verifier reasoning: All cited code exists and behaves as claimed in current HEAD (e410136). backend/main.py:109-116 `\_cors\_origins\_from\_env` returns `origins or [""]`; main.py:123-129 adds CORSMiddleware with allow_origins from that fn, allow_methods=['*'], allow_headers=['*'], allow_credentials=False. main.py:152-156 TrustedHostMiddleware defaults to loopback hosts and matches on host.split(':')[0], so a direct fetch to http://127.0.0.1:<port> sends Host:127.0.0.1 and passes ? it only stops DNS-rebinding via attacker domains, not direct-

IP cross-origin fetches. backend/api/auth.py:125-127 resolve_tenant returns None (no auth) when edge_auth off (default), and /api/process (main.py:1355-1370) has no other auth. Mechanism is correct: wildcard CORS is load-bearing to (a) read /api/videos responses cross-origin and (b) pass the JSON-preflight for POST /api/process (main.py:1817 ? paid Gemini). /api/process 404 fast-fail (main.py:1390) needs a valid local path, but that is harvestable from the readable /api/videos index, making the cost-leak chain concrete. git log -15 -- backend/main.py shows no fix; wildcard default is live. The finding also correctly flags the internal inconsistency: bind host (main.py:174) and allowed hosts (main.py:146) both default to loopback ('widen consciously'), yet CORS alone defaults open ? an honest oversight. Proposed fix is sane and correctly scoped; not owner-gated (local-appliance secure default, no cloud-strategy/serving-go-live decision).

159

160 ### R1-13 [P2/S] doc-rot ? badminton-highlight-indexer

161 **BACKLOG.md lists three shipped items as open, including a red 'blocks something'

ByteTrack entry**

162 - anchor: `docs/BACKLOG.md:39` · tracked in: ? · finder: engine:audit-doc-reconcile

163 - evidence: BACKLOG.md:39 marks 'supervision ByteTrack deprecation' as red/open, but the migration shipped in #386: backend/pipeline/detectors/person_detector.py:124 imports trackers.ByteTracker and requirements.txt:29 pins 'trackers @ git+...@2.5.0'. BACKLOG.md:22 lists 'App-factory / lifespan' as open, but create_app() exists at backend/main.py:93 (#370). BACKLOG.md:23 says 'orphaned Gemini uploads are never cleaned', but backend/providers/gemini.py:59 defines cleanup_orphaned_files() with a startup orphan sweep (#380). Doc header says 'Last updated: 2026-06-23'.

164 - why it matters: This stale state has already caused real confusion: the project memory feeding this very audit listed D13 (ByteTrack) as a remaining item. A red-flagged 'blocks something' entry that is actually done distorts session-start prioritization for every future session.

165 - proposed fix: Docs-only PR: flip the three entries to done with PR links (#386, #370, #380), matching the checked-off convention already used at BACKLOG.md:24. Can ride in the same reconcile PR as the DOC_STATUS/README fixes.

166 - verifier notes: Fix scoping correction: only the line-39 ByteTrack entry can be flipped fully to done (and its '<0.30.0 cap' sentence deleted ? the cap is already lifted in requirements.txt:18). Entries at lines 22 and 23 are COMPOUND and only half-shipped: line 22 bundles 'App-factory / lifespan + lazy torch-cv2 imports' ? create_app() shipped (#370) but there is no lifespan in backend/main.py (module-level app remains at main.py:90) and person_detector.py:34 still top-level-imports cv2; line 23 bundles 'WSL temp/frame/cache + Gemini remote-file GC' ? the Gemini GC shipped (#380) but no retention/cleanup exists for staged videos, PNG frame dumps, or *_wasbcache (#407 is a free-space guard, not GC). Correct fix: check off line 39 with #386; on lines 22-23 mark the shipped clause done with its PR link and keep the residual halves as the open item. Sharper failure scenario: the doc's own line 3 declares it 'the single tracker... so nothing gets lost between sessions', and the project memory feeding this audit already re-listed D13 as remaining ? a future session acting on the red ? line-39 entry would re-do or wrongly block a supervision bump that is already safe.

167 - verifier reasoning: docs/BACKLOG.md:39 marks 'supervision ByteTrack deprecation' ? open and claims 'requirements.txt is pinned <0.30.0 as a stopgap' ? but backend/pipeline/detectors/person_detector.py:124 imports `from trackers import ByteTracker` (docstring lines 108-110 records the D13/P12 migration), requirements.txt:29 pins `trackers @ git+https://github.com/roboflow/trackers.git@2.5.0`, and requirements.txt:18 is now `supervision>=0.21.0` (the <0.30.0 cap the entry describes is already lifted). Commit 3624a8d (#386) shipped it. docs/BACKLOG.md:22 lists app-factory as open but create_app() exists at backend/main.py:93 (commit de09ec8, #370). docs/BACKLOG.md:23 says orphaned Gemini uploads 'are never cleaned' but backend/providers/gemini.py:59 defines cleanup_orphaned_files() with a startup orphan sweep (commit cd5b3d2, #380). Header docs/BACKLOG.md:8 reads 'Last updated: 2026-06-23'; git log shows the last BACKLOG-touching commit is b3c6c2b (#347), predating all three feature PRs. Repo is at origin/master (e410136). Not owner-gated: docs-only reconcile of factual shipped state.

168

169 ### R1-14 [P2/S] doc-rot ? badminton-highlight-indexer

170 **DOC_STATUS register + docs/README index are ~15 PRs behind the cleanup-audit and decision-snapshot trackers**

171 - anchor: `docs/DOC\_STATUS.md:47` · tracked in: docs/CODE_CLEANUP_AUDIT_2026-06-24.md §8 DoD (unchecked 'docs/README.md updated' item) · finder: engine:audit-doc-reconcile

172 - evidence: DOC_STATUS.md:47 says CODE_CLEANUP_AUDIT is 'Phase 1: P0/P1/P4/P6a shipped... Quick wins (Q4/Q5/Q6/Q8) next', but the audit doc itself (CODE_CLEANUP_AUDIT_2026-06-24.md:8, :180-205) shows Phase 1 COMPLETE plus Phase-2 items

P8/P9/P11-P15/P17 all merged (#373-#386). DOC_STATUS has no row at all for DECISION_SNAPSHOT_REGRESSION.md (created 06-25, merged as #383) despite \$0 saying to update on every lifecycle change; DOC_STATUS was last touched 2026-06-30 (#403) without fixing this. docs/README.md:41 still calls the audit 'DRAFT, owner-gated' and README.md:42 says the snapshot doc's 'decisions in \$2 need locking' though D1-D11 were LOCKED 2026-06-25 (DECISION_SNAPSHOT_REGRESSION.md:26). NEXT_STEPS.md:6 still says 'owner M0 design sign-off pending', contradicted by its own line 14 ('M0-M9 LARGELY SHIPPED... ADR-014 ratified').

173 - why it matters: DOC_STATUS is by convention the archival/lifecycle board and README is the ramp-up index; both are read at every session start. When they contradict the per-doc source-of-truth trackers, sessions either relitigate finished work or trust stale gating claims. The cleanup audit's own DoD (line 225) is blocked on exactly this README update.

174 - proposed fix: One docs-only reconcile PR: refresh DOC_STATUS.md:47 row (Phase 2 nearly done, remaining = P10 + P16+), add a DECISION_SNAPSHOT_REGRESSION row, fix docs/README.md:41-42 blurbs, and rewrite the NEXT_STEPS.md:6 'where we are' paragraph to the post-06-21 state.

175 - verifier notes: Two refinements to the fix scope: (a) when rewriting NEXT_STEPS.md:6, preserve the owner NO-GO block (lines 8) added by #404 verbatim ? it is a locked owner decision, not rot; (b) DOC_STATUS.md:33 (COMPUTE_DECOUPLED_SERVING row, "M1?M4 SHIPPED ... M5 truly-local ? M6 Cloud Run ? M8 ledger ? M9-auth next") is stale for the same reason ? NEXT_STEPS.md:14 says M5 #286/M6 #287/M8 #288/M9-auth #290 are merged ? so the same reconcile PR should refresh that row too. Sharper failure scenario: DOC_STATUS \$0 designates itself the board sessions read to decide "what needs attention"; a session acting on row 47 would branch to implement Q4 (builtin-defaults cache), which merged as #358 on 2026-06-25, producing a duplicate/conflicting PR.

176 - verifier reasoning: Every cited fact reproduces in the current working tree (master == origin/master, HEAD #413). (1) docs/DOC_STATUS.md:47 still reads "IN PROGRESS ? Phase 1: P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next", while docs/CODE_CLEANUP_AUDIT_2026-06-24.md:8 says "Status: PHASE 1 COMPLETE ? 14 items shipped" and its \$6 tracker (lines 180-205) shows P6b-P7 plus Phase-2 items P8/P9/P11/P12/P13/P14/P15/P17 all ? merged (#358-#386); only P10 and P16+ remain ? ? the row is ~17 merged PRs stale. (2) DOC_STATUS.md has no row anywhere (\$3a-\$3e read in full) for DECISION_SNAPSHOT_REGRESSION.md, which exists (created 2026-06-25, merged in #383 per git log of docs/README.md: commit 1283e18) ? violating DOC_STATUS.md:4's own cadence ("whenever a doc changes lifecycle state (created ...)"). (3) git log confirms DOC_STATUS.md was last touched by #403 (6e973af, the 2026-06-30 rally-cluster fix) without fixing either ? so NOT already fixed; the cleanup-audit row was last updated in #357. (4) docs/README.md:41 still calls the audit "? Tracked project (DRAFT, owner-gated)" and :42 calls the snapshot doc "DRAFT, owner-gated ? decisions in \$2 need locking", contradicted by DECISION_SNAPSHOT_REGRESSION.md:8 and :26 ("D1?D11 LOCKED 2026-06-25"). (5) docs/NEXT_STEPS.md:6 says "owner M0 design sign-off pending", contradicted by line 14 in the same file ("M0?M9 LARGELY SHIPPED (2026-06-21, ADR-014 ratified)"); #404 (the last NEXT_STEPS commit) added the NO-GO block but left the stale paragraph. (6) The audit's own DoD at CODE_CLEANUP_AUDIT_2026-06-24.md:225 has the unchecked "[] docs/README.md updated to reflect the project's terminal state" item, matching tracked_in. Not owner-gated: a docs-only reconcile records already-shipped, already-merged state ? no launch, corpus-promotion, strategy, spend, or counsel decision is being made (the #404 NO-GO block must simply be preserved verbatim). Concrete failure: a fresh session following the session-start convention reads DOC_STATUS:47, plans "Q4/Q5/Q6/Q8 next" (shipped 5+ days ago in #358-#361), or reads README:42 and re-litigates D1-D11 that are LOCKED ? exactly the relitigation failure the register exists to prevent. P2/S are honest: it burns ramp-up time and risks duplicate work but breaks no code.

177

178 ### R1-15 [P2/M] test-gap ? badminton-highlight-indexer

179 **M4 quality-floor gate still missing ? the one unblocked item keeping the golden-fixtures track from DONE**

180 - anchor: `docs/GOLDEN\_REGRESSION\_FIXTURES.md:156` · tracked in: docs/GOLDEN_REGRESSION_FIXTURES.md \$7 M4; docs/BACKLOG.md:60 · finder: engine:audit-doc-reconcile

181 - evidence: GOLDEN_REGRESSION_FIXTURES.md:156 (M4 row) is ?; verified tests/test_golden_quality_floor.py and eval_baselines/regression_baseline.json do not exist (ls: No such file). \$9 DoD (line 197) requires M4 before the track can flip DONE. BACKLOG.md:60 calls it 'recommended-next; unblocked... the cheapest step to flip the track to DONE' (advice standing since ~06-16).

182 - why it matters: Today the golden suite is characterization-only: it catches behavior CHANGES but has no committed F1 floor or windowing fingerprint, so a deliberate re-freeze after a SERVED retune could silently lock in worse quality. It is also the sole unblocked blocker to archiving the whole fixtures project.

183 - proposed fix: Implement M4 per the doc's own recipe: seed per-slug floors from the F1 already frozen in each fx_*/expected.json score (floor = current - tolerance), record a

windowing fingerprint in eval_baselines/regression_baseline.json, keep CONTROL-only, add to the golden-crossos CI matrix.

184 - verifier notes: Two sharpenings. (a) Failure scenario, concrete: after a SERVED windowing retune the owner re-freezes real fixtures via `--refresh-snapshot --reason ...`; the characterization gate parse-compares against the NEW expected.json so it passes by construction, and with no committed floor (and no windowing fingerprint to raise "incommensurable") an F1 drop from 0.59 to e.g. 0.30 on fx_r0x lands green in CI ? the doc itself flags this rubber-stamp hole at GOLDEN_REGRESSION_FIXTURES.md:141 and names the floor baseline as the mitigation. Severity context keeping it P2 (not P1): the owner-box nightly tripwire (eval_baselines/nightly_baseline.json + scripts/nightly_regression.py) partially covers quality regressions, but only on the local uncommitted corpus, not on the committed fixtures / contributor CI path. (b) Fix note: `backend/eval/regression.py:34` already reserves DEFAULT_BASELINE_PATH = eval_baselines/regression_baseline.json for the DB-backed full-chain regression ? the M4 implementation should either share that schema deliberately or pick a distinct filename to avoid two tools silently contending for one file. Also, the finding's title should not be read as "last blocker to DONE": the P0 video-name-rename remainder (BACKLOG.md:61, GOLDEN_REGRESSION_FIXTURES.md:151 status ?) is a second DoD blocker, though owner-gated ? the finding's "sole UNBLOCKED blocker" phrasing is accurate.

185 - verifier reasoning: Every factual claim reproduced in the current tree (master, up to date with origin/master per `git status -sb`). (1) docs/GOLDEN_REGRESSION_FIXTURES.md:156 ? M4 row ("Quality-floor gate + first `regression\_baseline.json`") status ?, PR "?", Depends on M1/M2 which are ? (lines 152-153), Counsel-gated "No" ? unblocked. (2) docs/GOLDEN_REGRESSION_FIXTURES.md:197 ? \$9 DoD first checkbox: "***P0, M1?M4, P1, DOC** are ? and merged" ? M4 gates DONE. (3) Artifacts absent: `ls eval\_baselines/` shows experiment_leaderboard.json, fixtures/, gemini_wrapper_2026-06-10.json, heuristic_lovo_n6.json, nightly_baseline.json, reelcount_manifest.json ? NO regression_baseline.json; `ls tests/ | grep -i golden` shows test_golden_fixtures.py, test_golden_fixtures_split.py, test_golden_real_fixtures.py ? NO test_golden_quality_floor.py; repo-wide grep for "quality_floor" hits only docs. (4) docs/BACKLOG.md:60 ? M4 entry marked "recommended-next; unblocked" and "the cheapest step to flip the track to DONE", exactly as cited. (5) Not already fixed: `git log --oneline -- docs/GOLDEN\_REGRESSION\_FIXTURES.md` tops out at 7a40434 (P3 fixtures, #317) ? no M4 commit; eval_baselines history (#202/#217/#239 nightly work) created `nightly\_baseline.json`, which is a DIFFERENT mechanism (its own `\_doc`: "Tied to the LOCAL golden corpus (not committed)", owner-box `scripts/nightly\_regression.py`) and does not satisfy M4's committed video-free floor. (6) Failure scenario is real per the doc's own text: \$6 line 141 admits `--refresh-snapshot` "can rubber-stamp a real regression" and names the separate floor-baseline update (= M4) as the mitigation; \$4a line 82 confirms fixtures pin raw (vt,mg,mwd) so only M4's windowing fingerprint would flag a post-retune re-freeze as incommensurable. Not owner-gated: M4 row says counsel-gated "No", BACKLOG says unblocked/revisit-anytime; none of the owner-gate categories apply (the P0 rename remainder at BACKLOG.md:61 is the owner-gated sibling, and the finding correctly scopes itself to "the one UNBLOCKED item").

186

187 ### R1-16 [P2/L] tech-debt ? badminton-highlight-indexer

188 **Decision-snapshot regression track: decisions locked 06-25, zero code shipped since; tracker P0 row also stale**

189 - anchor: `docs/DECISION\_SNAPSHOT\_REGRESSION.md:66` · tracked in:

docs/DECISION_SNAPSHOT_REGRESSION.md \$7 (P1-P8) · finder: engine:audit-doc-reconcile

190 - evidence: \$7 tracker (lines 63-73): P1-P8 all ?. Verified in code: grep 'build_stitch_plan' across backend/ and tests/ returns nothing (P2's pure stitch-plan seam was never factored out of backend/pipeline/stitcher/compiler.py), and tests/test_golden_fixtures.py has no stitch-plan snapshot. Minor rot: the P0 row (line 65) still shows ? '(this PR)' though the doc merged as #383 (commit 1283e18, 2026-06-25) and the header still says 'DRAFT'.

191 - why it matters: This gate exists precisely to bless genuine (non-AST-identical) refactors as quality-neutral ? i.e., exactly the class of change as the stalled #390 main.py split. Executing P1-P4 first would de-risk redoing the god-module split; leaving it a week-stale locked-decisions doc means refactors continue with no behavioral proof.

192 - proposed fix: Start \$7 P1+P2 (canonical decision-snapshot schema + factor pure build_stitch_plan() out of compiler.py) as the next Claude-doable engine work; flip the P0 row to ? #383 and status to IN PROGRESS in the same PR. Ideally land P1-P4 before re-cutting the main.py split.

193 - verifier notes: One factual correction to the evidence: P8 (line 73) is ? (blocked/owner-gated), not ? ? "P1-P8 all ?" should read "P1-P7 all ?, P8 ?". Substance unaffected: zero of nine tracker rows show any progress. Sharper failure scenario: #390 gets re-cut and merged as a "quality-neutral" god-module split with no behavioral proof ? the only

offline gate today is the segmentation-only characterization snapshot (test_replay_matches_committed_expected), so a subtle behavior change in stitch/concat-plan construction (clip order, cut points, padding) ships undetected until a GPU nightly or a user-visible reel regression. Effort L is honest for the proposed P1-P4 scope; the minimal P0-row/status flip alone is S and could be split out.

194 - verifier reasoning: Personally verified in the current working tree (master, clean). (1) docs/DECISION_SNAPSHOT_REGRESSION.md:8 still reads "Status: `DRAFT` ... LOCKED (2026-06-25) ... Last updated: 2026-06-25"; line 65 P0 row is "? | (this PR)" even though the doc merged as PR #383 (gh: MERGED 2026-06-25T17:47:44Z; sole commit 1283e18 touching the file ? `git log --oneline -15 -- docs/DECISION\_SNAPSHOT\_REGRESSION.md` returns exactly one commit, so nothing fixed it since). (2) Zero code shipped: grep for build_stitch_plan across the repo hits only the doc itself and docs/README.md; backend/pipeline/stitcher/compiler.py:24 contains only `class VideoCompiler` with no pure plan-builder seam; grep -i stitch under tests/ returns 11 files NOT including tests/test_golden_fixtures.py, confirming no stitch-plan snapshot exists; repo-wide grep for stitch_plan|decision_snapshot matches only docs. Tracker rows at lines 66-72 (P1-P7) are all ? with PR "?". (3) Why-it-matters context holds: gh shows PR #390 ("refactor(D1/P10): extract orchestration.py from main.py (god-module split, step 1)") is still OPEN ? exactly the non-AST-identical refactor class this gate (D11, line 40) is designed to bless. (4) Not owner-gated: the only owner-gated rows are P0 (already satisfied ? line 82 "[x] D1?D11 locked by the owner (2026-06-25)") and P8 (line 73); P1-P7 are marked "Gated? No", and the proposed fix touches only non-gated rows and none of the listed gate categories.

195

196 ### R1-17 [P2/M] tech-debt ? badminton-highlight-indexer

197 **Fetched-input scratch copies are never reclaimed anywhere (process jobs, compile jobs, /api/assets, cloud_run marker reads) ? 'OS temp reclamation' premise is false on Windows**

198 - anchor: `backend/storage/\_\_init\_\_.py:89` · tracked in: ? · finder: engine:correctness

199 - evidence: storage/__init__.py:80-90: acquire_local_input docstring says the scratch copy 'is left under the system temp dir ... and relies on OS temp reclamation'. Call sites that leak one full-video copy per run: _execute_processing (main.py:1168-1171), _execute_compile (main.py:1683-1686), create_asset (main.py:791-799), get_source (main.py:646). cloud_run.py:219-222 additionally leaks a cr-read-* mkdtemp per marker/report read. No janitor exists (grep for rally-input-/cleanup found none); engine issue #301 covers only the upload_dir retention sweep, not temp scratch.

200 - why it matters: Slow-burn disk exhaustion on exactly the appliance the product ships as; also inflates the free-space guard's false rejections.

201 - proposed fix: Wrap remote-input acquisition in a per-job scope that deletes the scratch dir in a finally (worker already owns the job lifecycle), or route scratch under output_base(cfg)/tmp with an age-based sweep at startup; make cloud_run._read_json use TemporaryDirectory.

202 - verifier notes: Two sharpenings. (a) get_source (main.py:646) cannot use a simple finally-delete: the FileResponse streams from the scratch file after the handler returns, so deletion there needs the age-based-sweep option (or a keyed scratch cache reused across views) ? the finder's second proposed option covers it; per-job finally scopes fit _execute_processing/_execute_compile/create_asset. (b) Add a fifth leaking call site to the fix scope: the CLI single-shot path at main.py:2056. Sharper failure scenario: with input_backend=gdrive-api/gcs, each processed match leaks one full multi-GB copy into %TEMP% per process job, another per compile, one per /api/assets source_uri registration (fetched solely to hash), and one per /api/videos/{id}/source view ? until require_free_bytes(tempfile.gettempdir()) at main.py:789/1402 starts rejecting all further remote ingests at stage="fetch" even though the disk is full of the tool's own dead scratch.

203 - verifier reasoning: Personally verified in the current tree. (1) backend/storage/__init__.py:80-82 docstring says the scratch copy "is left under the system temp dir ? and relies on OS temp reclamation"; line 89 does tempfile.mkdtemp(prefix="rally-input-") and line 90 fetches into it ? no deletion. (2) Every claimed call site leaks: _execute_processing main.py:1168-1171 (no rmtree anywhere in main.py ? grep confirmed rmtree exists only in eval/golden_real_fixtures.py, pipeline/stitcher/compiler.py, pipeline/detectors/*); _execute_compile main.py:1683-1686; create_asset main.py:791-799 fetches a full video copy just to MD5-hash it (main.py:805) then returns at 865-875 without deleting local_working; get_source main.py:646 fetches per-request and serves it via FileResponse (line 660) ? a per-view full-video leak for remote sources. Plus a fifth site the finder missed: CLI single-shot main.py:2056. cloud_run.py:219-222 _read_json mkdtemps "cr-read-" per marker/report read, no cleanup. (3) No janitor: tools/cleanup_caches.py scans only WSL ~/clips, the output/tree, and security.upload_dir (docstring lines 3-5, classify() lines 52-97) ? never tempfile.gettempdir(); its "tmp*" rule (line 79-80) applies only inside those trees. (4) Not fixed: git log for storage/__init__.py tops out at b6214e5 (repo-wide ruff format); gh issue 301

confirmed to cover only the T11 upload-retention scheduled job, not temp scratch. (5) Guard interplay real: require_free_bytes(tempfile.gettempdir(), ?, stage="fetch") at main.py:789 and main.py:1402 ? leaked copies shrink exactly the headroom the fetch guard checks, so accumulation converts into false fetch rejections. Windows does not auto-reclaim %TEMP% by default, so the docstring premise is false on the shipping platform. Severity P2 honest: affects only remote-input flows (local paths are identity passthrough, __init__.py:85-88), slow-burn, no data loss. Not owner-gated ? pure disk hygiene, no strategy/spend/ToS/corpus decision.

204

205 **### R1-18 [P2/S] critical-fix ? badminton-highlight-indexer**

206 ****GDriveApiStorage.fetch_to_local buffers the entire file in RAM**

(get_media().execute()); put() deletes the old object before the new upload succeeds**

207 - anchor: `backend/storage/gdrive\_api.py:144` · tracked in: ? · finder:

engine:correctness

9. user (2026-07-02T04:46:25.772Z)

480 **### R2-5 [P1/S] ? badminton-highlight-indexer**

481 ****Engine CLAUDE.md 'Current state' section is ~3 weeks false ? claims platform/owned-model work 'has not started'****

482 - verdict: CONFIRMED

483 - reasoning: Personally reproduced against the working tree and git history. (1)

C:\Users\avidu\Projects\kshelsutra-guru\badminton-highlight-indexer\CLAUDE.md:30-31 states "***No platform/owned-model implementation has started.**"; lines 34-37 say the "committed next PLATFORM slice = async job model" and the owned-model track "starts at a greenlit Phase-0 milestone... Do not start implementation without explicit owner approval"; lines 41-42 call `StorageBackend` + `ComputeBackend` registries **"Future:"**. git blame shows lines 30-37 were last written 2026-06-10 (commit 62fb2fb) and never touched since, despite CLAUDE.md itself being edited 6+ times after (e735fd4 06-13 ... af1ccbe 06-30 ? all other sections). (2) Current-tree reality contradicts every one of those claims: backend/storage/ contains a StorageBackend ABC + local/gcs/gdrive/gdrive_api/s3/capabilities implementations; backend/compute/base.py + cloud_run.py exist (ComputeBackend); backend/api/job_worker.py + tests/test_async_jobs.py exist ? the "committed next slice" async job model already shipped (8bcf6ce #339 "make /api/compile async (202 + poll)", 2026-06-22); backend/flywheel.py + backend/worker + backend/eval/served_gate_a.py exist. Owned-model track: a555887 (2026-06-20) "M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed"; 4134721 archives DECOUPLED_COMPUTE as "DELIVERED (M0-M2 + M3.5 on GCP)"; 334ad0f (#336) M11 data-flywheel; b72e6de (#319) training worker fix. 314 commits landed since 2026-06-13, through 2026-07-01 (e410136), including a recorded launch NO-GO decision 2026-06-30 (4a7bfbe #404) ? i.e., the project reached launch-consideration, the opposite of "not started". (3) No fix landed: git log -- CLAUDE.md shows the last commit touching it (af1ccbe, 2026-06-30) added only the bit-flip-PR convention + Python 3.10 line; the Current state section is untouched since 06-10. Bonus staleness in the same file: lines 18-19 claim rally_gate.py Gate A is "not wired into production" while backend/eval/served_gate_a.py and c8a493d "served Gate-A litmus" (#401) indicate a served Gate-A path now exists.

484 - notes: Sharper failure scenario: CLAUDE.md is auto-loaded project memory for every fresh agent session in this heavily agent-driven repo. A fresh session reads "No platform/owned-model implementation has started" + "Do not start implementation without explicit owner approval" and will (a) tell the owner the async job model is an un-started, gated future slice and propose designing/implementing what already exists (backend/api/job_worker.py), (b) treat StorageBackend/ComputeBackend as "Future" and re-design duplicates of backend/storage/ and backend/compute/, (c) misreport project status upward (e.g., in handoffs/audits) as pre-implementation when Gen-0 weights shipped and a launch NO-GO was formally recorded 06-30. This also directly violates the workspace's own "keep it honest ? reflects real, shipped state" convention. Correctly-scoped fix (S, one doc-only PR): rewrite CLAUDE.md "Current state" (lines 29-37) to reflect reality as of 2026-07-01 ? platform slice (async jobs, storage/compute backends, capabilities/assets endpoints) SHIPPED; owned-model M0-M2+M3.5 DELIVERED on GCP, Gen-0 weights pushed; launch NO-GO 2026-06-30 with pointer to #404/docs; fix line 41-42 ("Future" ? "implemented, see backend/storage/, backend/compute/"); fix lines 18-19 Gate-A "not wired" claim against backend/eval/served_gate_a.py; re-point "committed next slice" at whatever docs/NEXT_STEPS.md currently prioritizes. Not OWNER_GATED ? this is a factual truth-sync of docs to shipped code, the same class as the repo's prior "audit truth-sync" commits (62fb2fb, 9736a3b); no spend or strategy decision required. Round-1 claim of "~3 weeks false" is accurate: section frozen 2026-06-10, contradicting implementation landed from ~2026-06-20 onward.

485

486 **### R2-6 [P1/M] ? sports-data-collector**

487 ****Golden-corpus registry rot: all 6 owned 'curated' videos point at a deleted absolute**

local_path and the 4.9GB source videos are single-copy with no documented backup**

488 - verdict: CONFIRMED

489 - reasoning: Registry rot half CONFIRMED by direct observation. sports-data-collector/data/sports_metadata.db (git-tracked; rows landed in commit 9e6f204 "#19 ingest the n=6 owned golden corpus", DB last touched 9f1e792 with no path fix since): videos table has 8 status='curated' rows; the 6 owned ones (adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles; license_type=Proprietary, is_commercial=1) all carry local_path under 'C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled*.MP4' ? that directory does not exist. The folder moved to 'C:\Users\avidu\Projects\khelsutra-guru\Annotation Setup\' whose 'Golden Labelled\' now contains ONLY the six .rallies.csv label files; zero copies of the MP4s exist anywhere in the workspace (searched). The committed export artifact data/eval_export/manifest.json repeats the same 6 dead absolute paths. Code consequence at sports-data-collector/src/cli/curate.py:679-684: export of a curated video with missing local_path emits md5=None plus a warning, and src/core/manifests.py:23-26 documents md5 as "the indexer's join key" ? so any fresh export of the entire owned golden corpus loses the deterministic MD5 join to badminton-highlight-indexer. Data-loss half CONFIRMED as single-copy but the "no documented backup" wording is wrong: (a) verified live via read-only 'gcloud storage ls' (2026-07-02): gs://khelsutra-rally-corpus/videos/ has 10 objects and NONE of the six originals, while labels/ DOES hold all six (2026-06-08_adarsh_avi_singles.rallies.csv etc.) and trajectories are in the bucket ? annotation work-product is safe, source footage is not; (b) Google Drive golden folder (G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15) holds only the 7 proxy clips, not the original six; (c) the sole remaining copy is documented at F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\ per badminton-highlight-indexer/scratch/copy_videos_to_gcs.ps1:31-36 and docs/DATA_IN_GCS.md:153 (\$6 Phase-2) ? and the F: drive is NOT currently mounted (only C: and G: exist), so even that copy's integrity is unverifiable today, and no MD5s of the six were ever persisted anywhere (eval_export/manifest.json predates the md5 field from PR #51; no proxy-registry rows).

490 - notes: Corrections to the round-1 claim: (1) Size is ~37.9 GB, not 4.9 GB ? per-clip sizes annotated in scratch/copy_videos_to_gcs.ps1:31-36 (10.9+7.7+11+6.7+1.4+0.19 GB); the ~4.9-5.2 GB figure matches the three unrelated Roora pilot videos under 'Annotation Setup/Archive/.../Annotation Videos/'. (2) "No documented backup" is false ? badminton-highlight-indexer/docs/DATA_IN_GCS.md \$5/\$6/\$8 explicitly documents the F: archive location and a Phase-2 migration plan (tracker at doc line 246: 14/15 videos still unmigrated); but "single-copy" stands: exactly one copy of the six source videos exists, on an intermittently-mounted external HDD, unverifiable right now, with no persisted hashes. (3) Downgrade from potential P0 to P1 because labels + trajectories for all six ARE in gs://khelsutra-rally-corpus (verified), so GPU-free eval/litmus and synth-train survive a drive failure; what would be permanently lost is the proprietary source footage (no re-extraction via --trajectory-source detector, no re-annotation, no re-rendering). Sharper failure scenarios: (a) F: HDD failure = permanent loss of ~38 GB of irreplaceable owned golden footage; (b) today, 'curate export' on the owned corpus emits md5=None for all six (curate.py:679-684), silently breaking the indexer MD5 join contract added in PR #51. Correctly-scoped fix (M, cross-repo): mount F:, verify + hash the six MP4s (persist MD5s into the DB/manifest immediately ? S and decision-free), generate 1080p proxies, upload via the existing scratch/copy_videos_to_gcs.ps1 or the rclone cloud path (Phase-2, already planned), then reimport/annotate local_path (or add a gs:// video_uri column) in sports_metadata.db and refresh data/eval_export/manifest.json. Partially OWNER_GATED sub-decision only: DATA_IN_GCS.md \$7.4 leaves proxies-only vs full-4K migration scope OPEN and the upload is real (though tiny, cents/month) GCS spend; the hash-and-record step and the DB path fix need no owner decision.

491

492 ### R2-7 [P2/S] ? non-git

493 **Roora vendor pilot record (paid deliverables + corrected golden labels + 3 videos) is a single unversioned local copy at risk of loss**

494 - verdict: CONFIRMED

495 - reasoning: CONFIRMED for the paid deliverables + corrected golden labels; the "3 videos" component of the claim is REFUTED (Drive copies exist). Evidence: (1) Single-copy artifacts verified on disk: raw paid CVAT deliverables at 'C:\Users\avidu\Projects\khelsutra-guru\Annotation Setup\Archive\Roora Pilot Review - 6 June 2026/' ? badminton/{badminton, badminton (1), badminton (2)}/Annotations/*.xml (~2.3 MB per-frame boxes), pickleball/*.zip (3 zips, 260 KB), tabletennis/*.zip (3 zips, 213 KB) ? plus the human-corrected golden labels 'Annotation Setup/golden_labels_badminton.csv' (2.5 KB, mtime 2026-06-06) and the rescued multisport goldens 'Annotation Setup/Trajectories/roora/{pickleball,tabletennis}_rallies.csv'. (2) Not in git: sports-data-collector tracks only the AS-DELIVERED parse (data/roora_badminton_canonical.csv, verified via git ls-files + diff ? it still has rally 2 =

0:02.503?0:32.533, rally 7 end 1:51.612, no rallies 7.5/16.5), while golden_labels_badminton.csv holds the corrections (rally 2 start?0:29, rally 7 end?1:38, missed rallies 7.5@1:46 and 16.5@3:34 added, winner?out fixes) ? those corrections exist NOWHERE else (find over C:/Users/avidu/Projects + Downloads returned only this one file). (3) Not in GCS: recursive 'gcloud storage ls -r gs://khelsutra-rally-corpus/**' contains zero objects matching roora|pilot|pickle|tabletennis|sample_long|golden_labels; labels/ holds exactly the 15 canonical corpus clips (pilot video not among them); archive/ holds only a droplet archive. (4) Not on Drive: 'G:/My Drive/Sharing' has guidelines/briefs and 'Golden Annotations' (MahadevpuraSingles only) ? no CVAT exports. (5) Outside every backup convention: badminton-highlight-indexer/docs/DATA_IN_GCS.md §5 gap table (lines 117?131) covers only the 15-clip corpus and never mentions the pilot artifacts (and its line 121/142 path 'C:\?Annotation Setup\Collect\Trajectories\' is stale ? the folder moved, no Collect dir exists); the F: backup (memory checkpoint 2026-06-22) covered only the 7 new-clip labels; docs/archives/roora-vendor/README.md archives only guidelines+ADR, not pilot results; F: is currently unmounted. REFUTED part: the 3 pilot videos have byte-size-identical cloud copies at 'G:/My Drive/Sharing/Annotation Videos/Pilot Videos/' (Sample Long Video - Baadminton.MP4 = 2,882,861,900 B; Video 2 - Pickleball.mp4 = 1,435,276,306 B; Video 3 - Tabletennis.mp4 = 871,944,715 B ? all match local), and the local 'Annotation Videos/desktop.ini' references GoogleDriveFS.exe proving Drive File Stream provenance. Owner's own memory audit had flagged the precursor ("Roora multisport CSVs live ONLY in %TEMP%? one cleanup from loss"); the rescue moved them to Annotation Setup but they remain a single unversioned C:-disk copy.

496 - notes: Correction to the finding: the at-risk payload is NOT the 4.9 GB folder ? the videos are safe on the owner's Google Drive. What is genuinely one C:-disk failure (or one folder-cleanup) from loss is ~3 MB of paid, partly irreplaceable work product: the raw vendor CVAT exports for all 3 sports (box geometry the tracked canonical CSVs discard), the human frame-by-frame correction record (golden_labels_badminton.csv + rally_review.csv ? hours of review labor and the calibration evidence behind the vendor feedback), the pickleball/tabletennis golden CSVs backing the recorded TT-0.909/PB-0.308 multisport result, and the pilot review MD/PDF record. Vendor re-export from CVAT job 4075945 is not under owner control. Failure scenario: C: SSD failure or a cleanup of the non-git 'Annotation Setup' folder silently destroys the only record of the paid pilot and its corrected labels; nothing in DATA_IN_GCS.md's migration plan would ever pick it up because the pilot artifacts are absent from its §5 gap table. Fix (S, not owner-gated for the metadata): one 'gcloud storage cp -r' of 'Annotation Setup/Archive/Roora Pilot Review - 6 June 2026/' (excluding Annotation Videos/, already on Drive) + 'Annotation Setup/{golden_labels_badminton.csv,rally_review.csv,Trajectories/roora/}' to gs://khelsutra-rally-corpus/archive/roora-pilot-2026-06/ (archive/ prefix precedent already exists; ~3?10 MB, negligible cost), then add a Roora-pilot row to DATA_IN_GCS.md §5 and fix its stale 'Annotation Setup\Collect\Trajectories' path (lines 121/142). Including the 3 pilot videos in the bucket is the only owner-gated piece (DATA_IN_GCS §7 open decision 4, ~5 GB ? \$0.10/mo) and is optional given the Drive copies.

497

498 ### R2-8 [P1/S] ? badminton-highlight-indexer

499 **Upload retention sweep tool exists but nothing schedules it ? unbounded upload growth + unenforced 7-day privacy retention on any hosted deployment (#301)**

500 - verdict: CONFIRMED

501 - reasoning: Tool exists: badminton-highlight-indexer/tools/cleanup_caches.py (T11 retention sweep, --uploads-retention-days default 7.0 at ~line 407; classify() purges uploaded videos + .partial-* in the uploads zone; landed cb7394b/#299). Nothing schedules it: workspace-wide grep for cleanup_caches hits only the tool/tests/docs and an error string (backend/pipeline/detectors/base.py:162); zero references in .github/workflows/nightly.yml, deploy/cloudrun/, deploy/digitalocean/, scripts/, and zero cron/timer/sweep references in khelsutra/deploy/droplet/ (khelsutra-engine.service has no companion timer). Backend startup only sweeps Gemini remote orphans (backend/main.py:1808 _maybe_sweep_gemini_orphans), not local uploads. Issue #301 is OPEN (viewed via gh today); no scheduling commit in git log between cb7394b and master e410136. Deployment is LIVE: khelsutra/deploy/DEPLOY_ALPHA.md line 3 "Status: LIVE (deploy live-verified on the droplet 2026-06-22)"; uploads land via backend/api/uploads.py into security.upload_dir (/srv/khelsutra/incoming/uploads per runbook §2.2).

502 - notes: Sharper failure scenario: the alpha droplet is live and accepting uploads behind Cloudflare Access; without any scheduled `python -m tools.cleanup\_caches --apply`, a tester's footage uploaded day 0 is still on disk day 30+ (unenforced 7-day privacy retention, an explicit owner policy per the tool docstring, 2026-06-21), and output/uploads grows unboundedly until require_free_bytes (backend/api/uploads.py:93) starts rejecting new uploads (service degradation on the small CPU droplet). Mitigations that keep this P1 not P0: runbook notes testers had not yet exercised the upload CUJ (OTP login + Gemini key rotation pending as of

2026-06-22), and the owner can run the sweep manually. Fix scope (S): do NOT re-implement the sweep ? ship a systemd timer + oneshot unit (or cron line) alongside
khelsutra/deploy/droplet/khelsutra-engine.service that runs `python -m tools.cleanup\_caches --apply --uploads-retention-days 7 --uploads-dir <security.upload\_dir>` daily with --json output captured to journald; default-OFF/opt-in flag so local single-user installs are unaffected; document enable+verify steps in DEPLOY_ALPHA.md; one smoke test asserting the wrapper passes the right flags (tool logic already unit-tested in tests/test_cleanup_caches.py). Issue #301's acceptance criteria already spec this exactly. Partial owner gate: shipping the timer unit + docs is not gated, but *enabling* it on the live droplet is an ops action requiring the owner's SSH access, and #301 flags one open design call (whether apply should also skip uploads referenced by non-terminal jobs rows) ? recommend defaulting to the time-window-only behavior and letting the owner ratify.

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505 ## Owner-gated (flagged, no action)

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507 - [P2] badminton-highlight-indexer: **P0-rename remainder: person/venue-attributable video names still committed in code and baselines** ? Defect verified in current tree: backend/eval/rally_seg_eval.py:52-54 commits MULTICOURT_CLIPS = frozenset({"mahadevpura_2", "GX010128", "Badminton_BXH_2", "Boxhill_Doubles"}) (venue-attributable), and eval_baselines/heuristic_lovo_n6.json per_video keys are adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles (person first names + venues). git log on rally_seg_eval.py (latest 0c9c693 #395) and on the baseline (#77) shows no rename ever landed. docs/GOLDEN_REGRESSION_FIXTURES.md:151 and docs/BACKLOG.md:61 track this exact remainder as DEFERRED per the owner's safe-scope decision (2026-06-16) with trigger "before any open-sourcing / public commit" ? a product/strategy owner gate ? and the coordinated fix requires renaming the owner-held gitignored output/human_lovo_manifest.json in lockstep or the gen0 LOVO training floor breaks. Repo is currently PRIVATE (gh repo view: Khelsutra/badminton-highlight-indexer, isPrivate=true), so no live public exposure today. Real, unfixed debt, but remediation is explicitly owner-gated on the open-sourcing decision and owner-side data coordination.

508

509 - [P2] khelsutra (avidullu/khelsutra): **Issue #114 stale-open: CF Access guest-login debugging against a deployment that no longer exists** ? Debt is real but the fix is an owner tracker call on the owner-gated alpha/serving track. Verified: (1) gh issue view 114 ? OPEN, createdAt == updatedAt == 2026-06-23T10:25:52Z, 0 comments; body is CF-dashboard-only triage of the Access allowlist for the droplet alpha and explicitly says "Deferred by owner 2026-06-23". (2) Cited anchor exists: deploy/DEPLOY_ALPHA.md:106 is the Allow-policy/Emails-allowlist step (line 105 is the app-domain step ? off by one). (3) Deployment gone: memory session-handoff.md:19 ? droplet claude-do-rally-test/168.144.126.176 deleted 2026-06-25, "the prior 'LIVE alpha on the droplet' ... is GONE ? alpha is paused"; api.khelsutra.guru CNAME/CF-Access flagged for cleanup. (4) Not already fixed: git log -- deploy/DEPLOY_ALPHA.md shows nothing relevant after 2026-06-23 (latest 6e4dbaf PR #113 is the unrelated GPU-VM doc); no in-repo reference to #114. (5) Nothing in-repo can fix or verify it ? the Access policy lives in the Cloudflare dashboard and the origin behind app./api. no longer exists. Fix requires owner disposition (close vs retitle), squarely under the alpha/serving go-live gate.

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512 ## Critic additions (round 1 completeness pass)

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514 - [P1/M] critical-fix: **Golden-corpus registry rot: all 6 owned 'curated' videos point at a deleted absolute path, and the 4.9GB source videos are single-copy with no documented backup** (`sports-data-collector/data/sports\_metadata.db:0`)

515 - evidence: Query of the git-tracked registry: videos rows adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles (all is_commercial=1, status=curated) have local_path='C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\...' and video_url=NULL. `ls /c/Users/avidu/Projects/Annotation Setup` -> 'No such file or directory'; the data now lives at 'khelsutra-guru/Annotation Setup/Golden Labelled' (folder moved, registry never updated). src/cli/curate.py:224 registers owned videos with `video\_url=None, local\_path=video\_path` ? raw operator-supplied absolute path, no normalization or later integrity check. ShuttleSet rows are also CWD-relative with mixed separators ('./data/videos/shuttleaset_1.mp4'). 'Annotation Setup/Archive' is 4.9G of source match videos; grep for gs://|bucket|backup across sports-data-collector docs/README returns nothing, while docs/MANUAL_INGESTION_POSITIONING.md:13 declares the collector the 'single canonical system-of-record' for golden rally labels.

516 - fix: Add a `curate verify` (or extend cleanup.py) that resolves every registered local_path and flags dangling rows; re-point the 6 golden rows to the current 'khelsutra-guru/Annotation Setup/Golden Labelled' location; store paths relative to a configurable corpus root instead of absolute machine paths; and back the 4.9GB Archive up to the existing GCS bucket, recording the gs:// URI in video_url so the registry has a durable pointer.

517

518 - [P2/S] ci-infra: **sports-obsreport has no CI at all ? combined with the engine's skipped reporting tests, zero reporting-stack code is gated anywhere** (`sports-obsreport/pyproject.toml:7`)

519 - evidence: `ls sports-obsreport/.github/workflows` -> no such directory (every other repo in the umbrella has workflows). The repo has 10 test suites (tests/test_customer_message.py ... test_spec.py) and a dev extra with pytest+pytest-cov, but nothing runs them on push/PR to github.com/Khelsutra/sports-obsreport (HEAD = v0.1.3, main). The engine consumes it as a tagged dependency (badminton-highlight-indexer/pyproject.toml:68) and the engine's own 17 obsreport-gated tests never run in CI (already-confirmed finding) ? so no test anywhere gates the reporting stack.

520 - fix: Add a minimal .github/workflows/ci.yml to sports-obsreport: checkout, pip install -e .[dev], pytest --cov on push/PR. This closes the library half of the reporting-stack gap independently of the engine's obsreport-lane work.

521

522 - [P3/S] hygiene: **Engine .git bloated to 281MB by ~229MB of dangling raw match-video blobs; 6.5MB yolov8n.pt is permanently in pushed history and .gitignore has no guard against re-staging raw videos** (`badminton-highlight-indexer/.gitignore:49`)

523 - evidence: `git count-objects -vH`: 4716 loose objects, 266.33 MiB loose vs only 2.61 MiB in packs; du .git = 281M. Four loose blobs are raw MP4s (ftypisom headers): 22.4MB, 31.7MB, 84.5MB, 90.2MB ? all unreachable even via `git rev-list --all --reflog` (dangling from an aborted `git add`), never gc-pruned. Separately, yolov8n.pt (6,549,796 bytes) is reachable history on origin/master (added 5257fd5, removed d19557e 'replace AGPL ultralytics YOLO'), so every clone carries the AGPL-era weights forever. .gitignore covers *.pt (line 5) but only two narrow mp4 patterns (lines 49-50: backend/static/highlights_*.mp4, *_compiled.mp4) ? nothing stops the raw-match-video staging accident from recurring.

524 - fix: Run `git gc --prune=now` in the local checkout to reclaim ~260MB, and add a broad guard (e.g. `\*.mp4` with explicit `!` exceptions for any tracked fixtures, or ignore the video scratch dirs) to .gitignore. Optionally note yolov8n.pt in docs as accepted history weight; rewriting pushed history is not worth it at 6.5MB.

525

526 - [P3/S] tech-debt: **obsreport dependency URL still points at the pre-transfer avidullu org ? installs survive only via GitHub's transfer redirect** (`badminton-highlight-indexer/pyproject.toml:68`)

527 - evidence: pyproject.toml:68: "obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3" (echoed in requirements.txt:35). The repo actually lives at Khelsutra/sports-obsreport ? `gh repo view avidullu/sports-obsreport` resolves to nameWithOwner Khelsutra/sports-obsreport, i.e. a transfer redirect. The redirect breaks the moment any repo is re-created under the old name, and like the already-flagged trackers pin this is a mutable tag, not a commit SHA. The local clone's own remote already uses the Khelsutra URL, so the pyproject reference is the only stale spot.

528 - fix: Update the dependency URL in pyproject.toml (and the requirements.txt comment) to github.com/Khelsutra/sports-obsreport.git and pin the v0.1.3 commit SHA instead of the tag, matching whatever convention the trackers-pin fix adopts.

529

530 - [P3/S] hygiene: **1.7GB of stale yt-dlp partial-download debris in data/videos and placeholder junk rows in the committed corpus registry** (`sports-data-collector/src/crawlers/badminton\_crawler.py:231`)

531 - evidence: data/videos contains _diag.mp4.part (354MB), _t.mp4.part (516MB), _v.mp4.part (849MB) plus Frag*.part and .ytdl markers, all dated 2026-05-26 ? 1.7G total of dead partial downloads (du -ch *.part *.ytdl). The crawler's failure cleanup (badminton_crawler.py:231-246) removes only `local\_path`, never yt-dlp's .part/.part-Frag*/.ytdl side files. The tracked sports_metadata.db also carries two placeholder rows with video_url='https://youtube.com/watch?v=example' (staging_match_non_comm, shuttlestet_12) ? test fixtures committed into the canonical registry.

532 - fix: Delete the stale *.part/*.ytdl files; extend the crawler's failure cleanup (or cleanup.py) to glob and remove yt-dlp fragment side files for the failed target; purge or clearly mark the two 'watch?v=example' placeholder rows out of the committed registry.

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535 ## Unverified backlog (dropped at the round-1 cap; lower-ranked or duplicate)
536
537 - [P2] sports-data-collector: Stale doc pointers after archive moves (cross-repo
DECISIONS.md path + 6 in-code CODE_HEALTH_PLAN references)
538 - [P2] sports-data-collector: curate.py is a 1531-line monolith (CLI + interactive
review + import/export/proxy/licensing)
539 - [P2] sports-data-collector: Golden-label store is a binary SQLite file tracked in git
? unmergeable across the owner's concurrent sessions
540 - [P1] rally-annotator (Khelsutra/rally-annotator; local checkout
C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Content script runs a 3 Hz full-DOM
shadow-piercing scan on every page the user visits, even with the panel hidden
541 - [P1] rally-annotator (Khelsutra/rally-annotator; local checkout
C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Locked decision LD-i8 (web language
persisted + seeded from browser language) is documented as shipped but not implemented ? panel
always boots in English and forgets the choice on reload
542 - [P2] rally-annotator (Khelsutra/rally-annotator; local checkout
C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Issue #26 open and unaddressed: CSV
carries zero provenance ? labels are unattributable to their source video except by filename
543 - [P2] rally-annotator (Khelsutra/rally-annotator; local checkout
C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): E2E never reloads a page: the two
headline persistence behaviors (resume numbering from storage, locale persistence) have no end-
to-end proof, and entrypoints/ are outside unit coverage
544 - [P2] rally-annotator (Khelsutra/rally-annotator; local checkout
C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Issue #22 still fully pending: all six
translation catalogs remain machine-draft ? the declared quality gate before any public launch
545 - [P2] rally-annotator (Khelsutra/rally-annotator; local checkout
C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): 'Annotation Setup' is the un-versioned
annotation-data workspace (not a code duplicate) and its Roora pilot-review artifacts appear to
exist nowhere else
546 - [P1] sports-obsreport: Collector pins obsreport as plain PyPI requirement while the
name is unclaimed on PyPI (dependency-confusion window; install broken today)
547 - [P2] sports-obsreport: Engine (and both repos' comments) reference obsreport via the
stale pre-transfer org URL avidullu/ instead of Khelsutra/
548 - [P2] sports-obsreport: sports-obsreport has zero CI ? tags/pushes are cut with no
automated test or lint gate
549 - [P2] sports-obsreport: Engine-obsreport integration is never exercised in any CI ? the
~12 reporting tests only ever run on dev machines
550 - [P1] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root
C:/Users/avidu/Projects/khelsutra-guru): Roora vendor pilot record (paid deliverables +
corrected golden labels + 3 videos) is a single unversioned local copy
551 - [P2] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root
C:/Users/avidu/Projects/khelsutra-guru): khelsutra-screencast: 8 ad-hoc QA scripts and doc-cited
DoD evidence are unversioned; the 5 cited checkpoint screencasts are absent from the canonical
dir
552 - [P2] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root
C:/Users/avidu/Projects/khelsutra-guru): DATA_IN_GCS.md \$5 gap table contradicts its own \$7:
trajectories/labels marked 'not in bucket' though migrated 2026-06-22
553 - [P2] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root
C:/Users/avidu/Projects/khelsutra-guru): Stale-open #175: OneDrive golden-data sharing plan
never executed and largely superseded by the GCS bucket
554 - [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra,
sports-data-collector, rally-annotator, sports-obsreport: Vendored SPA bundle in engine is 10
web/ commits stale ? engine-served UI lacks the fps display its own API just shipped
555 - [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra,
sports-data-collector, rally-annotator, sports-obsreport: Engine PR #390 (D1 main.py split) is
now CONFLICTING against master and going staler
556 - [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra,
sports-data-collector, rally-annotator, sports-obsreport: Engine CLAUDE.md 'Current state' is ~3
weeks false ? claims platform/owned-model work 'has not started'
557 - [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra,
sports-data-collector, rally-annotator, sports-obsreport: D0 droplet retired 2026-06-25 but
runbooks in BOTH repos still say the deploy is LIVE on it
558 - [P2] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra,
sports-data-collector, rally-annotator, sports-obsreport: sports-obsreport has zero CI ? no
.github directory at all despite shipping versioned releases

559 - [P2] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: Lint/type gates exist only in the engine ? sports-data-collector CI is pytest-only, plus GH-Actions version drift across repos

560 - [P2] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: sports-data-collector working tree is dirty: uncommitted .gitignore WIP + a full stale nested clone of itself

561 - [P0] badminton-highlight-indexer (Khelsutra): #340/#330: /api/process still re-bills Gemini on re-submit of an already-processed video ? no idempotency short-circuit (live P0 cost bug)

562 - [P1] badminton-highlight-indexer (Khelsutra): PR #390 (D1 main.py split, step 1) is CONFLICTING and stale ? master diverged 9 commits on main.py since it was cut

563 - [P1] badminton-highlight-indexer (Khelsutra): #332: NSFW pre-flight gate still absent ? non-owner alpha users can route arbitrary video to Gemini (ToS/abuse + paid-garbage risk)

564 - [P2] badminton-highlight-indexer (Khelsutra): #327: path-jail still blocks gs://-style ingest in hosted mode ? validate_input_path runs before scheme resolution

565 - [P2] badminton-highlight-indexer (Khelsutra): #388 premise is half-stale: the three job-worker functions already live in backend/api/job_worker.py (since PR #234, 06-18) ? remaining scope is shim removal + test patch-target migration

566 - [P2] badminton-highlight-indexer (Khelsutra): #384: coverage floor 70 and offline guard lanes still duplicated verbatim between ci.yml and nightly.yml

567 - [P2] badminton-highlight-indexer (Khelsutra): #301: upload retention sweep tool exists but nothing schedules it ? unbounded upload growth on any hosted deployment

568 - [P2] badminton-highlight-indexer (Khelsutra): #349 (+#331 overlap): WASB streaming decode still forced single-threaded ? L4 GPU ~80% idle; #331 additionally needs a refresh (droplet baseline retired, its lever 3 already shipped)

569 - [P2] badminton-highlight-indexer (Khelsutra): #328: per-video runlog/trace endpoint still missing ? obsreport exists server-side but is not user-accessible

570 - [P1] Khelsutra/badminton-highlight-indexer: No timeout on any serving-path ffmpeg subprocess ? one hung ffmpeg permanently wedges the single-worker job queue

571 - [P1] Khelsutra/badminton-highlight-indexer: MD5 segmentation reuse still unimplemented ? every re-submit of a byte-identical video re-bills Gemini and re-pays the full CPU pipeline

572 - [P1] Khelsutra/badminton-highlight-indexer: WASB streaming decode still single-threaded (num_workers forced to 0), no NVDEC anywhere ? L4 GPU ~80% idle during detection

573 - [P1] Khelsutra/badminton-highlight-indexer: Upload retention sweep still manual ? 7-day privacy retention window unenforced without an operator (#301 open)

574 - [P2] Khelsutra/badminton-highlight-indexer: Remote-input scratch copies (rally-input-*/rally-worker-*) are never reclaimed ? by-design temp leak outside all cleanup tooling

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577 ## Unverified P3 pool

578

579 - badminton-highlight-indexer: regression.py silently passes when no baseline exists (audit B8, still open) (critical-fix, `backend/eval/regression.py`)

580 - badminton-highlight-indexer: Chunked upload: concurrent/proxy-retried PUT of the same part index passes the sequence check twice and interleave-appends ? corrupt assembly detected only by part-count mismatch, forcing a full multi-GB re-upload (test-gap, `backend/api/uploads.py`)

581 - badminton-highlight-indexer: A failed mark_job_done/mark_job_failed write leaves a job in 'running' forever (writers swallow to False, worker ignores the return) ? pollers hang until the next restart's stale sweep (test-gap, `backend/api/job\_worker.py`)

582 - badminton-highlight-indexer: 4 stale remote branches from merged/superseded PRs never deleted (hygiene, `.git`)

583 - badminton-highlight-indexer: D13 is DONE but yolo_hybrid.py docstring still claims tracking uses supervision's MIT ByteTrack (licensing-accounting text now wrong) (doc-rot, `backend/pipeline/segmenters/yolo\_hybrid.py`)

584 - badminton-highlight-indexer: ablation PDF exporter (backend/eval/ablation_pdf.py) has zero CI coverage ? its 2 tests always skip for missing reportlab/pypdf (test-gap, `tests/test\_ablation.py`)

585 - badminton-highlight-indexer: CI test output hides skips and slow tests: -q with no -ra / --durations anywhere (hygiene, `run\_all\_tests.py`)

586 - badminton-highlight-indexer: ruff target-version pinned to py38 while the package requires Python >=3.10 and CI runs 3.13 (hygiene, `ruff.toml`)

587 - badminton-highlight-indexer: JWKS fetch does not enforce https on cf_team_domain ? a http:// misconfig would fetch signing keys over cleartext (security, `backend/api/auth.py`)

588 - badminton-highlight-indexer: GPU-box runbook instructs pasting the raw Gemini key

```

inline in a shell command (lands in root bash history) (hygiene, `deploy/gcp/README.md`)
589 - badminton-highlight-indexer: eval_baselines/reelcount_manifest.json hardcodes
gs://khelsutra-rally-corpus URIs outside the buckets.env single-source (tech-debt,
`eval_baselines/reelcount_manifest.json`)
590 - khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/: Open UX issues #99 and
#103 look overtaken by the 2026-07-01 PR wave but were never reconciled (hygiene,
`khelsutra/web/src/components/VideoLibrary.tsx`)
591 - khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/: web/ SPA has no linter
(no ESLint config or lint script) ? react-hooks correctness classes are unguarded (hygiene,
`khelsutra/web/package.json`)
592 - khelsutra (avidullu/khelsutra): Issue #102 fully open: no &debug= URL-param handling
exists in the SPA (test-gap, `web/src/App.tsx`)
593 - khelsutra (avidullu/khelsutra): Issue #72 fully open: Telugu locale not started
(hygiene, `web/src/i18n/index.ts`)
594 - sports-data-collector: ingest.py records a fake placeholder video URL by default
(hygiene, `src/cli/ingest.py`)
595 - rally-annotator (Khelsutra/rally-annotator; local checkout
C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Local git remote still points at pre-
transfer avidullu/rally-annotator instead of canonical Khelsutra/rally-annotator (hygiene,
`C:/Users/avidu/Projects/khelsutra-guru/rally-annotator/.git/config`)
596 - sports-obsreport: README status section is three phases and three releases stale (doc-
rot, `C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport/README.md`)
597 - sports-obsreport: _plural() mishandles common nouns (e.g. item_noun="match" renders "3
matches") in customer-facing summaries (critical-fix, `C:/Users/avidu/Projects/khelsutra-
guru/sports-obsreport/src/obsreport/render/markdown.py`)
598 - non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root
C:/Users/avidu/Projects/khelsutra-guru): 0.5 GB duplicate video object in the corpus bucket
awaiting the prescribed verify-dupe-then-delete (hygiene, `badminton-highlight-
indexer/docs/DATA_IN_GCS.md`)
599 - khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-
data-collector, rally-annotator, sports-obsreport: 13 remote + 7 local branches across the
workspace are merged/closed debris (hygiene, `khelsutra`)
600 - khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-
data-collector, rally-annotator, sports-obsreport: SPA comments hard-code engine main.py line
numbers that are already wrong and about to get worse (doc-rot,
`khelsutra/web/src/api/types.ts`)
601 - khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-
data-collector, rally-annotator, sports-obsreport: Agent/contributor conventions exist only in
the engine: CLAUDE.md and PR templates missing or inconsistent elsewhere (hygiene, `badminton-
highlight-indexer/CLAUDE.md`)
602 - badminton-highlight-indexer (Khelsutra): #175: golden-data sharing issue references a
design doc and OneDrive folder that do not exist in the workspace ? stale, needs revalidation
(doc-rot, `docs/GOLDEN_REGRESSION_FIXTURES.md`)
603 - badminton-highlight-indexer (Khelsutra): #172: ship-gate F1@tIoU target remains
provisional ? code correctly guards, ratification is pure owner work blocked on corpus growth
(tech-debt, `training/gen0/harness.py`)
604 - Khelsutra/badminton-highlight-indexer: Duplicate ffprobe duration logic:
compiler._get_video_duration re-implements utils.video.get_video_duration (hygiene,
`backend/pipeline/stitcher/compiler.py`)
605 - Khelsutra/badminton-highlight-indexer: Issue #331's per-phase perf baseline was
measured on the retired DO droplet ? re-baseline before spending optimization effort (doc-rot,
`docs/COMPUTE_DECOUPLED_SERVING/README.md`)

```

10. assistant (2026-07-02T04:46:56.390Z)

Audit evidence gathered. Now let me set up the isolated worktree from origin/master.

11. user (2026-07-02T04:47:00.439Z)

```

Preparing worktree (new branch 'docs/reconcile-trackers-2026-07-02')
branch 'docs/reconcile-trackers-2026-07-02' set up to track 'origin/master'.
HEAD is now at e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe
details) (#413)
docs/reconcile-trackers-2026-07-02
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)

```

12. assistant (2026-07-02T04:47:06.807Z)

Worktree is on my branch at origin/master HEAD (e410136, matching the audit's verified HEAD).
Now let me read all target files.

13. user (2026-07-02T04:47:09.175Z)

```
1      # Documentation Status & Archival Register
2
3      > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-06-21.
4      > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.
5      > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7      ---
8
9      ## 0. How to use this
10     - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
§3 is the full register.
11     - **Marking something complete?** Update the doc's own **§7 progress tracker** (the
source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2
archival checklist.
12     - **This register does NOT duplicate completeness detail** ? it indexes lifecycle +
archive-readiness and points to each tracked doc's §7.
13
14     ## 1. Lifecycle legend
15     `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent
reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal
record of a run/launch), **CONVENTION** (a standing rule/guide).
16
17     ## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git
mv`)*
18     Before a foundational/tracked doc moves to `docs/archives/`:
19     1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts
`[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it
first.)
20     2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped,
where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21     3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`,
`docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
22     4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the
evolution stays traceable.
23     5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state
ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this
in the doc's header).
24     6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.
25
26     > **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5
delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward
WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014
lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.
27
28     ## 3. The register
29
30     ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the
completeness source-of-truth)
31     | Doc | Status | Archive-ready? | Next action |
32     |---|---|---|---|
33     | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED**
(#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21) | No | M5 truly-local
? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual:
counsel/prices/OAuth/real-run within the D17 cap) |
```

34 | `RALLY\_QUALITY\_RESEARCH.md` | Foundation, ****tracked & ACTIVE**** (\$7) | No | Drives the R-series; complete when \$7.1 single quality number is attributed |

35 | `R2\_EVAL\_METRIC\_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |

36 | `R1\_MILESTONE\_LAUNCH\_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |

37 | `RALLY\_DETECTION\_GAP\_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of the R/quality cluster |

38 | `RALLY\_DETECTION\_FIXES\_PLAN.md` | ****DONE**** ? findings A/B/C resolved; P2b promoted `inactivity\_temporal\_density` default-ON (#396) | With the R-cluster (deferred per its header; gate = R2 tolF1 ablation + \$7.1 number) | Archive with the R-series when the track wraps |

39 | `RALLY\_DETECTION\_CODE\_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |

40 | `PER\_VIDEO\_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |

41 | `MULTI\_SIGNAL\_FUSION\_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |

42 | `EXPERIMENT\_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |

43 | `PROMINENT\_COURT\_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track selection design |

44 | `EXECUTION\_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |

45 | `GOLDEN\_REGRESSION\_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |

46 | ****Data layer**** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` . `GOLDEN\_SET\_PHASE1\_VIDEOS` . `GOLDEN\_VIDEOS` . `VIDEO\_RECORDING\_GUIDELINES` . `GOLDEN\_DATA\_SHARING`) | LIVING (data, ****no-ML****) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data\_pipeline/README.md`. Roora ***vendor*** docs archived ? `archives/roora-vendor/`. |

47 | `CODE\_CLEANUP\_AUDIT\_2026-06-24.md` | ****IN PROGRESS**** ? Phase 1: P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next. | No | Q4?Q5?Q6?Q8 ? P6b ? P2 (segmenters) ? P3 (extra=forbid) ? P5 (app-factory) ? P7 verify |

48 | `DESKTOP\_APP\_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |

49 | `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |

50 | `PERSONALIZATION\_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |

51 | `UI\_ALPHA\_EXECUTION\_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |

52 | `I18N\_PLAN.md` | Design (P0); ****in implementation**** (adapted) | No | Executed in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated |

53 | `POST\_166\_...RISK\_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |

54

55 **### 3b. Foundational / strategy references (LIVING ? archive only if ****superseded****, not when "done")**

56 | Doc | Status | Note |

57 | ---|---|---|

58 | `PLATFORM\_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING |

59 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |

60 | `VIDEO\_LOCALITY\_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |

61 | `HOSTING\_PLAN.md` | PROPOSAL | ****?** partly overtaken by ADR-013 + the shipped site ? verify (finding #5)****** |

62 | `UI\_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half ****in implementation**** ? `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc) |

63 | `STORAGE\_SHARING\_MODEL.md` | DRAFT (co-design pending) | ? |

64 | `ANALYZERS\_AND\_RECONCILERS.md` | DESIGN (owner-gated) | ? |

65 | `ROADMAP.md` / `CODE\_MAP.md` / `HOW\_RALLY\_DETECTION\_WORKS.md` / `KHELSUTRA\_PRODUCT\_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

66

67 ### 3c. Reports & terminal records (archive candidates once their track wraps)

68 | Doc | Status | Note |

69 |---|---|---|

70 | `REAL\_FOOTAGE\_VALIDATION\_REPORT.md` | validation complete | Motivates R2; archive with the R-cluster |

71 | `RALLY\_DETECTION\_QUALITY\_REPORT.md` | report | Quality snapshot; part of R-cluster |

72 | `QUALITY\_ITERATIONS.md` | LIVING empirical log | ****Keep**** ? the paper-backbone ablation log ([[paper-coauthor-aim]]) |

73 | `CODE\_AUDIT\_AND\_TEST\_HARDENING.md` · `HARDENING\_LOOP\_HANDOFF.md` · `DEFERRED\_HARDENING\_PLAN.md` | COMPLETED ? ****top-level pointer stubs**** (archived copies exist) | ****? confirm intentional stubs (per #165) vs finish-archiving (finding #4)**** |

74

75 ### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

76 `PROJECT\_DOC\_TEMPLATE.md` · `NEXT\_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION\_LAYER.md` · `SETUP\_NEW\_MACHINE.md` · ****`CREATING\_REELS.md`**** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · ****`CONVERT\_GCS\_VIDEOS\_TO\_HIGHLIGHTS.md`**** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET\_WSL\_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

77

78 ### 3e. Already archived (`docs/archives/`)

79 `past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) · `past\_projects/low-regret-moves/` · ****`field-pmf/`**** (field-ops + customer-feedback, 2026-06-21) · ****`roora-vendor/`**** (annotation-vendor engagement, 2026-06-21) · `PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED\_HARDENING\_PLAN-COMPLETED-2026-06-14.md` · `HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL\_READINESS\_REVIEW.md` · `MONETIZATION\_AUDIT.md` · `TEST\_RUN\_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/`.

80

81 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

82 ****ADR-001 ? ADR-014**** (14, all ratified). The compute/serving lineage: ****ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014****. ADR-014 was ****RATIFIED 2026-06-21**** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).

83

84 ## 4. Open due-diligence findings / actions (2026-06-21 review)

85

86 > ****? Doc-restructuring done (2026-06-21, owner-directed):**** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML ****data layer**** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

87 1. ****? RESOLVED (2026-06-21) ? `PER\_VIDEO\_WORKSPACE.md` supersedes `PER\_VIDEO\_OUTPUT\_LAYOUT.md`.**** Folded forward the `video\_slug` naming option (workspace \$8.5) + the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

88 2. ****? RESOLVED (2026-06-21) ? `NEXT\_STEPS.md` refreshed**** with a current 2026-06-21 state + a ****cross-track prioritized pending list****; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.

89 3. ****R-series / quality cluster is ACTIVE**** (`R1`, `R2`, `RALLY\_DETECTION\_GAP\_CLOSURE`, `RALLY\_QUALITY\_RESEARCH`, `QUALITY\_ITERATIONS`, `REAL\_FOOTAGE\_VALIDATION\_REPORT`, `RALLY\_DETECTION\_QUALITY\_REPORT`). ****Do NOT archive yet**** ? archive as a ***cluster*** once R2 is promoted-or-rejected as the gate + \$7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)

90 4. ****Confirm the COMPLETED top-level stubs**** (`CODE\_AUDIT\_AND\_TEST\_HARDENING`, `HARDENING\_LOOP\_HANDOFF`, `DEFERRED\_HARDENING\_PLAN`) are intentional pointer-stubs (per #165) ? or finish-archive them so there's one copy.

91 5. ****`HOSTING\_PLAN.md` / `UI\_PROPOSAL.md` are PROPOSAL but partly overtaken**** by ADR-013 (DO dropped) + the shipped khelsutra site. ****? `UI\_PROPOSAL.md` banner added (2026-06-21)**** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md` localization track is likewise in implementation there. ****`HOSTING\_PLAN.md` banner still pending**** (ADR-013 already noted in it).

92 6. ****? RESOLVED (2026-06-21) ? `COMPUTE\_DECOUPLED\_SERVING` M0 approved + M1?M4 SHIPPED**** (#279/#280/#282/#283); all \$8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 \$100/?10k cap).

93
94 ## 5. Maintenance
95 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at
each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md`
archive policy and [[working-agreement-tracked-project-docs]].
96

14. user (2026-07-02T04:47:09.929Z)

1 # Next steps (live) ? refreshed 2026-06-21
2
3 > ****How to read:**** this top section is the ****current cross-track state + prioritized
pending list**** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed
****owned-model & rally-quality track**** plan (2026-06-13) is ****preserved below, still valid**** ? it
is now ***one of several*** active tracks. Live blow-by-blow: memory `current-state.md`. Doc
lifecycle/hygiene: `docs/DOC\_STATUS.md`.
4
5 ## ? Where we are (2026-06-21)
6 ****DECOUPLED_COMPUTE** delivered on GCP + archived****** (#270); ****compute = GCP only****
(ADR-013, DO dropped). ****COMPUTE_DECOUPLED_SERVING**** subproject spun up (#272, ****ADR-014****) ?
one pure core + local/cloud profiles; ****owner M0 design sign-off pending****. ****Per-video
workspace P0**** shipped (#264); ****doc-status register + archival due-diligence**** in place (#273).
The ****khelsutra site + per-rally UI**** ship in parallel (sibling track). The ****rally-quality /
owned-model**** track (below) is the core ML work, gated on ****growing the golden corpus****.
7
8 > ****? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).**** The rally-detection product is
****not**** cleared for an external/commercial launch; we stay in ****development****. This is the
expected default for the dev phase. Two unblockers gate a future GO: **** (1) **** set + meet the
****ship-gate target (#172)**** ? currently unset, needs corpus growth (and the R2 tolF1 ablation,
item 8); **** (2) **** resolve the ****WASB shuttle-detector weight provenance**** (commercial-clean
licensing). P2b (#396) was a do-no-harm quality increment, ****not**** a launch gate ? merging dev
PRs ? launching. Re-decide only when both unblockers clear.
9
10 ## ? Prioritized pending ? all tracks
11 ****P0 ? highest leverage / unblocks the most****
12 0. ****? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few
recordings lying around ? highlights, from a page, no SSH".**** Today the cloud flow is ****SSH/CLI-
only**** (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`); ****M12 does NOT fix it.**** Needs (a separate khelsutra
`web/` SPA + deploy track, NOT an engine seam): **** (i) **** deploy the engine as a reachable
****service**** (M7 ? Cloud Run service, ? owner GCP spend for a ***persistent*** endpoint), **** (ii) ****
the ****SPA ingest/progress screen**** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ?
poll `/api/jobs` ? reel; ZERO engine code), **** (iii) **** flip ****M9 edge-auth ON**** (merged #290,
default-OFF) to gate alpha tenants, **** (iv) **** a ****compute PICKER in the UI**** ("your machine vs
our cloud", D13/D15). The engine seams make it ***possible***; the SPA makes it ***usable***. Full
record: [[ui-driven-cloud-serving-gap]].
13 0b. ****? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable
batteries-included app ? `docs/PACKAGING\_PLAN.md`.**** The LIGHT-tier engine is bundle-ready: `pip
install` ? `indexer --app` serves the ****bundled SPA**** single-origin, ****torch-free****, with app-
home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades
(P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). ****? Next (fresh focused pushes):**** **** (1) **** P3 ****GPU
validation + screencast**** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]` ; ?
native-WASB 0-rally tuned-config blocker), **** (2) **** P2b-install (uv/script + truly-local default
config), **** (3) **** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then
P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` /
`[[next-session-packaging-app]]`.
14 1. ? ****COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED**** (2026-06-21, ADR-014
ratified). M1?M4 (#279/#280/#282/#283), ****M5 #286, M6 #287, M8 #288, M9-auth #290**** all merged
(default-OFF); the ****real M7 L4 run**** validated the worker (22 rallies, VM deleted, \$0). All \$8
gating Qs locked (D7?D17). ****Owner residual**** (not blocking Claude's default-OFF code): counsel
ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12
real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a
****\$100/?10k cap**** (D17).
15 2. ****[owner, ongoing] Grow the golden corpus**** ? multi-court badminton + ?3
matches/sport (TT first). The single highest-leverage move; the quality track ***and*** full-corpus
training feed on it.

16 3. ****[owner + Claude + GPU] Full-corpus training (n?5)**** ? a real `weights/<ver>/` (the
`0.1.0-l4` bundle is n=2 plumbing-only).

17

18 ****P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)****

19 4. ****[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12****
(M5/M6/M8/M9-auth ? merged this session). ****M11**** = `backend/flywheel.py` capture?store?shadow-
eval?goldenize plumbing reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` +
`backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked
`consent\_attested` (? privacy-sensitive ? give it fresh focus). ****M12**** =
`backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme
(regex already allows the hyphen; add to `\_KNOWN` + `\_backend\_class` + the StorageConfig
Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://`
untouched. ? ****M9-consent cols = a new v7 migration**** (M8 took v6). M9-consent ToS/DPA text =
owner/counsel.

20 5. ****[Claude] Per-video workspace P1?P6**** (P1 v6 DB migration ? P2 worker ? P3 per-video
routing of `detector/+`tmp/` \*[the `output/chunks_` de-hardcode base = ? done via #282]* ? P4
? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-
ledger migration ? author the version bump once.

21 6. ? ****Ops-Agent codified (#291)**** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM
helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN
active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

22 7. ****[owner] Set the ship-gate target (#172)**** ? needs corpus growth.

23 8. ****[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate**** (the
ablation) ? needs GPU-extracted golden features.

24

25 ****P2 ? later / gated / deferred****

26 9. Rally-quality ****P3/P4**** (person-fusion LOV0, audio cue) ? needs GPU golden features.

27 9b. ? ****DONE (#396):**** the R4 density fix (finding B / P2b in
`RALLY\_DETECTION\_FIXES\_PLAN.md`) is ****promoted to default-ON****. Golden-set A/B at the ****SERVED****
windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so
`inactivity\_temporal\_density` default was flipped to True. (Landed flag-gated default-OFF in
#394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the
rally tail open.

28 10. ****M0.4 scorer swap**** ? TemporalMaxer (per the 2026-06-17 external review).

29 11. ****Multisport**** (table-tennis first, then pickleball).

30 12. Serving ****M8/M9**** ? per-video cost ledger, edge auth, DPA/consent (within
`COMPUTE\_DECOUPLED\_SERVING`; owner/legal-gated).

31 13. ****WASB-C3**** weight provenance (gates *sharing* a trained model) ? legal.

32 14. ****Rename repo off "badminton"**** (owner, BACKLOG).

33 15. Owner housekeeping: retire the shared ****DO droplet**** + ****DO token****; collector
****md5**** export (collector repo).

34 16. ****PMF field validation**** (owner-side, business track).

35

36 ****Doc-hygiene (tracked in `docs/DOC\_STATUS.md` S4):**** ? #1 per-video reconcile + ? #2
this refresh; ****pending:**** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary
confirm); confirm the COMPLETED top-level stubs; verify `HOSTING\_PLAN`/`UI\_PROPOSAL` currency.

37

38 ---

39

40 **## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)**

41

42 ****Updated:**** 2026-06-13 (****rally-detection quality track****: fusion + experiment-harness
design docs merged,

43 owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the ****UI-alpha track opened****:

44 ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see

45 `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`; ****PR-1** async jobs (#16) MERGED 2026-06-11, PR #87**;
****PR-2**

46 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +

47 compile?browser-download); next: PR-3 identity. Same day: the ****8?9 quality sweep**
#90?#98 landed** ? ruff/mypy/

48 coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into

```

49     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
50     **Authoritative roadmap:**
51     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
52     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
53
54     ## ? Next-session / GPU-box pickup (2026-06-15)
55     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
56
57     **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `.rallies.csv` labels are not yet here):
58     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
59     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
60     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
61
62     **Also queued (owner-gated / pending input):**
63     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
64     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
65     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
66
67     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
68
69     ## Where we are (one paragraph)
70     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
71     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
72     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
73     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
74     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
75     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
76     battery is done** (see table) ? the moat is quantified.
77
78     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
79     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
80     |---|---|---|
81     | Heuristic (velocity windowing) | 0.657 | 0.157 |
82     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
83     | Two-stage WASB+Gemini refine | 0.83 | ? |

```

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84 | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
85
86 **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
87 wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
88 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
89 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
90 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
91 Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
92
93 ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
in progress)
94 **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
#112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
ONE PR off `master`.
95
96 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
#156/#157/#160/#161/#163 + #165 + final records). ** See archived
`docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules
+ STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
done; clean slate.
97
98 - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion
layer) .
99 `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
100 `docs/ROORA_LABELING_GUIDELINES.md` (v8) . `docs/HOW_RALLY_DETECTION_WORKS.md`
(2-layer mental model).
101 - **Key finding (no re-investigation needed): ** the owner's "held-shuttle counted as a
rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already
implements the net-crossing **"Gate A" ( `apply_gate(..., min_crossings=2)` ) that kills
service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no
`min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via
fusion_hybrid) was the single cheapest, highest-confidence quality win. ** (P2 FusionSegmenter
specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists
`net_crossings` + optional person features, optional served Gate A/B via config knobs;
adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest
finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in
with knob for safety, and the leaderboard revert note.)
102 - **Current status (2026-06-14): ** P1 (person cue extraction to reusable
`PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment
harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-
ups): ** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos;
eager-person-compute optimisation); then P4 audio via hit_detector.
103 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-
OFF, promote only on measured LOVO): **
104 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served
path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served
neutral on production windowing per litmus ? kept opt-in with knob for safety.)
105 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant
(players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the
eager compute optimisation note/landing. Pure + harness (testable here).
106 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
107 - **Discipline (from EXPERIMENT_HARNESS.md): ** every new cue **flag-gated, default
OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit
0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
108 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the
heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden
features + `rallies.csv` labels, which are not yet present in this checkout.
109 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are

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exhausted; full details + citations in RALLY_QUALITY_RESEARCH.md

\$5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):

- 110 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as "the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."
- 111 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
- 112 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
- 113 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
- 114 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve "occluded end-of-rally states" (the #1 remaining gap vs Gemini), not just spectator noise.
- 115 6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

116

117 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

118

119 ## Prioritized next steps

- 120 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
- 121 Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target
- 122 lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are
- 123 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per
- 124 `VIDEO\_RECORDING\_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:
- 125 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
- 126 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
- 127 2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court
- 128 reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93
- 129 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with
- 130 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
- 131 3. **M0.4 next increment (\$0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the
- 132 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the
- 133 LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free
- 134 max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review
- 135 (`RALLY\_QUALITY\_RESEARCH.md` \$5.6) ? better fit for the cheap/local/fast mandate.
- 136 4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across
- 137 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated
- 138 by C3 (provenance multiplies per sport).
- 139 5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
- 140 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
- 141 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:

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142     `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
143     physical-visit approach; reframed product-feedback-first per owner spec):**
144     `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
145     `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
146     rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
147     capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
148     decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
149     6. **Housekeeping:** ~rotate the Gemini API key~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
150     deletion in AI Studio); **rename repo off "badminton"** (owner action,
151     in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
152
153     ## Pointers / artifacts
154     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
155     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
156     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
157     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
158     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
159     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
160     (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
161     labels: `...\Collect\Golden Labelled\`.
162     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
163     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
164     with a MISLEADING "credits depleted" 429; ramp gradually.
165     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
166     PLATFORM slice = async job model (single-tenant).
167

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15. user (2026-07-02T04:47:20.417Z)

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1     - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? **? Foundation doc** for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
**adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with
falsifiable success criteria**, and ($7) a **tracked project plan** ? work-item checklist,
sequencing, the single quality **number** attributed to the effort, and a **definition of done**
(project completes when all tiers land + the number is recorded). The plan that the
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.
2     - [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md) ? **Tracked execution
doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before
launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini
both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local
over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still
moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-
hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A
wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not

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research); §7 is the source of truth.

3 - PROMINENT_COURT_DETECTION.md ? **Design + experiment** (2026-06-19, default-OFF, owner-gated flip) **for the multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` **before** windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion\_features.point\_in\_polygon` + `FusionConfig.court\_polygon`. §8 progress tracker is the source of truth.

4 - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint** (2026-06-17) **of rally-detection quality**: the honest measured state (over-segmentation milestone tolerance-F1 0.09170.323 at n=11, p=0.002; merge-rate 0.69470.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

5 - EVALUATION.md ? How we measure and improve quality.

6 - EXPERIMENT_HARNESS.md ? **Design** (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

7 - QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

8 - ANALYZERS_AND_RECONCILERS.md ? **Design** (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

9 - PERSONALIZATION_PLAN.md ? **Design** (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization** feature on a generalization-first baseline: a thin per-user **tuning** layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; "a tool that gets better the more you help it"). Records the **locked owner decisions** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

10

11 - GOLDEN_REGRESSION_FIXTURES.md ? **Design** (owner-gated, NOT started): **video-free, non-attributable regression fixtures** so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).

12

13 **## ? Documentation status & governance**

14 - DOC_STATUS.md ? **living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.

15

16 **## Product / platform plans**

17 - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **Active tracked project** (2026-06-21, `DRAFT` owner-gated): **the productionization successor to DECOUPLED_COMPUTE** (its deferred M3/M4). **One pure core** (DETECT?WINDOW?STITCH) that never branches on deployment profile, with a **ComputeBackend** (where it runs) + **StorageBackend** (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU,

in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for \$0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). \$7 M0?M10 tracker is the source of truth.

18 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred** ? the "Cloud Run + GPU serving" subproject ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past\_projects/DECOUPLED\_COMPUTE/` (incl. `DEPLOY\_REPORT\_GCP\_2026-06-20.md`).

19 - PACKAGING_PLAN.md ? **Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP\_APP\_PLAN.md` narrative.** \$7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

20 - DESKTOP_APP_PLAN.md ? **Superseded narrative (by `PACKAGING\_PLAN.md`;** D10 = self-hosted webserver, NOT a desktop app). **Original scoping doc (2026-06-18, `design`):** package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical **no-GPU** enthusiast laptop (WASB is ~3.4x real-time on a laptop 2070S)? \$7 tracker = **P0 compute-feasibility spike** ? P1 native detector ? P2 app shell ? P3 packaging/installers. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

21

22 ## Practical / ongoing

23 - CREATING_REELS.md ? **End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a \$6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root [README.md](../README.md); packaging roadmap ? PACKAGING_PLAN.md.

24 - CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? **Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, \$0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (`verify\_compute\_path.py`) ? \$0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? ../deploy/cloudrun/README.md.

25 - SETUP_NEW_MACHINE.md ? **The canonical "spin up a box** ? ready for training & inference" runbook. **Copy-pasteable, human-followable, verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [deploy/gcp/README.md](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/\*` scripts are provider-agnostic and reused on GCP.

26 - DATA_IN_GCS.md ? **The corpus source-of-truth map:** where every piece of data lives in `gs://kheIsutra-rally-corpus` (canonical layout · current contents + drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:/F:/Drive` ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.

27 -

```

[`deploy/gcp/nightly_regression/README.md`](../deploy/gcp/nightly_regression/README.md) ? **?
Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change
any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full
WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status:
[`NIGHTLY_ENGINE_REGRESSION.md`](NIGHTLY_ENGINE_REGRESSION.md)). The lighter CPU cached-
trajectory tripwire is `scripts/nightly_regression.py` (local Windows Scheduled Task, PR #202).
28 - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
29 - Evaluation harness and calibration drivers (in `backend/eval/`)
30
31 ## Data layer (collection · annotation · golden sets ? **no ML**)
32 - [data_pipeline/](data_pipeline/README.md) ? **the isolated data pillar**: how
footage is recorded (`VIDEO_RECORDING_GUIDELINES`), golden sets are created + which videos
(`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS`), and data is
shared/ingested (`GOLDEN_DATA_SHARING`). Deliberately **no segmentation/stitching/model** detail
? that's the ML pillar (top level + `COMPUTE_DECOUPLED_SERVING/`). The **annotation-vendor
(Roora)** engagement is archived under `archives/roora-vendor/`.
33
34 ## Conventions & templates
35 - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a **tracked
project doc**: status header · decisions-locked · progress tracker (??/?) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7
and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
36
37 ## Historical documents
38 Completed plans, old research, ADRs, and checkpoints live in [archives/](archives/).
39
40 - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? **COMPLETED**
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)
41 - [CODE_CLEANUP_AUDIT_2026-06-24.md](CODE_CLEANUP_AUDIT_2026-06-24.md) ? **? Tracked
project (DRAFT, owner-gated):** the next-wave cleanup project ? 33 audit findings across bugs,
tech debt, operational risks, and quick wins. Phase 1 = pure refactorings (typed-config
migration, app-factory, segmenter validator fix); Phase 2+ = logic fixes. §7 progress tracker is
the source of truth; sibling of the completed hardening sweep above.
42 - [DECISION_SNAPSHOT_REGRESSION.md](DECISION_SNAPSHOT_REGRESSION.md) ? **? Tracked
project (DRAFT, owner-gated ? decisions in §2 need locking):** a deterministic **quality-neutral
proof** ? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat
plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-
AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel**
(non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`)
from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`.
§7 tracker is the source of truth.
43
44 - **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ?
[archives/field-pmf/](archives/field-pmf/) (archived 2026-06-21; PMF validation still an
active *business* thread ? see `NEXT_STEPS.md`).
45 - **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ?
[archives/roora-vendor/](archives/roora-vendor/) (archived 2026-06-21).
46 - Superseded: per-video output layout ? [archives/PER_VIDEO_OUTPUT_LAYOUT-
SUPERSEDED-2026-06-21.md](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) ( ?
`PER_VIDEO_WORKSPACE.md`).
47
48 See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.

```

16. user (2026-07-02T04:47:21.206Z)

```

1 # Backlog ? deferred items to revisit
2
3 Single tracker for work we consciously deferred, so nothing gets lost between sessions.

```

Each item records **what**, **why** it's deferred, and **the trigger to revisit**. Every entry traces to a real decision/PR/doc (no speculative work).

4

5 > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all of these to **GitHub issues** instead (or in addition).

6 > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.

7

8 _Last updated: 2026-06-23._

9

10 ---

11

12 ## Batch / declarative orchestration

13 - ? **One-shot "serving startup-script"** for the GPU VM ? `create\_gpu\_vm.sh` already injects a metadata `startup-script` (`deploy/gcp/ops-agent-startup.sh`), but it only installs the **Ops Agent**; standing up the live engine on the VM (`khsutra/deploy/GPU\_VM\_SERVING.md`) is still manual. Propose a `deploy/gcp/serving-startup.sh` fed via `--metadata-from-file=startup-script=?` (or pasted into the GCP console) that, on boot: installs ffmpeg/cv2 libs ? pulls the engine source (from GCS) + builds the venv ? **(optional)** the WASB native env (`gpu\_setup.sh` + weights) ? writes the serving `config.json`/`engine.env` ? installs `cloudflared` with the tunnel token ? starts `khsutra-engine`. So a GPU serving box comes up **hands-off** (reuses the `deploy/gcp/nightly\_regression/run\_on\_vm.sh` bootstrap pattern). **Why deferred** ? the design hinge is **SECRETS**: the Gemini key / CF-Access AUD / tunnel token must **not** sit in a plaintext startup-script (instance metadata is broadly readable by anyone with `instances.get`); they should come from **Secret Manager** (VM SA ? `secretAccessor`) or be injected post-boot. Needs that small secrets-from-metadata design + which config is baked vs fetched ? not just a script (owner asked 2026-06-23). **Revisit**: when the GPU-VM serving path is used often enough to automate.

14 - ? **One YAML manifest drives the whole bucket?reels run** ? today, converting `gs://` videos into highlight reels is a sequence of API calls (see [`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`])(`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`)). Propose a declarative **batch manifest** (`pipeline.yaml`): **inputs** (bucket prefix/glob or explicit list), **compute** target (local/cloud), **detection** (segmenter, skip_ai_handoff), per-video **selection** rules (all / top-N longest / min duration / a query string), and **output** (bucket prefix or local dir + watermark locale) ? driven by one `indexer --batch pipeline.yaml` runner that loops process?query?select?compile?deliver, **idempotent/resumable** (skips already-done `video\_id`s), reusing the existing API + the compute-picker + the compute-path provenance. **Why deferred**: needs a small design ? the manifest schema, the selection grammar, idempotency/resume, and partial-failure handling ? i.e. a tracked project doc per `PROJECT\_DOC\_TEMPLATE.md`, not a quick add (owner agreed 2026-06-23: ship the runbook doc now, backlog the YAML driver). **Revisit**: when hands-off batch conversion is wanted ? start with a design doc.

15

16 ## Owned-model / multisport (source: `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`, `NEXT\_STEPS.md`)

17 - ? **Rename the repo off "badminton"** ? owner-emphatic "ASAP" action; the project is multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. **Revisit**: owner action, before broad external sharing.

18 - ? **Quantify cost-to-Gen-2** (the largest open risk) ? benchmark real VRAM/token/\$ for the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 678 GPUs) **before** standing up any owned serving stack. **Revisit**: before Phase-2 commitment.

19 - ? **Define the ship-gate F1@tIoU target** ? "very high P/R" must be operationalized (floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). **Revisit**: at M0.4 harness.

20

21 ## Engineering hardening deferred from the 2026-06-10 audit sweep

22 - ? **App-factory / lifespan + lazy torch-cv2 imports** ? `uvicorn backend.main:app` leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice (#16). **Revisit**: when the async slice starts.

23 - ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps, `\_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy. **Revisit**: before any volume/batch processing.

24 - ? **shlex-quote WSL config values** ? **DONE 2026-06-11 (#97)**: `detectors/base.py:wsl_tilde_quote` quotes everything after a leading `~/`, so quoting and tilde expansion coexist; applied to every `~`-capable arg in both runners; `tests/test_wsl_tilde_quoting.py` pins the command shapes (no quoted tilde can ever reach`

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`_wsl_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the
default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations
(repo_dir/conda_env) if config ever becomes per-tenant.
25
26    ---
27
28    ## Monetization / licensing
29    *(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)*
30
31    - ? Ultralytics Enterprise pricing ? get a quote only if the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently not
needed (swap done). Quote-only pricing. Revisit: if detector quality forces reconsidering
YOLOv8.
32    - ? Codec patent thresholds (AVC/HEVC/AAC) ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds with legal counsel before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. Revisit: before commercial launch.
33    - ? Per-checkpoint weight/dataset licenses ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the code license ? the distributed weights/dataset terms.
Either train our own or confirm each checkpoint. Revisit: when adopting any borrowed
weights.
34    - ? Real GPU $/video & latency ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. Revisit: when choosing the production LLM step.
35
36    ## Detector swap follow-ups
37    *(source: PR #8, ADR-005)*
38
39    - ? supervision ByteTrack deprecation ? `sv.ByteTrack` is deprecated as of
supervision 0.28.0 and removed in 0.30.0; `requirements.txt` is pinned <0.30.0 as a
stopgap. Migrate to ByteTracker from the separate trackers package before lifting the
cap. Revisit: before bumping supervision past 0.30.
40    - ? Real-video validation of the swap ? the torchvision + ByteTrack Stage-1 is unit-
tested with mocks but not yet validated on actual footage for detection/tracking quality.
(ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
expect tuning.) Revisit: when a real test clip + the pipeline are run together.
41    - ? yolo_hybrid naming ? the registry key, yolo_* config keys, and yolo-
hybrid-* DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. Revisit: if/when a clean break is worth the migration.
42
43    ## Golden-set subproject
44    *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
45
46    - ? Annotation guideline ? DONE (2026-06-02): authored as the Roora brief +
illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time
/ include every rally (incl. faults & lets) / shot = one contact; ending_reason
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
47    - ? Internal gold key ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. Unblocked. Revisit: anytime.
48    - ? GS-3 ? HumanVendorProvider ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). Vendor selected for the initial pilot: Roora
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). Revisit: once the
Roora pilot confirms quality, then build the adapter.
49    - ? GS-4 oracle model ? AutomatedProvider interface is stubbed (this work) with a
deterministic fake. Wire a concrete fast-tier auto-labeler later. Oracle rule: it MUST be
a different model lineage than production, or the regression test is blind to shared failures.
Revisit: when a production LLM step is chosen (pick the oracle's lineage to differ).
50    - ? GS-5 ? regression triggers ? wire regress() into CI (pre-merge gate), on-
demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path.
Revisit: after GS-4.
51    - ? Vendor pilot (Roora) ? run the paid pilot with Roora as a single-vendor
pilot first (not a multi-vendor bake-off): assess quality-per-rupee against the $4 bar;
escalate to the iMerit/Taskmonk fallback only if it misses. Revisit: the pilot doubles as
calibration once the gold key exists.

```

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52     - ? **Phase-1 video collection** ? gather the round-1 golden corpus per
`docs/GOLDEN_SET_PHASE1_VIDEOS.md`: **static self-recorded** footage (mirror deployment, not
broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with
**? held out by match**. **Unblocked**
53     - ? **Train/eval separation guardrails** (was `DEFERRED_HARDENING_PLAN` #13; design
preserved here on archival 2026-06-14) ? train & eval videos live in **separate catalogs**
today, kept apart by convention only; nothing **enforces** that a video used to train the learned
segmenter (`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the
eval/regression set, so leakage would silently inflate the headline metrics. **Concrete design**
(from the archived DEFERRED plan §2): split unit = **match, never clip** (hold ~? of each sport
as eval, per `GOLDEN_SET_PHASE1_VIDEOS.md`); add a `split: "train"|"eval"` field to the
collector's `curate export` manifest (cross-repo contract, ADR-002); **[DONE ? pure-logic core,
PR #171]** `backend/eval/splits.py` now exists with `load_split(manifest) -> (train_ids,
eval_ids)` keyed by **match id** (not file MD5 ? re-encodes share a match but differ in
`compute_video_id`) + `assert_disjoint(train_ids, eval_ids)` that raises on overlap, with
`tests/test_splits.py` (overlapping ids ? `assert_disjoint` raises). **STILL DEFERRED:** wire
the guard into the training entrypoint (`distill_local`/`training`) and into `run_regression.py`
(baseline computed strictly on the eval split); add the collector's `curate export` manifest
`split` field (above); the integration test (a deliberately-leaked manifest fails the training
entrypoint). Full rationale/risks: [`archives/DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md`](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md) §2.
**Revisit:** dedicated design session with the user (needs a collector-side manifest change).
54
55     ## Golden regression fixtures (video-free, non-attributable regression suite)
56     *(source: `docs/GOLDEN_REGRESSION_FIXTURES.md` §7 tracker / §9 definition-of-done)*
57
58     Core suite **shipped** (M1·M2·M3·MX·P1·P2·P3·DOC ? ? synthetic + 6 de-attributed real
`fx_r01`?`fx_r06` fixtures replay cross-OS, leak-guarded, CONTROL-only). The two items below are
the **gating remainder** of the doc's §9 definition-of-done (it flips to `DONE`/archive only
when **P0 + M1?M4 + P1 + DOC** are ?); 01/02 are explicitly optional. Tracked here so the track
isn't silently treated as "done" while DoD items remain.
59
60     - ? **M4 ? quality-floor gate + first `regression_baseline.json`** *(recommended-next;
unblocked).* Assert per-fixture **and** mean F1 ? a committed floor
(`tests/test_golden_quality_floor.py` + `eval_baselines/regression_baseline.json`). **First
move:** seed the per-slug floor from the F1 already frozen in each `fx_*/expected.json` `score`
(real fixtures sit at **0.0?0.59**, so floor = `current ? tolerance`, not a global bar).
**Gotchas:** record a **windowing fingerprint** in the baseline (else a SERVED retune silently
passes); keep it **CONTROL-only** (learned/numpy-BLAS is non-deterministic across CPUs); add it
to the `golden-crossos` CI matrix. **Revisit:** anytime ? it's the cheapest step to flip the
track to DONE.
61     - ? **P0 remainder ? functional video-name rename** *(deferred per owner safe-scope,
2026-06-16).* Rename the functional names (`eval_baselines/*.json` keys,
`rally_seg_eval.MULTICOURT_CLIPS`, quality-log names) ? opaque ids. **Why it's hard:** the keys
feed `training/gen0/harness.py`'s LOVO floor matched to the **gitignored
`output/human_lovo_manifest.json` + data files** ? rename them together or the floor breaks;
also collides with the hot quality track. **First move:** put the names in a gitignored
`.leakscan_private` and run `python -m tools.leak_scan --names .leakscan_private` to enumerate
every occurrence, then plan the coordinated rename. **Revisit:** before any open-sourcing /
public commit.
62     - ? **01 ? protobuf + deterministic serialization** *(optional).* `rally.fixture.v1`
proto3 + codegen + `protoc`/`git diff --exit-code` guard. **Revisit:** only if cross-OS **byte-
stable** snapshots are wanted (JSON parse-compare suffices today).
63     - ? **02 ? Tier-1 key-gated path** *(optional).* `scripts/fetch_golden.py` (OneDrive
token) and/or SOPS+age, `RALLY_GOLDEN_DIR` wired, Tier-1 CI gated on a secret (never on fork
runners). **Revisit:** only if sharing person/biometric streams (Tier-1) becomes necessary.
64
65     ## Recording best practices
66     *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
67
68     - ? **Validate recording hypotheses** ? empirically test which capture conditions
(resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote
the winners into user-facing "how to record" guidance. Only the static-camera rule is currently
evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure
per-condition deltas.

```

```

69
70     ## Code & test organization
71     *(source: owner review comments on PR #35, 2026-06-07)*
72
73     - ? tests/` reorganization to reduce fragmentation ? owner wants to isolate test
components on a grouping
74         metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new
`backend/pipeline/audio/` +
75         `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval
tests under `tests/eval/`).
76         The exact grouping metric is the owner's design choice (consolidate small files
vs. add per-area dirs ? the
77         "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a
dedicated refactor PR,
78         not guessed here. Revisit: when the owner specifies the grouping scheme. (The
`backend/eval/audio/` half of
79         the request ? comment 1 ? was done in PR #35.)
80
81     ## Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)
82
83     *(source: Commercial Readiness Review + structure/sequencing discussion with Grok as
second-in-command)*
84
85     - ? Docs-only checkpoint PR ? This update + the refreshed
`docs/archives/past_projects/low-regret-moves/LOW_REGRET_MOVES_PLAN.md` (since archived)
formally record the approved path before any code changes. The initial review docs (PR #37)
introduced the findings; this entry locks the execution decisions.
86     - ? Approved directory structure ? Adopt a layered approach under `backend/` for
professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-
file dirs):
87         - `backend/config/` (package ? start with this)
88         - `backend/api/` (FastAPI app, routes, static UI)
89         - `backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)
90         - `backend/infrastructure/` (database + external adapters)
91         - Keep `backend/eval/` prominent as a first-class concern.
92         - Logical grouping for providers, sports, annotations, utils, and a small `core/` only
when needed.
93     - ? Sequencing (Config first, then atomic refactor) ? Implement Deliverable 1
(Config) first, introducing the `backend/config/` package immediately. Perform the full
directory structure refactor as a separate high-pri atomic PR afterwards. This minimizes
conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.
94     - ? Deliverable 1 (Config) ? `backend/config/` package (`models.py` + `loader.py`)
instead of a single `config.py`. Typed `AppConfig`, central loader with built-in defaults + file
override, `extra="allow"` for migration. Update `setup.py` + `main.py`. Add tests. Land this
before the layout move.
95     - ? Subsequent items ? See the refreshed `docs/archives/past_projects/low-regret-
moves/LOW_REGRET_MOVES_PLAN.md` (archived) for the full ordered list (Error contracts, logging,
packaging, detector abstraction, etc.). The structure refactor is now explicitly tracked as its
own high-pri step right after Config.
96
97     Status legend reminder: ? = approved / in progress tracking, ? = do when triggered.
98
99     ## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)
100    *(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR
#11/#12; shared lib `obsreport`)*
101
102    - ? Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are
DONE. Remaining = items 3?9 in `docs/archives/past_projects/low-regret-
moves/LOW_REGRET_MOVES_PLAN.md` (archived).
103    - ? Per-video observability reports shipped in BOTH tools (a TECHNICAL report + a
CUSTOMER-friendly summary), via the soft-dependency lib `obsreport`
(github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes `Item 8 (Input-Quality UX) is LARGELY SUBSUMED by these reports (validation stage
+ footgun flags + verdict already surface quality; a clean customer summary already exists).

```

****Do NOT build a parallel mechanism**** ? instead EXTEND the report: add an audio-readiness stage/flag to `backend/reporting/` (harvest) + the obsreport spec. ****Revisit:**** when item 8 is picked up.

105 - ? ****Item 7 (unify hybrid segmenters) should land before the report's real AI-provider timing.**** The report records AI timing as `not\_captured`; once `wasb\_hybrid`/`tracknet\_hybrid` are unified there is ONE place to thread a recorder callback through `provider.analyze\_video` ? the report. ****Revisit:**** with item 7 (Claude to add the timing hook there).

106 - ? ****Item 3 (Error/Result contract) synergizes with the report.**** Today the report INFERS failures (e.g. `silent\_provider\_noop` = 0 segments + no API key). A structured Result/error from providers/segmenters would let the report read the real reason as a first-class flag. ****Revisit:**** with item 3.

107 - ? ****`obsreport` is a SOFT/commented dep**** in `requirements.txt` (works locally via editable install; pipeline no-ops if absent). ****Pin the `git+ssh@v0.1.3` tag**** in both repos once SSH/CI access to the private lib repo exists.

108 - ? ****`run\_all\_tests.py`**** added at repo root ? run the full suite (clear PASS/FAIL, pytest exit code) before sending PRs.

109 - ? ****Config-migration follow-ups (DONE ? config & identity consistency pass)**** ? the two typed-config bugs from

110 #39/#40 are fixed: (1) ****`/api/compile` now honors `--output-dir`**** and (2) ****`output\_dir` is a real field on**

111 ****`OutputConfig`**** (default "output"); the CLI overrides it on the typed model in ****`main()`**, retiring the

112 write-only ****`\_runtime\_overrides`** hack. Same pass also unified ****`video\_id`** MD5 derivation into

113 ****`backend/utils/hashing.py::compute\_video\_id`** and migrated all four ****`run\_\*.py`** scripts off raw ****`json.load`** onto

114 ****`backend.config.load\_config`**. See ****`CODE\_MAP.md`** §2.0/§6 (footguns retired).

115

17. user (2026-07-02T04:47:31.868Z)

```

1      <!--
2      PROJECT DOC ? tracked code-cleanup + audit remediation plan.
3      Follows the PROJECT_DOC_TEMPLATE.md conventions.
4      -->
5
6      # Code Cleanup ? 2026-06-24 Deep Audit Remediation
7
8      > **Status:** `PHASE 1 COMPLETE` ? 14 items shipped + D2 app-factory + test fixes .
**Owner:** Avi . **Created:** 2026-06-24 . **Last updated:** 2026-06-25 (DoD/line-count accuracy
fixes ? see Changelog)
9      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to
`docs/archives/past_projects/` when DONE)
10     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`.
11     > **Relation to existing docs:** Sibling of `docs/CODE_AUDIT_AND_TEST_HARDENING.md` (the
2026-06-10 hardening sweep).
12     >
13     > **Phase 1 summary:** All structural refactorings landed. `backend.api.job_worker`
lifted from mypy quarantine. Suite: **1463/1463 green**. `ruff`: all checks passed. `mypy`:
quarantined modules unchanged (ratchet held).
14
15     ---
16
17     ## 0. TL;DR
18
19     A deep, whole-repo audit on 2026-06-24 surfaced **35 items** (33 from fresh review + 2
absorbed from `docs/BACKLOG.md`) across bugs, technical debt, operational risks, and quick wins.
This doc organizes them into a phased cleanup project: **Phase 1 = pure refactorings** (no logic
change, all tests must stay green, coverage must not drop), **Phase 2+ = logic fixes +
hardening**. The first approved targets are **B2, D3, D2, B5** ? the **`Success`/`Failure`
contract for Gemini, typed-config migration completion, app-factory refactor, and the no-op
segmenter validator fix.
20
21     ---

```

```

22
23     ## 1. Problem & goal
24
25     The codebase has grown rapidly since the June 2026 scaffolding and has accumulated
several patterns that make it harder to extend safely:
26
27     - Silent failure masking ? `analyze_video()` returns `[]` on ANY error (missing API
key, network failure, model crash) indistinguishable from "video has no rallies."
28     - Dual config access ? new code uses typed `AppConfig` attribute access; old code
(validators, segmenters) uses legacy `.get("indexing", {}).get(...)` dict chains. The
`extra='allow'` silently keeps typo'd config keys.
29     - Optional module globals ? `db_instance` and `global_config` are `Optional[...] =
None`, set only in `main()`. Importing `app` without `main()` ? NPE on every endpoint. The mypy
quarantine on `backend.main` and `backend.api.job_worker` exists because of this.
30     - No-op validator ? `segmenter_must_be_registered` field validator is a pass-
through; a typo'd segmenter name in `config.json` passes validation and fails at runtime.
31
32     The goal is to fix these systematically, with zero behavior change in Phase 1, and a
strict no-regression gate (all tests pass, coverage never decreases).
33
34     ---
35
36     ## 2. Decisions locked
37
38     | # | Decision | Source / date | Implication |
39     |---|-----|-----|-----|
40     | D1 | Phase 1 = pure refactorings only. No logic or behavior change. All tests
must pass; coverage must not decrease. | Owner, 2026-06-24 | B2 (Success/Failure contract for
Gemini) is deferred to Phase 2 since it changes return types callers depend on. D3 (typed-config
migration), D2 (app-factory), and B5 (segmenter validator) are pure-refactoring-possible and are
the first targets. |
41     | D2 | First targets: D3 ? B5 ? D2 in that order (increasing risk). D3 (typed-
config migration) is the safest ? fix the validator callers one module at a time. B5 (segmenter
validator) is a one-liner. D2 (app-factory) is the largest but still a refactoring. B2
(Success/Failure in Gemini) is the first Phase-2 item. | Owner, 2026-06-24 | Each lands via a
separate, small PR branched from `master`; no PR stacking. |
42     | D3 | Docs-only PR first. This document lands as a single PR with no code
changes. The subsequent code PRs reference this doc. | Owner, 2026-06-24 | The doc is approval
for the approach; individual code PRs are approval for the implementation. |
43     | D4 | Coverage ratchet: the CI `coverage` lane must not decrease for any PR in
this project. | Convention from `CODE_AUDIT_AND_TEST_HARDENING.md` | Every code PR must include
updated/additive tests. Pure refactorings that remove dead code may increase coverage; a drop is
a hard fail. |
44     | D5 | Follow the CODE_MAP placement rules. No new top-level files without a
placement-rule match. Tools go in `tools/`, standalone run-drivers in `scripts/`, eval library
code in `backend/eval/`. | `docs/CODE_MAP.md` §0 | Keeps the repo navigable as it evolves. |
45     | D6 | mypy quarantine ratchet: when touching a quarantined module, try removing
its `ignore_errors = True` entry from `mypy.ini`. | `mypy.ini` | D2 (app-factory) is the biggest
opportunity ? if `backend.main` and `backend.api.job_worker` can leave quarantine, that's a
major milestone. |
46     | D7 | Absorb relevant BACKLOG items. Deferred hardening items from
`docs/BACKLOG.md` that overlap with this audit are absorbed into this project rather than left
in BACKLOG. Items that are feature-gaps (not code-cleanup) stay in BACKLOG. | PR #352 review,
2026-06-24 | D2 and O1 were already independently found; D13 and D14 are new absorptions. The
train/eval splits guardrail stays in BACKLOG (it needs a collector-side manifest change). |
47
48     ---
49
50     ## 3. Findings register (full audit)
51
52     ### 3a. Bugs & Defects
53
54     | ID | Severity | File(s) | Issue | Phase |
55     |---|-----|-----|-----|-----|
56     | B1 | HIGH | `backend/pipeline/segmenters/motion.py` | Fabricated data.

```


Random values for server/receiver/shot_count/events/intensity/avg_speed_kmh persisted as real data. | 2 |

57 | **B2** | **HIGH** | `backend/providers/gemini.py` | **Silent zero-segment returns mask API failures.** `analyze_video()` returns `([], metrics)` on ANY failure, indistinguishable from "no rallies." | 2 |

58 | **B3** | **MEDIUM** | `backend/pipeline/validators/pts\_continuity.py` | Hardcoded 10.0s gap threshold (comment says 8.0s), not config-driven. | 3 |

59 | **B4** | **MEDIUM** | `backend/pipeline/stitcher/compiler.py::\_parse\_time` | Unparseable timestamps silently default to 0.0 ? could stitch from time zero. | 3 |

60 | **B5** | **MEDIUM** | `backend/config/models.py::IndexingConfig` | `segmenter_must_be_registered` validator is a **no-op pass-through**. Typo'd segmenter passes validation. | **1** |

61 | **B6** | **LOW** | `backend/pipeline/validators/resolution\_fps.py` | Hard-fails if `format_check` didn't run first ? ordering is an import-time side-effect. | 3 |

62 | **B7** | **LOW** | `backend/eval/calibrate\_wasb.py::load\_golden\_gts` | Silently skips bad rows (opposite of `annotations.load_annotations` which RAISES). | 3 |

63 | **B8** | **LOW** | `backend/eval/regression.py` | No baseline ? `regressed=False` (silent pass). Library function diverges from `run_regression.py` exit 3. | 3 |

64

65 **### 3b. Technical Debt / Design Smells**

66

ID	Severity	Area	Issue	Phase
D1	HIGH	`backend/main.py`	Massive god-module (~1450 lines). CLI, API, worker orchestration, upload, debug-clip, trim, reingest all in one file. 2	
D2	HIGH	`backend/main.py` + `backend/api/job_worker.py`	Optional module globals (<code>db_instance</code> , <code>global_config</code>). NPE-prone; mypy-quarantined. 1	
D3	HIGH	`backend/config/models.py`	Dual config access. Typed Pydantic model + legacy <code>.get()</code> dict shim. <code>extra='allow'</code> silently keeps typo'd keys. 1	
D4	MEDIUM	`backend/config/loader.py`	Shallow top-level merge: a <code>config.json</code> section REPLACES the defaults dict for that section ? new fields missed on upgrade. 2	
D5	MEDIUM	`backend/main.py`	<code>print()</code> used in production backend code (~20 calls in <code>--debug-clip</code>). Bypasses logging. 3	
D6	MEDIUM	`backend/eval/`	Duplicate "annotations" modules: <code>backend.eval.annotations</code> vs <code>backend.annotations</code> . Different schemas, different error handling. 3	
D7	MEDIUM	Multiple	Hardcoded tuning constants duplicated by copy (BASELINE, TUNED, padding). 3	
D8	MEDIUM	`backend/pipeline/segmenters/base.py`	Registry instantiates <code>cls(None, {})</code> to read <code>.name</code> ? <code>__init__</code> must not touch <code>self.db</code> / <code>self.config</code> . Implicit contract. 3	
D9	LOW	`backend/pipeline/validators/__init__.py`	Import mutates global registry as side-effect. Removing import = de-registering validator. 3	
D10	LOW	`backend/reporting/spec.yaml` + `harvest.py`	Tight coupling via magic <code>when.path</code> strings. Rename a detail ? rule silently stops firing. 3	
D11	LOW	`backend/pipeline/stitcher/compiler.py`	Two-stage codec contract is implicit. Change a preset ? concat breaks. 3	
D12	LOW	`tools/generate_highlights.py`	Uses <code>sys.path.insert(0, ...)</code> hack instead of proper package install. 3	
D13	MEDIUM	`requirements.txt`, `pyproject.toml`	Absorbed from BACKLOG. <code>supervision</code> ByteTrack is deprecated as of 0.28.0 and removed in 0.30.0 . The <code><0.30.0</code> pin in <code>requirements.txt</code> and <code>pyproject.toml</code> is a stopgap ? migration to the standalone <code>trackers</code> package is needed before the pin can be lifted. 2	
D14	LOW	`tests/` directory	Absorbed from BACKLOG. 90+ test files in a flat <code>tests/`</code> directory. Owner wants reorganization by subsystem (e.g. <code>tests/eval/`</code> , <code>tests/audio/`</code>) to reduce fragmentation. The exact grouping metric is the owner's design choice. 3	

83

84 **### 3c. Operational Risk**

85

ID	Severity	Area	Issue	Phase
O1	HIGH	`backend/providers/gemini.py`	Orphaned remote files on process death. No GC for uploaded videos on Gemini File API. 2	

```

89      | **Q2** | MEDIUM | `backend/main.py` | Single `ThreadPoolExecutor(max_workers=1)` ? one
stuck job blocks all. | 2 |
90      | **Q3** | MEDIUM | `backend/infrastructure/database.py` | SQLite WAL + busy_timeout but
concurrent writers (CLI + server) risk `SQLITE_BUSY`. | 3 |
91      | **Q4** | LOW | `backend/pipeline/detectors/tracknet_runner.py` | `weights_path=''` ?
silent zero results. No startup weights-exist validation. | 3 |
92      | **Q5** | LOW | `backend/utils/single_instance.py` | Windows byte-range locks may not
work on network drives (Google Drive, SMB). | 3 |
93
94      ### 3d. Quick Wins (Low Risk, High Value)
95
96      | ID | Area | Issue | Phase |
97      |----|-----|-----|-----|
98      | **Q1** | `backend/config/models.py` | Fix the no-op `segmenter_must_be_registered`
validator (same as B5). | **1** |
99      | **Q2** | `backend/pipeline/segmenters/motion.py` | Fix return type annotation:
`List[Dict]` ? `Result[List[Dict[str, Any]]]`. | 1 |
100     | **Q3** | `run_eval.py`, `run_regression.py`, `scripts/` | Deduplicate
`_default_db_path()` helper into a single shared utility. | 1 |
101     | **Q4** | `backend/config/loader.py` | Cache builtin defaults dict at module level
(avoid double-instantiation of `AppConfig`). | 1 |
102     | **Q5** | `requirements.txt` vs `pyproject.toml` | Add comment explaining why
`torch`/`torchvision` are in requirements.txt but not `pyproject.toml`. | 1 |
103     | **Q6** | `ruff.toml` | Enable `"I" (isort) rule ? auto-fixable, consistent import
ordering. | 1 |
104     | **Q7** | `mypy.ini` | Type-clean `motion.py` (small, legacy) as a ratchet exercise. |
2 |
105     | **Q8** | `backend/main.py` | Add comment explaining lazy `import cv2` in `--debug-
clip` (cv2 not in LIGHT tier). | 1 |
106
107     ### 3e. Absorbed from BACKLOG
108
109     These findings were already recorded in `docs/BACKLOG.md` as deferred items from the
2026-06-10 hardening sweep. They are absorbed into this project for execution:
110
111     | My ID | BACKLOG item | BACKLOG status | Absorbed as |
112     |-----|-----|-----|-----|
113     | **D2** | "App-factory / lifespan + lazy torch-cv2 imports" | ? ? "Belongs with the
async-job slice (#16)" ? #16 is now done; unblocked | Phase-1 target (P5) |
114     | **D1** | "WSL temp/frame/cache + Gemini remote-file GC" | ? ? "staged videos, PNG
frame dumps, `_wasbcache`, and orphaned Gemini uploads are never cleaned" | Phase-2 target. The
WSL half is partially handled by `keep_frames=false`; the Gemini half is new. |
115     | **D13** | "`supervision` ByteTrack deprecation" | ? ? pinned `<0.30.0` as stopgap;
migration to `trackers` package needed | New finding in this doc (see S3b) |
116     | **D14** | "`tests` reorganization to reduce fragmentation" | ? ? owner's design
choice on grouping metric needed first | New finding in this doc (see S3b) |
117
118     **Not absorbed** (kept in BACKLOG as feature-gap, not code-cleanup):
119     - ? **Train/eval separation guardrails** ? `splits.py` exists but needs collector-side
manifest change; this is a cross-repo feature, not a code-cleanup item.
120
121     ---
122
123     ## 4. Design / architecture ? Phase 1 approach
124
125     ### 4a. D3 ? Typed-config migration (first target)
126
127     **What:** Convert all call sites that use `config.get("indexing",{}).get(...)` to typed
`AppConfig` attribute access.
128
129     **Reuse map:**
130     - `backend/config/models.py` ? `AppConfig`, `IndexingConfig`, `ValidationConfig` (single
source of truth for defaults + types)
131     - `backend/config/__init__.py` ? re-exports `load_config`, `AppConfig`
132     - `backend/pipeline/validators/*.py` ? 6 validators that consume `config: Dict[str,

```

```

Any]`
133     - `backend/pipeline/segmenters/motion.py` ? uses `self.config.get("indexing",{})`
134     - `backend/reporting/__init__.py` ? already accepts both typed and dict; normalizes via
`model_dump()`
135
136     **Plan:** Convert one module at a time, bottom-up. Validators first (they receive config
as a dict, change the contract to `AppConfig`), then segmenters, then remove the `.get()` compat
shim and flip to `extra='forbid'`.
137
138     **Test strategy:** Every module has existing tests. The config models have
`test_config.py` that pins defaults. No new test behavior needed ? existing tests should pass
byte-identically.
139
140     ### 4b. B5 ? Segmenter validator fix (second target)
141
142     **What:** Make `segmenter_must_be_registered` actually validate against
`segmenter_registry`.
143
144     **File:** `backend/config/models.py`, field validator on
`IndexingConfig.default_segmenter`.
145
146     **Risk:** Near-zero. A typo'd segmenter in `config.json` that currently "works" (passes
validation, fails at runtime) would now fail at startup. This is the DESIRED behavior ? catching
config errors early.
147
148     ### 4c. D2 ? App-factory refactor (third target, largest)
149
150     **What:** Replace `db_instance: Optional[Database] = None` + `global_config:
Optional[AppConfig] = None` module globals with a `create_app(config) -> FastAPI` factory that
injects state via `app.state`.
151
152     **Reuse map:**
153     - `backend/main.py` ? FastAPI app definition, all endpoints
154     - `backend/api/job_worker.py` ? resolves `db_instance` / `global_config` through `import
backend.main as _main`
155     - `tests/test_api_endpoints.py` ? already uses `TestClient`; would switch to
`create_app(test_config)`
156     - `tests/test_async_jobs.py` ? already monkeypatches `backend.main`
157
158     **Plan:** This is the highest-risk Phase-1 item but still a pure refactoring. The key
insight: Starlette's `app.state` is the standard way to attach shared state; FastAPI's
dependency injection can access it. Every endpoint that accesses `db_instance` / `global_config`
would use `request.app.state.db` / `request.app.state.config` instead. The worker would receive
these explicitly rather than resolving through the module object.
159
160     **Test strategy:** All existing tests must pass byte-identically. The `TestClient` usage
changes from `from backend.main import app` to `from backend.main import create_app; app =
create_app(test_config)`.
161
162     ---
163
164     ## 5. Honest limits ? what Phase 1 does NOT do
165
166     - **Does NOT change any behavior.** The Gemini provider still returns `[]` on failure;
the PTS validator still has a hardcoded 10.0s gap.
167     - **Does NOT fix B2 (Success/Failure in Gemini).** That changes the return contract
callers depend on ? deferred to Phase 2.
168     - **Does NOT split `main.py` (D1).** The god-module split is deferred ? D2 (app-factory)
is a necessary precursor.
169     - **Does NOT address any operational risks (O1?O5).** Those need behavioral changes.
170     - **A green suite after Phase 1 does NOT mean the code is "clean."** It means the
structural debt targeted by D2/D3/B5 is resolved. The remaining 28 items still need attention.
171
172     ---
173

```

```

174    ## 6. Deliverables & progress tracker ? **source of truth**
175
176    Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking.
177
178    | ID | Deliverable | Depends on | Gated? | Status | PR |
179    |----|-----|-----|-----|-----|----|
180    | **P0** | **Docs-only PR ? this project doc** + index in `docs/README.md` | ? | Owner
approval | ? | [352](https://github.com/Khelsutra/badminton-highlight-indexer/pull/352) |
181    | **P1** | **WASB cache test fix (Windows)** ? 2 pre-existing failures | ? | No | ? |
[353](https://github.com/Khelsutra/badminton-highlight-indexer/pull/353) |
182    | **P4** | **B5: Fix `segmenter_must_be_registered` validator** | P0 | No | ? |
[354](https://github.com/Khelsutra/badminton-highlight-indexer/pull/354) |
183    | **P1** | **D3-a: Convert validators to typed `AppConfig`** (6 files) | P0 | No | ? |
[355](https://github.com/Khelsutra/badminton-highlight-indexer/pull/355) |
184    | **P6a** | **Q3: Deduplicate `_default_db_path`** | P0 | No | ? |
[356](https://github.com/Khelsutra/badminton-highlight-indexer/pull/356) |
185    | **P6b** | **Q4: Cache builtin defaults** | P0 | No | ? |
[358](https://github.com/Khelsutra/badminton-highlight-indexer/pull/358) |
186    | **P6c** | **Q5: requirements.txt torch comment + Q6: isort (I) + Q8: cv2 comment** |
P0 | No | ? | [359](https://github.com/Khelsutra/badminton-highlight-indexer/pull/359)
[360](https://github.com/Khelsutra/badminton-highlight-indexer/pull/360)
[361](https://github.com/Khelsutra/badminton-highlight-indexer/pull/361) |
187    | **T1** | **Tier 1: Q2 (motion return type) + D11 (codec docstring)** | P0 | No | ? |
[362](https://github.com/Khelsutra/badminton-highlight-indexer/pull/362) |
188    | **T2** | **Tier 2: B6 (resolution_fps fallback + warning log)** | P0 | No | ? |
[363](https://github.com/Khelsutra/badminton-highlight-indexer/pull/363) |
189    | **T3** | **Tier 3+4: D12 (sys.path hack) + B3 (PTS gap config) + O5 (lock note) + D7
(centralize TUNED)** | P0 | No | ? | [364](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/364) |
190    | **P6d** | **~D7: Centralize hardcoded tuning constants~ ? folded into T3 | P0 | No | ?
| [364](https://github.com/Khelsutra/badminton-highlight-indexer/pull/364) |
191    | **P2** | **D3-b: Convert segmenters to typed `AppConfig`** (15 files: 5 segmenters + 4
detectors + worker + eval + tools) | P1 | No | ? |
[366](https://github.com/Khelsutra/badminton-highlight-indexer/pull/366) |
192    | **P3** | **D3-c: Remove `.get()` compat shim; flip `extra='forbid'`** (18 files,
+87/-169) | P2 | No | ? | [367](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/367) |
193    | **P5** | **D2: App-factory refactor** ? replace module-level Optional globals with
`create_app(config, db)` ? FastAPI | P3, P4 | No | ? |
[370](https://github.com/Khelsutra/badminton-highlight-indexer/pull/370) |
194    | **P7** | **Phase-1 verification:** full suite green, coverage ? 84%, mypy ratchet |
P1 P6 | No | ? | 1463/1463 pass (#371); `backend.api.job_worker` lifted from mypy quarantine |
195    | ? | ? | ? | ? | ? | ? |
196    | **P8** | **B2: Error-flagging in Gemini provider metrics** (Phase 2) | P7 | No | ? |
[373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |
197    | **P9** | **B1: Remove fabricated data from motion segmenter** ? random
server/receiver/shot_count/speed ? `None`; events dropped | P7 | No | ? |
[385](https://github.com/Khelsutra/badminton-highlight-indexer/pull/385) |
198    | **P10** | **D1: Split `main.py` god-module** (~1450 lines ? smaller modules) | P7 | No
| ? | ? |
199    | **P11** | **O1: Gemini remote-file GC** ? per-call `_safe_delete` (#373) + startup
orphan sweep scoped to our `UPLOAD_TAG`-tagged files, grace-gated, HTTP-timeout-bounded | P7 |
No | ? | [380](https://github.com/Khelsutra/badminton-highlight-indexer/pull/380) |
200    | **P12** | **D13: ByteTrack deprecation** ? migrated `sv.ByteTrack` ?
`trackers.ByteTrackTracker` (git dep; PyPI wheel broken) | P7 | No | ? |
[386](https://github.com/Khelsutra/badminton-highlight-indexer/pull/386) |
201    | **P13** | **D4: Deep config merge** ? section-level replace ? recursive merge | P7 |
No | ? | [373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |
202    | **P14** | **O2: Single-executor risk** ? documented with mitigations in job_worker.py
| P7 | No | ? | [374](https://github.com/Khelsutra/badminton-highlight-indexer/pull/374) |
203    | **P15** | **Q7: Type-clean `motion.py`** ? ratchet exercise | P7 | No | ? |
[373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |
204    | **P17** | **B2 (cont.): finish the `Success`/`Failure` return contract** ? the 4
callers that dropped provider error metrics (`trajectory_hybrid`, `yolo_hybrid`,

```

```

`gemini_refine.refine_window`, `gemini_refine.full_video_rallies`) now surface
`metrics["error"]` loudly instead of silently treating a failure as "0 rallies". (Full
`Success`/`Failure` return-type migration of the hybrids is follow-up.) | P8 | No | ? |
[#378](https://github.com/Khelsutra/badminton-highlight-indexer/pull/378) |
205 | **P16** | Phase 3 items (B4/B7/B8/D5/D6/D8-D10/D14/O3/O4) | P15 | No | ? | ? |
206
207 ---
208
209 ## 7. Open questions
210
211 1. **Q: Should D1 (split `main.py`) be Phase 1 or Phase 2?` ? Leaning Phase 2: D2 (app-
factory) is a natural precursor, and D1 is lower risk after the globals are gone. Owner to
confirm.
212 2. ~~Q: Should Phase 1 include enabling `I` (isort) in ruff.toml?~~ ? ? Shipped in
#360.
213
214 ---
215
216 ## 8. Definition of done
217
218 - [x] P0: This doc approved and merged to `master`
219 - [x] P1?P3: Typed-config migration complete ? no `.get()` dict access remains in
backend code; `extra='forbid'` on `AppConfig`
220 - [x] P4: `segmenter_must_be_registered` actually validates against the registry
221 - [x] P5: `create_app()` injects shared state via `app.state` (handlers read
`request.app.state`); `backend.api.job_worker` removed from mypy quarantine. (The Optional
module globals `db_instance`/`global_config` are **retained as a transitional back-compat shim**
and `backend.main` stays mypy-quarantined ? full removal is follow-up, see D1/P10.)
222 - [x] P6: Quick wins Q1?Q6, Q8 landed
223 - [x] P7: Full test suite ? 1463/1463 pass (#371); `ruff` + `mypy` clean;
`backend.api.job_worker` lifted from quarantine
224 - [x] All PRs are single-purpose, branched from `master`, with descriptive commit
messages
225 - [ ] `docs/README.md` updated to reflect the project's terminal state
226 - [ ] Final status flipped to `DONE` and this doc moved to
`docs/archives/past_projects/`
227
228 ---
229
230 ## 8.1 ? Next items ? Phase 2 pick-up (prioritized)
231
232 All items below are **Claude-doable**, small-PR-friendly, and unblocked. Pick up in
order:
233
234 ### Immediate (HIGH impact, low risk)
235
236 | # | ID | What | Why |
237 |---|---|---|---|
238 | ~1~ | ~**B2 / P17**~ | ~**Finish** the `Success`/`Failure` return contract~ ?
**? DONE (this PR).** The 3 callers that dropped provider error metrics now surface
`metrics["error"]` loudly. **Follow-up remains:** migrate the hybrid segmenters' `process_video`
return type from bare `List[Dict]` to the ABC's `Result[...]` (`Success`/`Failure`) ? currently
suppressed with `type: ignore[override]` in `trajectory_hybrid.py` (motion.py is already fully
migrated). |
239 | ~2~ | ~**Q1 / P11**~ | ~Gemini remote-file GC~ ? **? DONE (#380).** Per-call
`_safe_delete` (#373) + startup orphan sweep scoped to `UPLOAD_TAG`-tagged files, grace-gated,
HTTP-timeout-bounded. |
240
241 ### Near-term (MEDIUM impact)
242
243 | # | ID | What | Why |
244 |---|---|---|---|
245 | ~3~ | ~**B1 / P9**~ | ~Remove fabricated data from `motion.py`~ ? **? DONE
(#385).** Random server/receiver/shot_count/speed ? `None`; events dropped. |
246 | ~4~ | ~**D13 / P12**~ | ~ByteTrack deprecation migration~ ? **? DONE.**

```

```

`sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel broken ? filed Roboflow
issue). |
247 | 5 | **D1 / P10** | Split `main.py` god-module | ~1450 lines. D2 (app-factory) was the
necessary precursor ? now done. Extract: CLI handlers ? `backend/cli/`, upload routes ? already
in `backend/api/uploads.py`, debug helpers ? `scripts/`. |
248
249     ### Later (deferred)
250
251 | # | ID | What |
252 |---|---|-----|
253 | 6+ | **P16+** | B4, B7, B8, D5, D6, D8?D10, D14, O3, O4 ? Phase 3 items |
254
255     > **Removed from this list ? already shipped (see the §6 tracker); they were stale
entries left from before Round 3/4 landed:** D4/P13 deep config merge (#373 ?
`loader.py::_deep_merge`), O2/P14 single-executor risk documented (#374 ? `job_worker.py`),
Q7/P15 `motion.py` type-cleaned (#373 ? no longer in the `mypy.ini` quarantine list).
256
257     ---
258
259     ## 9. References
260
261     - **Audit source:** 33 findings from a deep whole-repo read on 2026-06-24, plus 2
absorbed from `docs/BACKLOG.md` (D13, D14) for a total of 35. Two additional BACKLOG items (D2,
O1) were already independently found and are cross-referenced in §3e.
262     - **Prior hardening sweep:** `docs/CODE_AUDIT_AND_TEST_HARDENING.md` (completed
2026-06-14) ? the previous audit-driven hardening round. This project is the next wave.
263     - **BACKLOG.md** `docs/BACKLOG.md` ? deferred items from the 2026-06-10 sweep; §3e maps
the absorbed items.
264     - **CODE_MAP.md** `docs/CODE_MAP.md` ? placement rules, gotchas, and the cross-cutting
footguns list.
265     - **mypy.ini** ? quarantine list and the ratchet instruction.
266     - **ruff.toml** ? lint rules; `I` (isort) deferred.
267     - **PROJECT_DOC_TEMPLATE.md** `docs/PROJECT_DOC_TEMPLATE.md` ? the template this doc
follows.
268
269     ---
270
271     ### Changelog
272     - `2026-06-24` ? Initial doc created from 33-item deep audit.
273     - `2026-06-24` ? Absorbed 2 findings from `docs/BACKLOG.md` (D13, D14). Cross-referenced
D2 and O1 in §3e.
274     - `2026-06-24` ? Phase 1 begun. P0 (#352) ? #353 ? P4/B5 (#354) ? P1/D3-a (#355) ?
P6a/Q3 (#356). Status ? IN PROGRESS.
275     - `2026-06-24` ? **Round 2.** Quick wins (#358 Q4, #359 Q5, #360 Q6 isort, #361 Q8) ?
Tier 1 (#362 Q2+D11) ? Tier 2 (#363 B6) ? Tier 3+4 (#364 D12+B3+O5). 11 items shipped total.
276     - `2026-06-24` ? **Round 3.** P2/D3-b segmenters (#366) + P3/D3-c extra='forbid' (#367).
18 items shipped.
277     - `2026-06-24` ? **Round 4 (final).** P5/D2 app-factory (#370) + test fixes (#371).
Status ? PHASE 1 COMPLETE. Suite 1463/1463 green. §8.1 added with prioritized Phase 2 pick-up.
278     - `2026-06-25` ? **§8.1 reconciled with the §6 source-of-truth (docs-only).** Removed
three already-shipped items that §8.1 still listed as deferred future work ? D4/P13, O2/P14,
Q7/P15 (all merged in #373/#374) ? and split the unfinished half of B2 out of the done P8 into a
new **P17** (the provider return contract + the `trajectory_hybrid`/`gemini_refine` callers that
still drop `metrics["error"]`). Grounded against `loader.py::_deep_merge`, `mypy.ini`,
`backend/results.py`, and `backend/providers/gemini.py` + its callers. No code change.
279     - `2026-06-25` ? **Accuracy fixes (docs-only).** (1) §8 DoD P5 said the module globals
were "replaced with `app.state`"; corrected ? `create_app()` *added* `app.state` and lifted
`job_worker` from quarantine, but `db_instance`/`global_config` remain as a transitional shim
and `backend.main` is still mypy-quarantined (`backend/main.py:264`, `mypy.ini`). (2) `main.py`
size "~1300" ? "~1450" (now 1448 lines) in §3b/§6/§8.1.
280     - `2026-06-25` ? **P17 done (B2 finish).** The 4 callers that dropped provider error
metrics (`trajectory_hybrid.py`, `yolo_hybrid.py`, `gemini_refine.refine_window`,
`gemini_refine.full_video_rallies`) now check `metrics["error"]` and log loudly instead of
silently treating a provider failure as "0 rallies" ? matching the pattern the `gemini`
segmenter already uses. (`yolo_hybrid` was caught during PR #378 review ? it bound `metrics` but

```

never read it.) Suite 1484/1484 green; ruff + mypy clean. Follow-up: migrate the hybrids' `process\_video` return type from bare `List[Dict]` to the ABC `Result[...]` contract (motion.py is already migrated; hybrids suppressed with `type: ignore[override]` for now).

281 - `2026-06-25` ? **P9 done (B1 finish).** Removed fabricated `random` data from the motion segmenter ? server/receiver/shot_count/avg_speed_kmh ? `None`, serve/smash/foul/winner events dropped entirely (motion was the only `add\_event` caller). PR #385 review caught a real regression: `run\_process\_and\_stitch.py` crashed on `None >= min\_shot\_count` (the `.get(?, 0)` default no longer applied) ? fixed with a None-guard matching the API path. Suite 1492/1492 green; ruff + mypy clean.

282 - `2026-06-26` ? **P12 CI fix (#386).** The D13/P12 PR failed the `full-suite` lane with `ModuleNotFoundError: No module named 'trackers'`. Two root causes: (1) the two new tests hard-imported the standalone `trackers` package, but it's a git-only dep (PyPI wheel is metadata-only ? roboflow/trackers#474) deliberately omitted from `pyproject.toml`, and the lane installs via `pip install -e .[dev]`; (2) the git ref `@v2.5.0` was wrong ? roboflow/trackers uses bare version tags (`2.5.0`), so pip failed with `pathspec 'v2.5.0' did not match`. Fixes: added a `pip install trackers @ git+?@2.5.0` step to both `ci.yml` and `nightly.yml` full-suite lanes (same out-of-band pattern as torch); fixed the tag in `requirements.txt`; and guarded the package-importing test with `pytest.importorskip("trackers")` (matching the repo's `obsreport` soft-dep convention). All 10 CI checks green.

283

18. user (2026-07-02T04:48:20.749Z)

```

1      # Data in GCS ? the corpus source-of-truth map
2
3      > **Scope:** this map covers the **training corpus only** (`gs://khelsutra-rally-
corpus`). **User-uploaded
4      > videos are a SEPARATE, already-shipped surface** ? `POST /api/assets` + the per-
request `medium_uri` +
5      > `GET /api/capabilities.storage` (see `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) ? and
must stay physically
6      > isolated from the corpus bucket (the **D1** compliance boundary: consent / no-minors /
not-training holds at
7      > the storage layer). The deploy layer's bucket names now live in one place
(`deploy/gcp/buckets.env`).
8
9      **Status:** LIVE / in-progress (migration not complete). **Owner-gated** decisions in
$7.
10     **Last inventoried:** 2026-06-22 against `gs://khelsutra-rally-corpus`.
11     **Goal (owner, 2026-06-22):** run **all** training + inference from the cloud with **all
data in GCS**,
12     so the local Windows box (and the intermittently-mounted external drive) is **no longer
a blocker**.
13
14     > **? Other sessions: this is where the data lives.** Pull from `gs://khelsutra-rally-
corpus`, not
15     > from anyone's `C:` / `F:` / Google Drive. If something you need isn't in the bucket
yet, it's in the
16     > **gap table ($5)** ? flag it / migrate it, don't hard-code a local path.
17
18     This doc is the **data map**. The *machine* runbooks are
[`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md)
19     (spin up a box) and [`deploy/gcp/README.md`](../deploy/gcp/README.md) (provision GPU +
train/infer).
20     The cloud engine itself was built by the (archived) DECOUPLED_COMPUTE project
21     (`docs/archives/past_projects/DECOUPLED_COMPUTE/`); this doc does **not** redesign it ?
it records
22     *where the bytes are* and *what's still missing*.
23
24     ---
25
26     ## 1. The bucket
27
28     | | |
29     |---|---|

```

```

30 | Bucket | gs://khelsutra-rally-corpus (single corpus bucket) |
31 | Project | gen-lang-client-0306026826 (billing *Khelsutra Billing*) |
32 | Region | asia-south1 (Mumbai), UBLA, co-located with the GPU VMs |
33 | Auth | SA rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com
    (roles/storage.objectAdmin); key at ~/gcp/rally-worker-sa.json (local only, **never
    committed**). Interactive: gcloud auth login. |
34 | Size today | ~2.06 GB (mostly past dry-run reels/outputs ? **not** the full corpus) |
35
36 The engine is backend-agnostic by URI (backend/storage/ref.py): gs:// (or
gs://),
37 s3://, gdrive://, or a bare local path. Pick the backend with config + creds, no
code change**.
38
39 ---
40
41 ## 2. Canonical layout (the target contract)
42
43 The cloud worker's input/output contract (backend/worker/__main__.py; lineage:
44 DECOUPLED_COMPUTE/04):
45
46 | Prefix | Holds | Who reads/writes it |
47 |---|---|---|
48 | videos/ | source videos / proxies (<clip>.mp4) | **input** to worker infer &
worker train |
49 | labels/ | golden rally labels (<clip>.rallies.csv) | **input** ground truth to
worker train + all eval |
50 | trajectories/ | cached WASB trajectory CSVs | ? **eval/litmus input** (GPU-free),
consumed via manifests/golden.json. **NOT** read by worker train (it synths or re-extracts)
|
51 | weights/<version>/ | trained Gen-0 bundles (weights.json + a run manifest.json)
| **output** of worker train |
52 | models/ | upstream pretrained weights (wasb_badminton_best.pth.tar) | **staged to
a local path** ($WASB_WEIGHTS / ~/models/?) for the detector ? not read from gs:// in
place |
53 | outputs/<video_id>/ | per-video inference outputs (intervals, etc.) | **output** of
worker infer |
54 | highlights/, promo/ | rendered reels + promo artifacts | output / demo |
55
56 > **Note on the manifest.** worker train does **not** read a pre-staged corpus
manifest.json ? it
57 > *builds* one on-box at train time from labels/ (+videos/), extracting trajectories
**synthetically
58 > by default** (--trajectory-source synth, CPU-mock) or for real via opt-in
--trajectory-source
59 > detector (needs videos/). A manifest.json only appears as an **output** inside
each
60 > weights/<version>/ bundle. The **eval/litmus** path is what wants a stable, path-
resolvable manifest
61 > ($7.3) ? and manifests/golden.json is currently consumed by **no code yet** (eval is
local-FS only;
62 > see Known blockers).
63
64 ---
65
66 ## 3. What's actually in the bucket today (2026-06-22)
67
68 > **Phase-1 landed 2026-06-22:** added trajectories/ (15), 15 canonical
labels/<date>_<name>.rallies.csv
69 > (= original-6 + GX010137/GX010141 + 7 new; the 7 old _proxy/bare dupes were
deleted), and
70 > manifests/golden.json. The table below is the **pre-Phase-1** snapshot; the
remaining gap is **videos**
71 > (14 of 15) ? see $6/$7b/$8.
72
73 | Prefix | Present | Drift / notes |

```



```

74 |---|---|---|
75 | `labels/` | 7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2_proxy`, `Boxhill
Doubles_proxy`, `GX020094_proxy`, `GX030094_proxy`, `mahadevpura-2_proxy` | ? **inconsistent
naming** (`_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`,
`GX010141` |
76 | `videos/` | 3: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4`
| ? corpus videos/proxies almost entirely absent |
77 | `trajectories/` | ? | ? **does not exist** (every trajectory is local-only) |
78 | `models/` | `wasb_badminton_best.pth.tar` | ? (? WASB-C3 weight-provenance still an
open legal item before any external share) |
79 | `weights/` | `0.1.0-l4/` | ? first real L4 train bundle |
80 | `outputs/` | one `/`, `parity/` | ? dry-run outputs |
81 | `highlights/`, `promo/` | reels + promo | ? |
82
83 **Drift summary:** the bucket has extra prefixes (`highlights/`, `promo/`, `models/`
alongside
84 `weights/`) and **inconsistent `labels/` naming**, and is **missing the bulk of the
corpus inputs**
85 (`videos/`, all `trajectories/`, most `labels/`). It reflects past *dry runs*, not a
complete corpus.
86
87 ---
88
89 ## 4. How the cloud consumes it
90
91 Provisioning + the verified commands live in
[`deploy/gcp/README.md`](../deploy/gcp/README.md) §3§5.
92 The data-facing entry points (`backend/worker/__main__.py`):
93
94 ```bash
95 # config (no secrets in config; auth via GOOGLE_APPLICATION_CREDENTIALS):
96 # { "storage": { "backend": "local", "gcs": { "project": "gen-lang-client-0306026826"
} } }
97
98 # INFERENCE on one cloud video -> outputs/<video_id>/ in the bucket
99 python -m backend.worker infer \
100 --video-uri gcs://khelsutra-rally-corpus/videos/<clip>.mp4 \
101 --out-uri gcs://khelsutra-rally-corpus --segmenter wasb_hybrid --config config.json
102
103 # TRAIN from the bucket (reads labels/ [+ videos/], extracts trajectories, builds
manifest on-box,
104 # runs training.gen0.harness, pushes weights/<version>):
105 RALLY_DETECTOR_IMPL=native python -m backend.worker train \
106 --corpus-uri gcs://khelsutra-rally-corpus --out-uri gcs://khelsutra-rally-corpus \
107 --version <v> --trajectory-source detector --segmenter wasb_hybrid --config
config.json
108 ```
109
110 A bare/relative path resolves under a configured `input_root`/`input_backend`
111 (`backend/storage/ref.py::resolve_input_uri`), so workflows can be written bucket-
relative.
112
113 ---
114
115 ## 5. The gap ? what's still local-only (why the box is still a blocker)
116
117 All of this is **tiny except the originals** ? the metadata that blocks cloud eval is <
25 MB total:
118
119 | Data | Where it is now | Size | In bucket? | Needed for |
120 |---|---|---|---|---|
121 | 20 WASB trajectory CSVs | `C:\?\Annotation Setup\Collect\Trajectories\` | **17.5 MB**
| ? none | eval/litmus (cached). *(No `cached` train source exists ? train synths or detector-
extracts.)* |
122 | golden manifest | `output/human_lovo_manifest.json` (C:-absolute paths) | 4.5 KB | ? |

```

```

eval corpus definition |
123 | original-6 labels | `F:\[Khelsutra]?\\Training\Golden Labelled\*.rallies.csv` | ~12 KB
| ? | train + eval |
124 | 7 new-clip labels | repo `output/*.rallies.csv` (gitignored) *** F: archive (backed
up 2026-06-22) | ~12 KB | ? partial/inconsistent | train + eval |
125 | source videos / proxies | F: archive + Google Drive golden folder | **100s of GB** | ?
3 only | infer + train (real WASB extraction) |
126
127 > The corpus *metadata* (trajectories + labels + manifest, ~25 MB) is what actually
blocks GPU-free
128 > cloud eval and the served Gate-A litmus ? and it's trivially small to migrate. The
**videos** are the
129 > only large migration, and they already exist on **Google Drive** (the golden-corpus
folder), so they
130 > can go **Drive ? bucket from a cloud box**, keeping the local machine out of the path
entirely.
131
132 ---
133
134 ## 6. Migration plan (close the gap)
135
136 **Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker).** Stage all
trajectories,
137 all labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`.
Pending the $7
138 naming decision. Sketch:
139
140 ```bash
141 # trajectories (decision-free; absent today)
142 gcloud storage cp "C:\?\Collect\Trajectories\*.csv" gs://khelsutra-rally-
corpus/trajectories/
143 # labels (after the $7.1 naming decision) -> labels/<canonical-name>.rallies.csv
144 # eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object ($7.3)
145 ```
146
147 **Phase 2 ? videos ? `videos/<name>.mp4`** (bucket name = the eval `name`, consistent
with
148 `trajectories/` + `labels/`). Sources (resolved 2026-06-22 ? see
`scratch/copy_videos_to_gcs.ps1`):
149
150 | Source | Clips | Size |
151 |---|---|---|
152 | Drive golden folder `?/Golden Label Videos Request - June 15/[Done] Video N/`
(**proxies**) | `mahadevpura_1/2`, `GX010128`, `GX020094`, `GX030094`, `Badminton_BXH_2`,
`Boxhill_Doubles` (7) | ~13.5 GB |
153 | F: archive (**full MP4s** ? proxy-first before upload) | `GX010137`, `GX010141`, +
original-6 (`adarsh_avi_singles`, `kushagra_singles`, `largetest_doubles`, `gbaaddy`,
`testlarge_short`, `mahadevpura_singles`) (8) | ~50 GB |
154
155 **Two ways to run it:**
156 - **Decoupled (preferred ? no local box):** `rclone` on a cloud VM with a Google-Drive
remote ?
157 `rclone copy gdrive:"<?/[Done] Video N>/<proxy>.mp4" :gcs:khelsutra-rally-
corpus/videos/<name>.mp4`
158 (one-time owner step: `rclone config` ? `gdrive` remote OAuth). Or `gdown <file-id>`
on the box for
159 the shared folder. This is the path that actually removes the local box.
160 - **Interim (today, owner box ? streams Drive/F: ? local ? GCS):** `pwsh
scratch/copy_videos_to_gcs.ps1
161 -Execute` (dry-run without `-Execute`; `-Only <name>` for one; skip-existing). Uses
local bandwidth.
162
163 **Priority:** the 7 Drive **proxies** first (already 1080p, sufficient for detection).
The 8 large F:
164 full MP4s should be **proxied to 1080p first** (cheaper storage + faster cloud decode)

```

```

before upload.
165     ? Two pre-existing non-canonical video objects (`videos/GX030094_proxy.mp4`,
`videos/mahadevpura-1.MP4`)
166     are superseded once `videos/GX030094.mp4` / `videos/mahadevpura_1.mp4` land ? verify-
dupe-then-delete
167     (same as the label cleanup). Track per-clip in $8.
168
169     **Phase 3 ? make eval/litmus read GCS (real code, NOT a config flip).** The eval drivers
170     (`served_gate_a.evaluate_video`, `calibrate_wasb.load_golden_gts`,
`TrackNetRunner.parse_trajectory_csv`)
171     open `spec["trajectory"]`/`spec["labels"]` with plain `open()` and gate on
`os.path.isfile()` ? they
172     have **no storage-layer fetch**, so a `gs://` manifest cannot be consumed today. Phase-3
must thread
173     `storage.fetch`/`resolve_input_uri` through those paths first. Then: re-probe
`fps`/`frame_width` from
174     the bucket videos (don't trust the copied values ? a wrong fps silently mis-aligns every
label), and
175     **re-measure the served Gate-A litmus on the GCS 15-clip set as a [[launch-report-
convention]] event**
176     (dual-set RAW+golden no-regression, not a silent eval-set swap ? see PR #236).
177
178     ---
179
180     ## 7. Decisions
181
182     1. **`labels/` naming ? LOCKED (owner, 2026-06-22):** `labels/<YYYY-MM-
DD>_<name>.rallies.csv`, where
183     `<name>` is the eval/manifest clip name and `<YYYY-MM-DD>` is the label's **authoring
date** =
184     the *earliest* mtime across all known copies (C: working copy + F: archive ? robust
to copy-resets;
185     a restored C: copy shows today, the F: archive keeps the real date). Fixes the
`_proxy` inconsistency
186     and disambiguates recycled GoPro names by date. *(Future, when videos are in the
bucket: optionally
187     add a `video_id`/MD5 sidecar for the strongest keying ? deferred, needs hashing each
video.)*
188     2. **Cache `trajectories/` ? YES, done 2026-06-22.** `trajectories/<name>.csv` cached so
cloud/CI eval +
189     the served Gate-A litmus run **GPU-free** (training can still regenerate via
`--trajectory-source detector`).
190     3. **Eval manifest in GCS ? done 2026-06-22.** `gs://khehsutra-rally-
corpus/manifests/golden.json`
191     (15 clips, `gs://` trajectory + label paths). The **local** litmus keeps its local-
path manifest;
192     flipping the litmus to read the GCS manifest is Phase 3.
193     4. **Originals migration scope ? OPEN** ? proxies-only vs full 4K, and how much of the
15-clip golden
194     set vs the wider archive. Source = Google Drive ? bucket from a cloud box (not the
local box).
195
196     > Phase-1 was run via a dry-run planner (resolve every clip's trajectory + label, stamp
the label's
197     > authoring date, build the `gs://` manifest) ? `gcloud storage cp` (planner kept in
gitignored
198     > `scratch/`; promote it to `scripts/` if corpus-sync recurs).
199     > ? **Cleanup DONE 2026-06-22:** the 7 pre-existing inconsistent label objects
(`*_proxy.rallies.csv`,
200     > bare `GX010128`/`mahadevpura-1`) were confirmed **byte-identical (MD5)** to the new
canonical names
201     > (content also on C:/F:) and **deleted** ? `labels/` now holds exactly the 15 canonical
objects.
202
203     > ? **gcloud gotcha (Phase-2):** `gcloud storage cp` treats `[...]` as a glob, so source

```

```

paths under
204 > `[Done] Video N/` (Drive) or `[Khelsutra] ?` (F:) fail with "matched no objects." The
owner-box
205 > script (`scratch/copy_videos_to_gcs.ps1`) stages such paths through a bracket-free
temp file; the
206 > `rclone` cloud path has no such issue.
207
208 ---
209
210 ## 7b. Known blockers ? must-fix before the DoD (2026-06-22 adversarial review)
211
212 1. **? RESOLVED 2026-06-22 ([#319](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/319))
213 ? dated labels broke `worker train --trajectory-source detector`.** *History:*
`cmd_train` derived the
214 join key as `fname.replace(".rallies.csv","")` ? `2026-06-22_GX020094`, then looked
up
215 `video_by_stem.get(name)` against bare video stems (`GX020094`), so every clip was
skipped and train
216 aborted with `<2 videos`. **Fix:** the label?clip-name derivation is now the helper
`_label_clip_name()`
217 in [backend/worker/`__main__.py`](../backend/worker/`__main__.py`), which strips both the
218 `.rallies.csv`/`.csv` suffix AND a leading `^\d{4}-\d{2}-\d{2}_` prefix, so a dated
label and a legacy
219 bare label both resolve to the bare `videos/<name>` stem (preserves the locked human-
readable naming).
220 Tested: parametrized helper cases + a dated-label?bare-video `cmd_train` join
regression + per-label
221 skip-branch coverage. *(`synth` train + all eval were unaffected ? they don't join
videos.)*
222 2. **Phase-3 eval-on-GCS is net-new plumbing, not a flag** ? see §6 Phase-3. Eval is
local-FS only.
223 3. **`manifests/golden.json` is consumed by zero code today** (train ignores it; eval
can't fetch
224 `gs://`) ? it's correct-but-dormant until blocker #2 lands. Add a tiny manifest-
validator + a CI job
225 that authenticates to GCS and runs the litmus against it, or the DoD ("CI reads only
from `gs://`)")
226 stays unverifiable.
227 4. **Gating infra (owner-only, not done):** GPU `GPUS_ALL_REGIONS` quota is still **0**
(no VM ever
228 created ? `deploy/gcp/README` §2); cloud-box secrets (SA JSON + rotated
`GEMINI_API_KEY` + WASB
229 weights staged from `models/`) aren't set up persistently. No real cloud train/infer
runs until these.
230 5. **Reproducibility:** the migration + copy helpers live in gitignored `scratch/`
231 (`migrate_golden_to_gcs.py`, `copy_videos_to_gcs.ps1`) ? promote to `scripts/` so
Phase-2/re-syncs
232 are repeatable from the repo, and add a referenced **proxy-generation** step for the
8 large F: originals.
233
234 ## 8. Progress tracker
235
236 | Item | Status |
237 |---|---|
238 | Bucket + SA + provisioning (deploy/gcp) | ? built |
239 | Cloud worker `infer`/`train` by URI | ? built (`backend/worker`) |
240 | This data-map doc | ? 2026-06-22 |
241 | Phase 1: trajectories ? `trajectories/` (15) | ? 2026-06-22 |
242 | Phase 1: labels reconciled ? `labels/<date>_<name>` (15) | ? 2026-06-22 |
243 | Phase 1: eval manifest ? `manifests/golden.json` | ? 2026-06-22 |
244 | Phase 1: remove 7 stale `_proxy`/bare label objects | ? 2026-06-22 (MD5-verified
dupes, deleted) |
245 | Blocker §7b#1: dated labels join in `worker train --trajectory-source detector` | ?
2026-06-22 ([#319](https://github.com/Khelsutra/badminton-highlight-indexer/pull/319)) ?

```

```

`_label_clip_name` strips the ISO-date prefix |
246 | Phase 2: corpus videos/proxies ? `videos/<name>.mp4` | ? mapping + script done;
**1/15** demo'd (`mahadevpura_1`); 14 remain ? run `rclone` on a cloud box (preferred) or the
owner-box script |
247 | Phase 2: proxy the 8 large F: full-MP4 clips before upload | ? |
248 | Phase 3: eval/litmus default to the GCS manifest | ? |
249
250 **Definition of done:** a fresh GPU box (or CI) can run inference, the eval suite, and
Gen-0 training
251 reading **only** from `gs://khelsutra-rally-corpus` ? no `C:` / `F:` / local-Drive path
required.
252
253 ---
254
255 ## Related
256 - [`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md) ? spin up a box (storage backends in
$7).
257 - [`deploy/gcp/README.md`](../deploy/gcp/README.md) ? GCP GPU provisioning + train/infer
($5).
258 - [`VIDEO_LOCALITY_MODEL.md`](VIDEO_LOCALITY_MODEL.md) .
[`PER_VIDEO_WORKSPACE.md`](PER_VIDEO_WORKSPACE.md) ? locality tiers + `outputs/<video_id>/`.
259 - `docs/archives/past_projects/DECOUPLED_COMPUTE/` ? the cloud-engine design lineage.
260

```

19. assistant (2026-07-02T04:48:47.189Z)

Now let me examine the Python files with potentially stale ByteTrack docstrings, and verify code facts.

20. user (2026-07-02T04:48:49.628Z)

```

1      """Person-detection rally cue ? torchvision detector + ByteTrack + motion windowing.
2
3      Extracted verbatim from `yolo_hybrid` (fusion P1, `docs/MULTI_SIGNAL_FUSION_PLAN.md`):
4      the **person cue is a swappable producer**, not logic baked into one segmenter. Today
5      `HybridYoloSegmenter` is the only consumer (it delegates Stage-1 here, byte-
identically);
6      next the fusion segmenter consumes the SAME producer over shuttle-seeded windows, and
7      later the learned classifier reads its persisted features. Keeping detection here (not
in
8      the segmenter) is what makes "add a cue = register a producer" true.
9
10     Behaviour is unchanged from the in-lined version ? only the home moved. The one
11     exception is the ByteTrack swap (D13/P12): ``make_tracker`` now builds a
12     ``trackers.ByteTrackTracker`` with its constructor params pinned to ``sv.ByteTrack``'s
13     behaviour (see that method's docstring), so the person-cue recall is preserved.
14     - `lazy_load_model` ? load a permissively-licensed torchvision detector
15     - `make_tracker` ? a fresh ByteTrack tracker per chunk (standalone
`trackers` pkg)
16     - `detect_and_track` ? one frame ? confident persons with persistent IDs
17     - `extract_action_windows_from_chunk` ? a chunk ? high-motion candidate windows
18     with the motion stats (`peak/mean_velocity`, `active_frame_count`, `track_id_count`)
19     that feed the candidate `stats` JSON and the learned fusion features.
20
21     The model libs (torch, torchvision, supervision) are imported lazily inside the methods,
and the
22     detector weights load only on first detection (`lazy_load_model`) ? so importing the
segmenter
23     registry never loads a detector model NOR requires torch (the LIGHT tier ships no torch
? P2e;
24     the person cue is HEAVY-only and never instantiated under Gemini/motion/cpu-mock), and
tests can
25     mock the seams. (cv2 stays top-level ? opencv-python is a hard dependency, always
installed.)
26     Licensing: torchvision detectors are BSD + COCO weights; ByteTrack is MIT (via the

```

```

27     standalone ``trackers`` package, Apache-2.0, since D13/P12; was
``supervision.ByteTrack``)
28     (ADR-005; ultralytics' AGPL YOLO was dropped).
29     """
30
31     import logging
32     from typing import Any, Callable, Dict, List, Optional, Tuple
33
34     import cv2
35
36     from backend.utils.geometry import get_velocity_normalised
37
38     logger = logging.getLogger(__name__)
39
40     # torchvision detection models are trained on COCO where the "person" category is
41     # class 1 (index 0 is the implicit background). NB: ultralytics YOLO used class 0 ?
42     # this off-by-one is the classic gotcha when swapping detectors. Keep it named.
43     _PERSON_CLASS_ID = 1
44
45     # Permissively-licensed torchvision detectors selectable via config
46     # `indexing.detector_model`. All are BSD-licensed (code + COCO weights).
47     _DETECTOR_FACTORIES = {
48         "fasterrcnn_resnet50_fpn": "fasterrcnn_resnet50_fpn",
49         "fasterrcnn_mobilenet_v3_large_fpn": "fasterrcnn_mobilenet_v3_large_fpn",
50         "retinanet_resnet50_fpn": "retinanet_resnet50_fpn",
51     }
52     _DEFAULT_DETECTOR = "fasterrcnn_resnet50_fpn"
53
54
55     class PersonDetector:
56         """Stage-1 person detection + tracking + motion-window extraction (a rally cue).
57
58         Owns the torchvision model + per-chunk ByteTrack tracker. Constructed with the run
59         ``config`` (reads ``indexing.{use_gpu, detector_model, detector_score_threshold,
60         min_action_window_duration}``); the model loads lazily on first use.
61         """
62
63         def __init__(self, config: Dict[str, Any]):
64             self.config = config
65             # Dynamic torchvision module (None until lazy_load_model); typed Any so the
66             # `self.model([tensor])` inference call type-checks without a quarantine.
67             self.model: Any = None
68             self.device: Optional[str] = None
69
70         def lazy_load_model(self) -> None:
71             if self.model is not None:
72                 return
73
74             # Imported lazily so the module imports without torch/torchvision present (the
LIGHT tier
75             # ships no torch ? P2e) and so the test suite (which pre-sets self.model) never
pulls weights.
76             import torch
77             from torchvision.models import detection as tv_detection
78
79             idx_cfg = self.config.get("indexing", {})
80             if isinstance(idx_cfg, dict):
81                 from backend.config.models import IndexingConfig
82
83                 idx_cfg = IndexingConfig(**idx_cfg)
84                 use_gpu = idx_cfg.use_gpu
85                 self.device = "cuda" if use_gpu and torch.cuda.is_available() else "cpu"
86                 if use_gpu and not torch.cuda.is_available():
87                     logger.warning(
88                         "use_gpu is True but CUDA is not available. Falling back to CPU."

```

```

89         )
90
91         model_name = idx_cfg.detector_model
92         if model_name not in _DETECTOR_FACTORIES:
93             logger.warning(
94                 f"Unknown detector_model '{model_name}'. Falling back to
{_DEFAULT_DETECTOR}."
95             )
96             model_name = _DEFAULT_DETECTOR
97
98         builder = getattr(tv_detection, _DETECTOR_FACTORIES[model_name])
99         logger.info(f"Loading torchvision detector '{model_name}' on {self.device}...")
100         # weights="DEFAULT" pulls the best available COCO-pretrained checkpoint.
101         self.model = builder(weights="DEFAULT")
102         self.model.eval()
103         self.model.to(self.device)
104
105     def make_tracker(self, fps: float) -> Any:
106         """Create a fresh ByteTrack tracker for one chunk (track IDs are chunk-local).
107
108         D13/P12: migrated from ``supervision.ByteTrack`` (deprecated, removed in 0.30.0)
109         to the standalone ``trackers`` package (``ByteTrackTracker``). Imported lazily
110         so
111
112         tests can mock this method without the package installed.
113
114         The constructor params are pinned to ``sv.ByteTrack``'s behaviour so the
migration
115         is behaviour-preserving. The new package's defaults differ materially ?
116         ``track_activation_threshold`` 0.7 (vs sv's ~0.35 effective floor),
117         ``high_conf_det_threshold`` 0.6 (vs sv's ~0.35), and
118         ``minimum_consecutive_frames`` 2 (vs sv's 1). Left at the new defaults, a person
occlusion)
119         would never spawn a track, silently regressing rally recall; and the first
sampled
120         frame of every track would be emitted as -1 (dropped). Pinning all three keeps
the
121         0.5 ``detector_score_threshold`` pre-filter the effective floor and yields an ID
on
122         the first matched frame. ``minimum_iou_threshold`` is left at the package
default
123         (0.1), which is roughly equivalent to sv's matching gate.
124         """
125         from trackers import ByteTrackTracker
126
127         frame_rate = max(1, int(round(fps)))
128         return ByteTrackTracker(
129             frame_rate=frame_rate,
130             track_activation_threshold=0.25, # sv default; <= detector_score_threshold
(0.5)
131             high_conf_det_threshold=0.35, # match sv's det_thresh so [0.5,0.7) can
spawn
132             minimum_consecutive_frames=1, # sv default: emit a real ID on the first
frame
133         )
134
135     def detect_and_track(
136         self, frame: Any, tracker: Any, score_thresh: float
137     ) -> List[Tuple[int, Tuple[float, float, float, float]]]:
138         """Detect persons in a frame and assign persistent track IDs.
139
140         Returns a list of (track_id, bbox) where bbox is (x1, y1, x2, y2) in pixels.
This replaces the old ultralytics ``model.track(...)`` call, which bundled

```

```

1:"""Two-stage rally segmenter: torchvision person detection + ByteTrack tracking
2-(Stage 1, local) feeding an AI provider for semantic boundaries (Stage 2).
3-
4-Stage-1 detection now lives in `backend/pipeline/detectors/person_detector.py`
--
10-service free of copyleft obligations (see docs/MONETIZATION_AUDIT.md and ADR-005),
11-detection uses torchvision's permissively-licensed (BSD) detectors and the
12-association/tracking that YOLO.track() used to bundle in is provided by
13:supervision's MIT-licensed ByteTrack.
14-
15-The registry key stays "yolo_hybrid" for back-compat with existing configs and the
16-DB `detector` tags, even though no YOLO is involved anymore.
--
54-     @property
55-     def description(self) -> str:
56-         return (
57-             "Two-stage pipeline: torchvision person detection + ByteTrack for fast "
58-             "candidate filtering, then an AI provider for high-precision semantic
59-             boundaries."
60-         )
61-
62-     return None
63-
64-     logger.info(
65-         f"Initializing Hybrid Segmenter (torchvision + ByteTrack) for video:
66-         {video_path}"
67-     )
68-     return provider
69-
70-

```

22. user (2026-07-02T04:49:13.054Z)

```

2198 backend/main.py
torch>=2.12.0
torchvision>=0.27.0
supervision>=0.21.0

```

D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the standalone `trackers` package. The PyPI wheel is BROKEN (ships metadata only, no source ? see + deploy/cloudrun/Dockerfile ? edit ALL of them in lockstep on a bump, or the lanes/b silently resolve different versions.

```

trackers @ git+https://github.com/roboflow/trackers.git@2.5.0
scipy>=1.10

```

```

--- create_app/lifespan ---
90:app = FastAPI(title="Sports Highlight Indexer API")
93:def create_app(config: AppConfig, db: Database) -> FastAPI:
2184:     create_app(global_config, db_instance)
--- gemini GC ---
59:     def cleanup_orphaned_files(self, grace_sec: int = 3600) -> int:
--- M11/M12 files ---
backend/flywheel.py
backend/storage/gdrive_api.py
--- cv2 import ---
34:import cv2

```

23. user (2026-07-02T04:49:15.254Z)

```

69c6266 docs(nightly): add person-cue Stage-1 stats lane to the engine-regression plan (design)
(#389)
2028f98 docs(nightly): add person-cue Stage-1 stats lane to the engine-regression plan (design)
3624a8d feat(D13/P12): migrate ByteTrack to standalone trackers package (#386)

```


1283e18 docs(project): Decision-Snapshot Regression plan ? a deterministic quality-neutral proof (#383)
cd5b3d2 feat(01/P11): Gemini remote-file GC ? startup orphan sweep (#380)
947a36b docs: update CODE_CLEANUP_AUDIT ? Phase 1 complete, Phase 2 prioritized (#372)
9cd7598 fix(tests): resolve remaining test signature mismatches from app-factory refactor (#371)
de09ec8 refactor(main): app-factory + typed-config migration (D2/D3/P2/P3/P5/P7) (#370)
334ad0f feat(serving): M11 data-flywheel skeleton ? capture?shadow-eval?promote (default-OFF + consent faked-deny) (#336)
aac7633 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF) (#290)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity test (default-OFF) (#286)
--- gdrive_api PR ---
b6214e5 chore: ruff format the whole repo (259 files) (#387)
f523d3a refactor: enable isort (I) in ruff, auto-fix 143 files (Q6) (#360)
819e395 feat(serving): M12 ? gdrive-api:// OAuth/API StorageBackend (D14 Northstar) (#338)
--- flywheel PR ---
b6214e5 chore: ruff format the whole repo (259 files) (#387)
334ad0f feat(serving): M11 data-flywheel skeleton ? capture?shadow-eval?promote (default-OFF + consent faked-deny) (#336)

24. user (2026-07-02T04:49:29.211Z)

```
{"mergeStateStatus":"DIRTY","mergeable":"CONFLICTING","state":"OPEN","title":"refactor(D1/P10):  
extract orchestration.py from main.py (god-module split, step  
1)","updatedAt":"2026-06-26T14:37:18Z"}
```

Archive
Golden Labelled
Trajectories
golden_labels_badminton.csv
rally_review.csv
ls: cannot access 'C:/Users/avidu/Projects/Annotation Setup': No such file or directory

25. user (2026-07-02T04:49:30.939Z)

126:[Omitted long matching line]
127:[Omitted long matching line]

26. user (2026-07-02T04:49:38.560Z)

```
110      ## 7. Deliverables & progress tracker ? **source of truth**  
111      Legend: ? Todo . ? In progress . ? Done . ? Gated. **One small PR per row**, off  
`origin/master`, no stacking. Default-OFF where it touches core callers.  
112  
113      | ID | Deliverable | Depends | Gated? | Status | PR |  
114      |----|-----|-----|-----|-----|----|  
115      | M0 | This design doc + project dir + `runlogs/` + ADR-014 | ? | owner sign-off | ? |  
**? APPROVED 2026-06-21** |  
116      | M1 | `deployment.profile` + `compute_target` config + preset bundles +  
precedence/auto-detect; unit tests (no behaviour change, profile='' default) | M0 | safe | ? |  
[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |  
117      | M2 | `input_backend`/`input_root`; wire main CLI/API input through  
`StorageRef.parse_uri` (local=identity); storage-fake tests | M1 | safe | ? |  
[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |  
118      | M3 | Configurable output/scratch base ? kill hardcoded `output/`  
(`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden  
+ stub-E2E byte-identical | M1 | ? default-OFF + Launch Report on flip | ? |  
[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282) |  
119      | M4 | `DeviceContext` + WASB **CPU-fallback**; unified healthcheck/device wiring | M1 |  
safe (cuda default unchanged) | ? | [#283](https://github.com/Khelsutra/badminton-highlight-
```

```

indexer/pull/283) |
120 | M5 | truly-local profile end-to-end (preset + startup checks); first committed
runlog (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? |
[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-
checks+L4 parity ?; runlog ? owner GPU) |
121 | M6 | Cloud Run GPU dispatch shim (control plane ? job via storage ? re-claim) +
containerized worker image (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner
(GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)
(shim+image+mocked ?; real build/run = M7 ? owner) |
122 | M7 | cloud-serving profile e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch
on Cloud Run ? reel via signed URL; DEPLOY_REPORT (posterity, sample runs + contact sheet) |
M6 | ? owner (spend) | ? | ? |
123 | M8 | Per-job cost ledger (v6 migration: `owner_id, gpu_active_seconds,
wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + pricebook + aggregation +
`/api/.../cost` | M7 | ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission
? owner/follow-up) |
124 | M9 | Edge auth (Cf-Access extraction) + per-tenant scoping on `user_id`;
DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? |
[#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT
verify + owner_id tag, default-OFF ?; exposure-safety boot posture in `startup.py` ? edge-
auth-on requires the `allowed_video_root` jail + CF team/AUD or the server refuses to start, see
changelog; consent cols + ToS/DPA ? owner/counsel) |
125 | M10 | `byo_worker` / zero-infra-BYO ? design-only, DEFERRED | M8,M9 | ? explicit
owner approval | ? | ? |
126 | M11 | Data-flywheel harness (D12): on a consented adult job, capture the
upload + Gemini reel/rally output ? reliably store under the per-video workspace ? shadow-
eval owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? promote good ones to the
golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the
owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent
gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | skeleton landed (see changelog):
`backend/flywheel.py` + `FlywheelConfig` (default-OFF) + gated `_maybe_run_flywheel` hook,
consent faked-deny; live activation ? owner/counsel (consent schema) + owned-model/golden
data |
127 | M12 | gdrive-api (OAuth) StorageBackend (D14 Northstar): cloud reads the
source from + writes the stitched reel back to the user's Google Drive next to the source**
(per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local
mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? |
backend landed (see changelog): `backend/storage/gdrive_api.py` `GDriveApiStorage` +
`gdrive-api://` scheme + ADC build; hermetic fake-Drive tests. Live per-user OAuth app ?
owner** |
128
129 Testing strategy is a first-class deliverable ?
[`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). Run logs / sample runs for posterity ?
[`runlogs/`](runlogs/).
130
131 ## 8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
132 ?? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:
133 1. ? Cold-start ? D7** (scale-to-zero, accept cold-start).
134 3. ? Pricebook ? D8** (versioned DB table; USD internal / INR display).
135 4. ? Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to
proxy).
136 5. ? DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data;
counsel-gated to M9).
137 6. ? Quality floor ? D11** + the data-flywheel harness D12 / M11 (Gemini-ON until
the owned model clears the floor; capture?store?eval?goldenize meanwhile).
138
139 ?? The remaining smaller picks are LOCKED (owner, 2026-06-21) ? see §2 D16:
140 2. ? Stitch placement ? co-located** in the detect job (= D4; one metered unit; the
job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).
141 7. ? RESOLVED ? D14:** cloud needs a gdrive-api (OAuth) StorageBackend** (read the
source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ?
distinct from the local mount backend (which stays for truly-local). New milestone M12.
142 8. ? Codec ? D16:** libx264 ENCODE (protect parity) + NVDEC DECODE-only
(speed, no output-byte change); NVENC-encode deferred behind a config knob.

```

143 9. ? ****Input cap ? D16:**** ****hard-reject`<2GB`**** for v1, as a config cap (raise to 4GB later in one line).
144 10. ? ****Directory name ? `COMPUTE\_DECOUPLED\_SERVING/` CONFIRMED final**** (already the name everywhere + in ADR-014); `CLOUD\_RUN\_GPU\_SERVING/` dropped.

27. user (2026-07-02T04:51:05.592Z)

```
1      """Two-stage rally segmenter: torchvision person detection + ByteTrack tracking
2      (Stage 1, local) feeding an AI provider for semantic boundaries (Stage 2).
3
4      Stage-1 detection now lives in `backend/pipeline/detectors/person_detector.py`
5      (`PersonDetector`, a reusable rally-cue producer ? fusion P1) so the SAME person cue
6      can be consumed by the future fusion segmenter; this segmenter just orchestrates
7      chunking, the AI-provider handoff, and DB population around it. Behaviour is unchanged.
8
9      Stage 1 originally used the AGPL-licensed ultralytics YOLO. To keep the hosted
10     service free of copyleft obligations (see docs/MONETIZATION_AUDIT.md and ADR-005),
11     detection uses torchvision's permissively-licensed (BSD) detectors and the
12     association/tracking that YOLO.track() used to bundle in is provided by
13     supervision's MIT-licensed ByteTrack.
14
15     The registry key stays "yolo_hybrid" for back-compat with existing configs and the
16     DB `detector` tags, even though no YOLO is involved anymore.
17     """
18
19     import concurrent.futures
20     import logging
21     import os
22     import time
23     from typing import Any, Dict, List, Tuple
24
25     from backend.pipeline.detectors.person_detector import PersonDetector
26     from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
27     from backend.providers import provider_registry
28     from backend.sports import sport_registry
29     from backend.utils.paths import output_base
30     from backend.utils.video import extract_video_chunk, get_video_duration
```

28. user (2026-07-02T04:51:07.898Z)

tools/cleanup_caches.py
b6214e5 chore: ruff format the whole repo (259 files) (#387)
3b36f9f docs(CLAUDE.md): truth-sync the stale Current-state section to shipped reality (audit R2-5)
f28af73 P2: engine free-space guard (HTTP 507 + WSL frame-stage BLOCK) (#407)
4a7bfbe docs: record the launch NO-GO decision (owner, 2026-06-30) (#404)

29. user (2026-07-02T04:51:20.059Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)

30. user (2026-07-02T04:51:33.471Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)

31. user (2026-07-02T04:51:45.447Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)

32. user (2026-07-02T04:51:57.366Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DOC_STATUS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

33. user (2026-07-02T04:52:02.692Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DOC_STATUS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

34. user (2026-07-02T04:52:18.202Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DOC_STATUS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

35. user (2026-07-02T04:52:19.488Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DOC_STATUS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

36. assistant (2026-07-02T04:52:28.606Z)

Now docs/README.md lines 41-42.

37. user (2026-07-02T04:52:35.050Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

38. user (2026-07-02T04:52:38.216Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

39. user (2026-07-02T04:53:09.235Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

40. user (2026-07-02T04:53:21.791Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

41. user (2026-07-02T04:53:25.165Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

42. user (2026-07-02T04:53:34.330Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-

indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

43. user (2026-07-02T04:53:38.807Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

44. user (2026-07-02T04:53:51.046Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

45. assistant (2026-07-02T04:53:56.426Z)

Now the stale ByteTrack docstring in yolo_hybrid.py (doc-comment only).

46. user (2026-07-02T04:54:01.815Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/backend/pipeline/segmenters/yolo_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

47. assistant (2026-07-02T04:54:29.644Z)

Now the DATA_IN_GCS.md \$5 gap-table reconcile against its own \$7/\$8.

48. user (2026-07-02T04:54:32.798Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

49. user (2026-07-02T04:54:57.219Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

50. user (2026-07-02T04:55:05.405Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

51. user (2026-07-02T04:55:17.338Z)

[check-md-links] clean ? no broken internal links in live docs.
LINKCHECK_EXIT=0

52. user (2026-07-02T04:55:17.886Z)

All checks passed!
test_fusion_hybrid.py
test_tracknet_hybrid.py
test_trajectory_hybrid_equivalence.py
test_yolo_hybrid.py

53. user (2026-07-02T04:56:08.708Z)

```
..... [100%]
8 passed in 2.40s
IMPORT_OK
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 1 source file
```

54. assistant (2026-07-02T04:56:13.829Z)

All green. Reviewing the final diff before committing.

55. user (2026-07-02T04:56:17.365Z)

```
M backend/pipeline/segmenters/yolo_hybrid.py
M docs/BACKLOG.md
M docs/CODE_CLEANUP_AUDIT_2026-06-24.md
M docs/DATA_IN_GCS.md
M docs/DOC_STATUS.md
M docs/NEXT_STEPS.md
M docs/README.md
backend/pipeline/segmenters/yolo_hybrid.py | 6 +++-
docs/BACKLOG.md | 8 +++---
docs/CODE_CLEANUP_AUDIT_2026-06-24.md | 11 ++++----
docs/DATA_IN_GCS.md | 36 ++++++-----
docs/DOC_STATUS.md | 9 ++++---
docs/NEXT_STEPS.md | 8 +++---
docs/README.md | 4 +++-
7 files changed, 46 insertions(+), 36 deletions(-)
docs/reconcile-trackers-2026-07-02
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
```

56. user (2026-07-02T04:56:26.012Z)

```
1
8:> **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is **not**
cleared for an external/commercial launch; we stay in **development**. This is the expected
default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the **ship-gate
target (#172)** ? currently unset, needs corpus growth (and the R2 tolF1 ablation, item 8);
**(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean licensing).
P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev PRs ?
launching. Re-decide only when both unblockers clear.
```

57. user (2026-07-02T04:56:36.591Z)

```
> **? LAUNCH DECI
```

58. assistant (2026-07-02T04:56:47.487Z)

NO-GO block confirmed as untouched context. Committing with explicit paths.

59. user (2026-07-02T04:57:02.458Z)

```
docs/reconcile-trackers-2026-07-02
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
[dfdd2d9] docs: reconcile stale trackers/blurbs with shipped
state (audit R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
7 files changed, 46 insertions(+), 36 deletions(-)
dfdd2d9 docs: reconcile stale trackers/blurbs with shipped state (audit
R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
```

60. user (2026-07-02T04:57:15.464Z)

```
dfdd2d9 docs: reconcile stale trackers/blurbs with shipped state (audit
R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
remote:
remote: Create a pull request for 'docs/reconcile-trackers-2026-07-02' on GitHub by visiting:
remote:      https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/docs/reconcile-
trackers-2026-07-02
remote:
branch 'docs/reconcile-trackers-2026-07-02' set up to track 'origin/docs/reconcile-
trackers-2026-07-02'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      docs/reconcile-trackers-2026-07-02 -> docs/reconcile-trackers-2026-07-02
```

61. assistant (2026-07-02T04:57:22.839Z)

Branch verified clean (only my commit) and pushed. Opening the PR.

62. user (2026-07-02T04:57:53.427Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/420>

63. assistant (2026-07-02T04:58:02.720Z)

PR opened. Final state check and summary.

64. user (2026-07-02T04:58:05.551Z)

docs/reconcile-trackers-2026-07-02...origin/docs/reconcile-trackers-2026-07-02

```
dfdd2d9 docs: reconcile stale trackers/blurbs with shipped state (audit
R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
```

65. user (2026-07-02T04:58:36.050Z)

Structured output provided successfully